# AoE2 Civ Explorer

## Code documentation

What each part of the code does, and why

10 November 2026 · 14 chapters and 57 subchapters · 652 functions and 12 objects

This document explains the code of AoE2 Civ Explorer. Each folder is a chapter, and each file in it a subchapter. In the subchapters on scripts, each function defined at the top level of its script is a section, numbered after its chapter and subchapter, so the fourth section of the first file in chapter 4 is 4.1.4. A section gives the lines the function takes up, says what it does and why, and lists the functions that call it and the functions it calls. A function handed on to be called later, such as an event handler, counts as called by the function that hands it on. Helpers defined inside a function are part of that function's text.

"Exported" marks a function that other scripts import, and "internal" one that only its own script uses. The subchapters on the page, its style sheets and the data are split into parts in the same way, each with its lines. Rules and requirements are those of RULES.md and REQUIREMENTS.md.

Every name of a folder, a file or a function links to its chapter, subchapter or section, and the page header always names the chapter and the subchapter. The contents list the chapters, each with its subchapters indented under it. The four UML diagrams after the contents show how the parts and the scripts use each other, and the index at the end lists every chapter, subchapter and function in alphabetical order.

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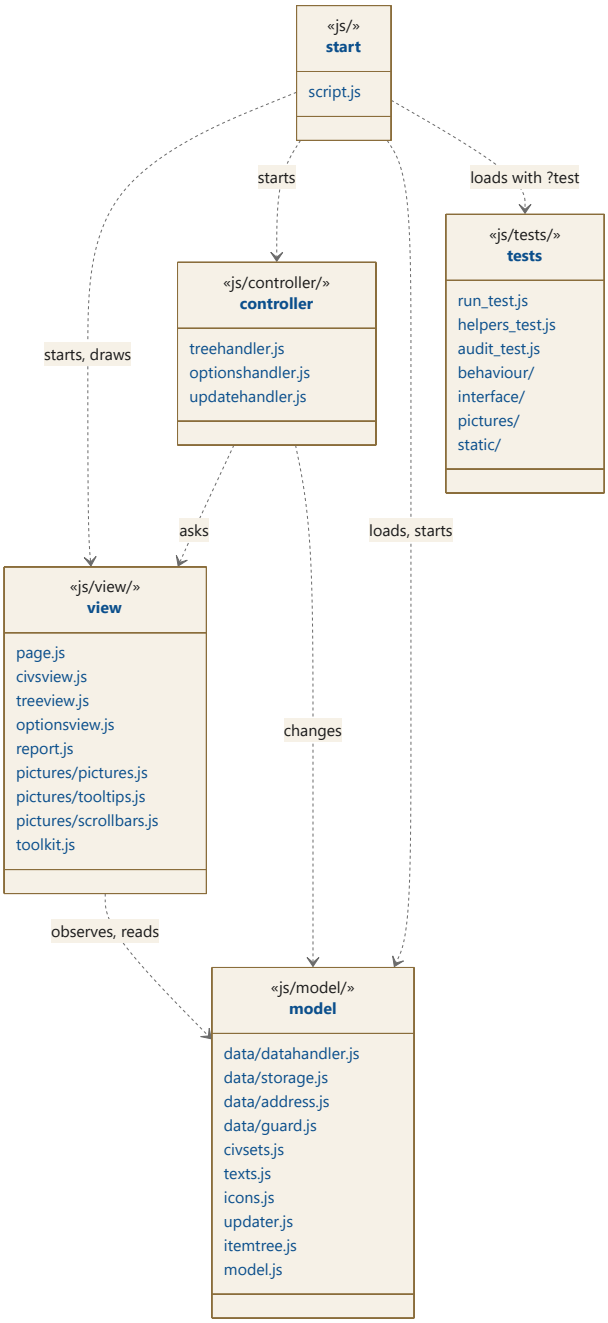
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# UML diagrams

Four class diagrams show how the scripts use each other. The first, the one in MAINTAINER.md, shows the code in its three parts, model, view and controller, file by file; the other three show the scripts of each part, function by function. In each, an arrow points from what uses to what is used. No arrow crosses or touches another, or runs under a box or a label: making this document checks every diagram for it, and stops if one does. Each box and each name in a box links to its chapter, subchapter or section.

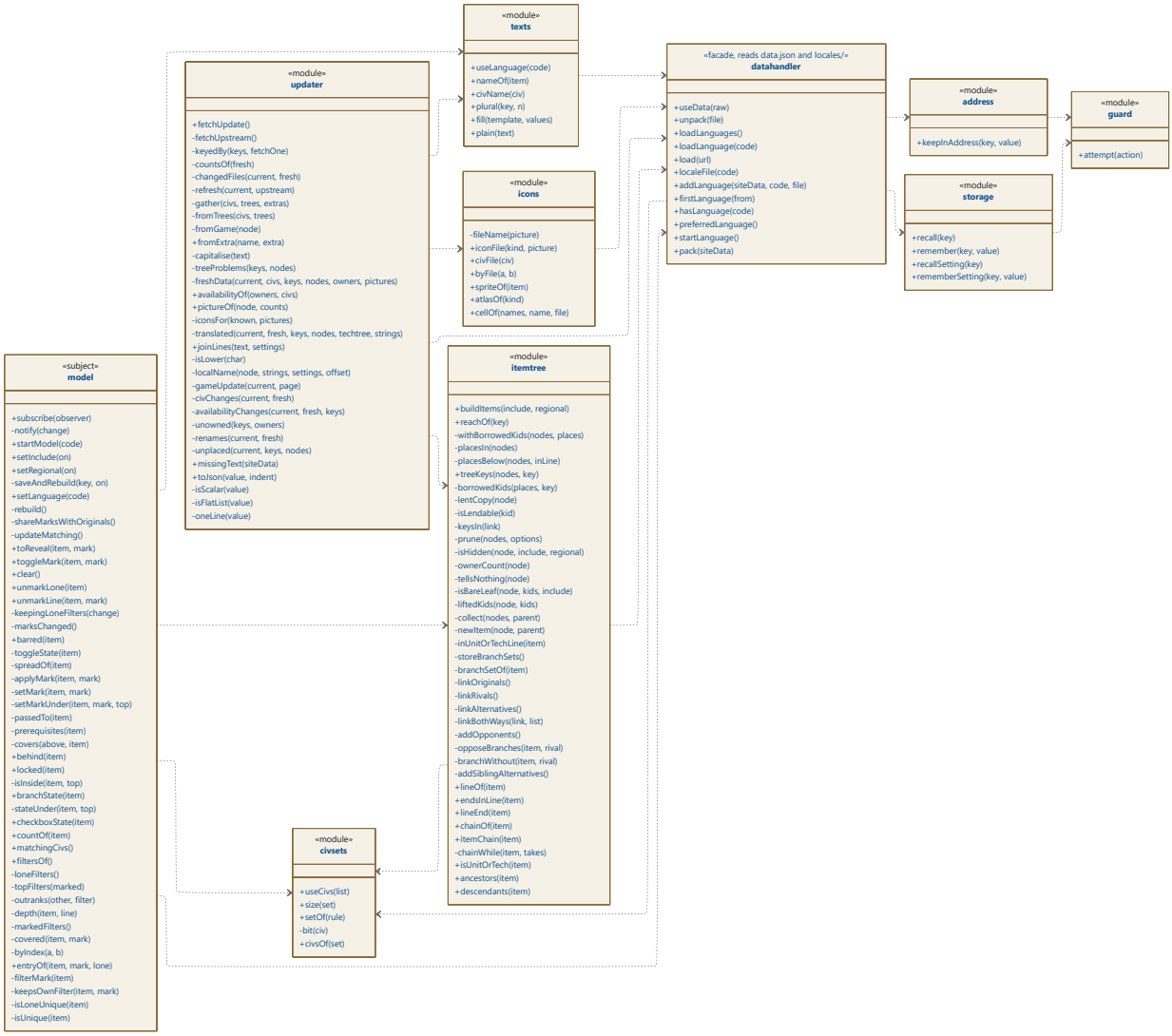
## Overview

Each box is a folder, with the scripts in it and in its subfolders, each named from the box's folder, such as data/datahandler.js; `start` is js/, whose script.js starts the page, and the box of the tests names its four folders of tests in place of their scripts. Each arrow points from a part to a part whose scripts it uses, and says what for. The tests use all three parts as well; those arrows are left out, because with them some arrows would have to cross. A box links to its folder's chapter, and a script to its subchapter.



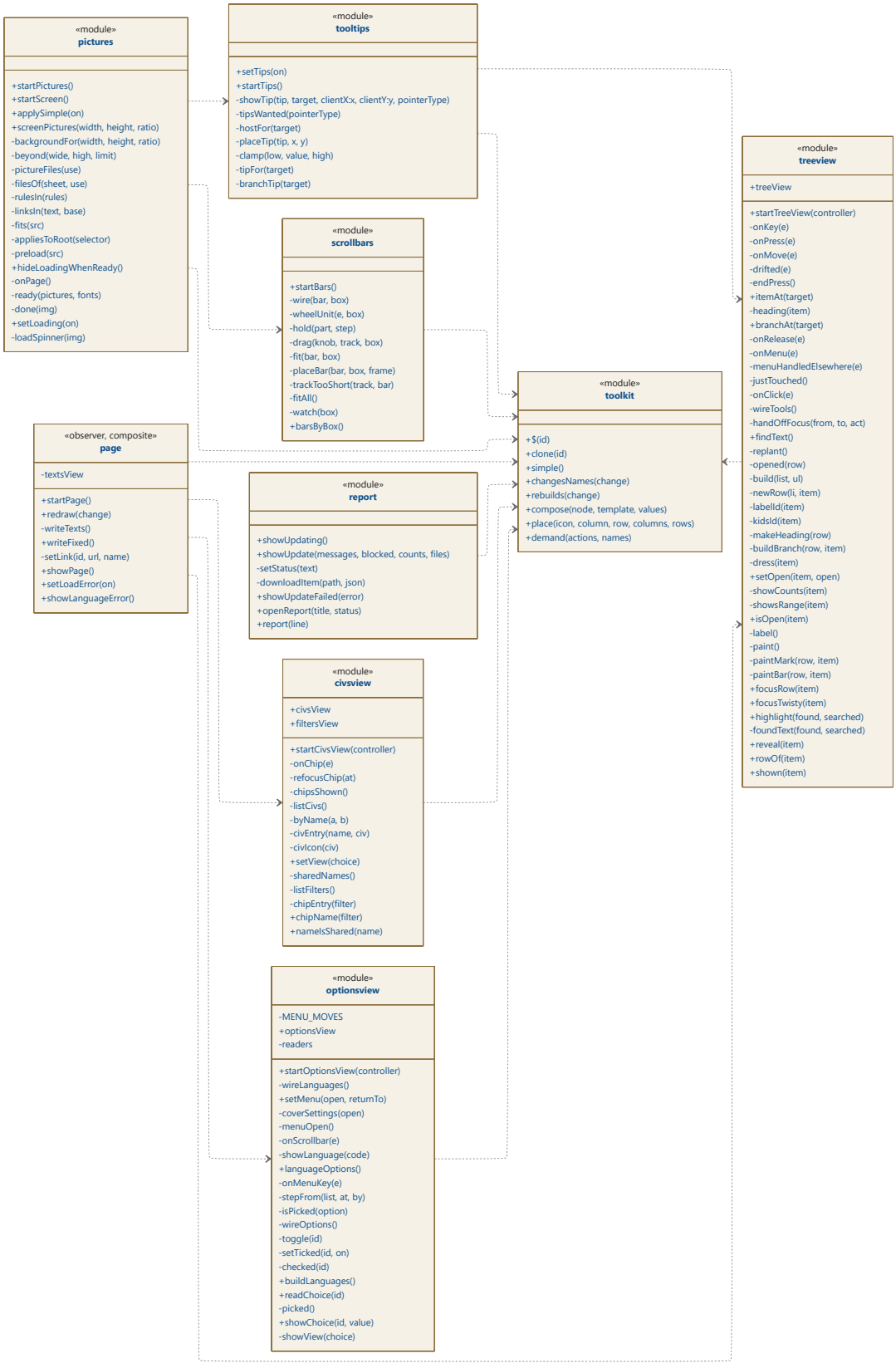
# Model

The scripts of js/model/ and js/model/data/, each with its functions and objects: + marks what a script exports, and − what it keeps to itself. The line under a script's name gives its role: its part in model, view and controller, which MAINTAINER.md explains, what it reads, or "module". An arrow points from a script to a script of the model that it imports; what the scripts use of the view and the controller is in the overview.



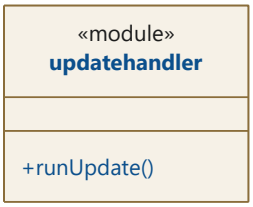
# View

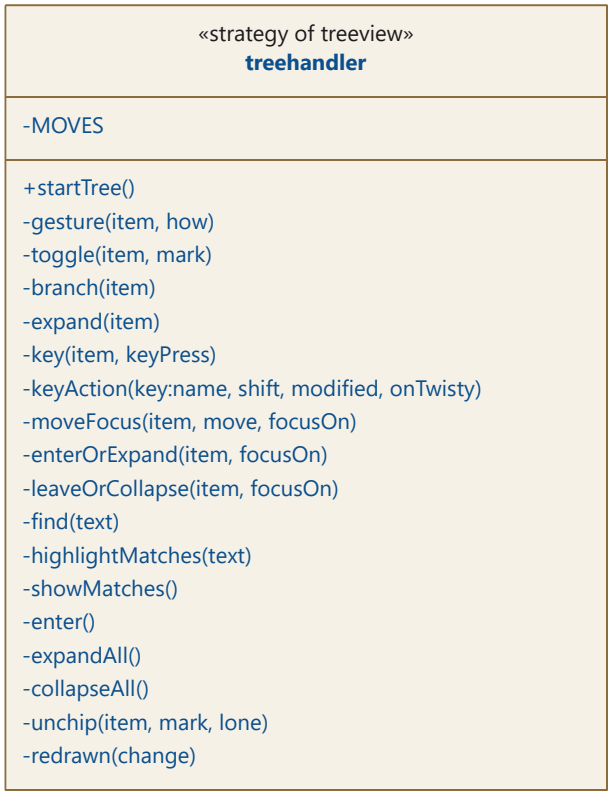
The scripts of js/view/ and js/view/pictures/, each with its functions and objects, drawn as in the diagram of the model: + for what a script exports, − for what it keeps to itself, its role under its name, and an arrow to each script of the view that it imports.

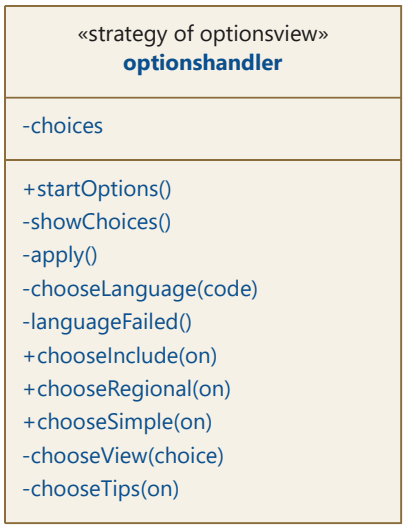


## Controller

The scripts of js/controller/, each with its functions and objects, drawn as in the diagram of the model. None of them imports another, so there are no arrows: what they use of the view and the model is in the overview.



# 1 Main folder

2 files 2386 lines

The page and its data: index.html, which the browser opens, and data.json, which the scripts load at start-up. The other files there, such as README.md, are not part of the code.

**Subchapters:** 1.1 index.html, 1.2 data.json

## 1.1 index.html

**Folder:** the main folder 109 lines 12 parts
**Loads:** css/base.css, css/pictures.css and js/script.js

The page's structure, and all of its HTML (Rule 2). It holds no text of its own: the scripts fill in every word in the chosen language, and copy its templates instead of writing HTML.

### 1.1.1 Head

lines 1–20

Sets the language, character set, viewport, light and dark colour schemes, title and description, and what link previews show: the `og:` tags give a title, a description, `assets/images/preview.jpg` and the page's address. It links base.css, pictures.css and script.js, as a module. It names no tab icon: browsers fetch favicon.ico from the main folder by themselves.

### 1.1.2 Loading screen

line 22

The screen shown without Simple UI until every picture has loaded, with its text. `setLoading()` adds the turning picture from its template.

### 1.1.3 Header

lines 23–26

Holds the title twice: the full sentence of Req. 2 for Simple UI, and "Civ Explorer" for the pictures (Req. 0). Then the cogwheel Options button (Req. 6), a text character with a hidden label. Hidden until the page starts.

### 1.1.4 Messages

lines 27–28

A noscript line for browsers without JavaScript, and a load error shown in English if data.json cannot be loaded, so a visitor is never left with a blank page.

### 1.1.5 Tree pane

lines 29–50

The start of the content, then the left pane, in the order of Req. 3: heading, Clear, Expand all and Collapse all, the Find field with its label, clearing cross and live match count, the must have and must lack legend, and the empty tree.

### 1.1.6 Civilisations pane

lines 51–64

The right pane (Req. 4): heading, live count, the Filters list (Req. 22), the civilisation list and the hint shown when it is empty. It scrolls on its own, so the page never does (Rule 12).

### 1.1.7 Footer

lines 65–66

Names and links the data source and the game update the data is from, which links its notes (Req. 5), then Microsoft's notice, in English, with links to its rules and to the project. The code writes the sentence around the links.

### 1.1.8 Options window

lines 67–83

A dialog (Req. 7, 8, 32): title, the language label, button and list, then the settings: Simple UI, Include Unique and Universal with its description, Include Regionals, Civ view and Tooltips; and Cancel and OK.

### 1.1.9 Language popup

lines 84–87

The popup that says a language could not be loaded: a modal dialog with the role `alertdialog`, so screen readers read its message at once, and an OK that closes it. `showLanguageError()` opens it.

### 1.1.10 Report window

lines 88–96

A dialog for the reports of the tests and the updater: title, live status, result list, download links and Close. Always in English, so it is marked `lang="en"`.

### 1.1.11 Templates

lines 97–106

Markup for tree rows, Filters buttons, civilisations, whose links open in a new tab, the loading picture, the icons of civilisations and of the tree, language options, report lines, downloads and the scroll bar. The code copies these instead of writing HTML (Rule 2).

### 1.1.12 Tooltip

line 107

The note that follows the mouse without Simple UI: over the tree, over every control, each naming its text with `data-tip`, and over a link, with its address. Hidden otherwise.

# 1.2 data.json

**Folder:** the main folder 2277 lines 12 keys

All of the page's data, in English: the civilisations, the tree, who has each item, its icon and the page's texts. `useData()` reads it at start-up, and the updater writes it through `pack()` and `toJson()`: short objects and lists on one line, the rest indented by two spaces. The other languages' texts are in `locales/`, one file each, laid out like the `overrides` and `text` here.

An item is named by its English name throughout: the tree, `availability`, `pictures`, `extras` and `text.items` use the same names as keys.

| Lines | Key | What is there |
|---|---|---|
| 2–11 | `source` | Where the data comes from: aoe2techtree.net and its data address, `helpOffset` (taken from an item's help string number to find its text, in `localName()`), the game update the data is from, with the address of its notes, which the updater reads and the footer links, and Microsoft's notice with the rules and the project it links. |
| 12–30 | `languages` | Every language by its code, with its name in itself, its language tag, and whether its words are spaced and its hyphens at line ends are merged. English comes first. |
| 31 | `civs` | The civilisations, in the order whose bits the sets use. A set is a number with one bit per civilisation (`bit()`). |
| 32–46 | `icons` | Where the pictures are: the file names of civilisations that differ from their names, and the folder of the tree's icons. |
| 35–39 | `groups` | The icon of each top-level group, by kind and picture. |
| 40–45 | `atlas` | For each kind, the pictures in its atlas, which `cellOf()` places by file name, and `cellIn()` in the tests too. |
| 47–686 | `tree` | The tree as nested objects: `{"group": …}` for a top-level group or a group in a building, `{"item": …}` for an item, with `kids` for what is under it and the flags `own`, `lines`, `excludes` and `replaces` where they belong. `rebuild()` reads it. |
| 687–1158 | `availability` | Who has each item: `"all"`, `{"has": […]}` or `{"lacks": […]}`, whichever is shortest (`availabilityOf()`). `setOf()` turns it into a set. |
| 1159–1630 | `pictures` | Each item's icon: its kind, its picture in that kind's atlas, and its reach, common, regional or unique, which Include Regionals and the outlines use. |
| 1631 | `ignore` | Names on aoe2techtree.net that the updater does not report as new items. |
| 1632–1641 | `extras` | Items aoe2techtree.net does not show in its tree, with their kind, picture, reach, name number and civilisations. `fromExtra()` turns each into availability and a picture. |
| 1642 | `overrides` | Names to use instead of the game's own, where it gets one wrong. |
| 1643–1671 | `tools` | The texts of the tests' report and of the updater, in English only, with `{}` placeholders that `fill()` fills in. |
| 1672–2276 | `text` | The page's texts in English, in four parts. |
| 1673–1738 | `ui` | The page's own texts, by the name `data-text`, or for a tooltip `data-tip`, gives in index.html, counted ones as `{"one": …, "other": …}` for `plural()`. |
| 1739–1745 | `groups` | The name of each group. |
| 1746–1803 | `civs` | The name of each civilisation. |
| 1804–2275 | `items` | The name of each item. |

# 2 css/

2 files 466 lines

The two style sheets index.html links: base.css, for Simple UI and the layout either way, and pictures.css, for every rule with a picture or a font, which Simple UI turns off.

**Subchapters:** 2.1 base.css, 2.2 pictures.css

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## 2.1 base.css

**Folder:** css/ 204 lines 12 parts
**Loaded by:** index.html

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How the page looks with Simple UI, and its layout either way: colours, sizes and layouts for big and small screens, in light and dark mode. Every rule with a picture or a font is in pictures.css instead.

### 2.1.1 Colours

lines 1–33

Named colours for light mode, redefined for dark mode: text, lines, marks, hover, Find's yellow, the green and red of the marks, and the dark box behind the white dash; and the 44-pixel target size and the font. The tests measure their contrast.

### 2.1.2 Base

lines 34–46

Box sizing, `hidden` that always hides, a full-height page that never scrolls (Rule 12), the system font with 1.5 line spacing (WCAG 1.4.8), a 3 px focus outline (WCAG 2.4.13), and a class that hides text from sight but not from screen readers.

### 2.1.3 Header and language

lines 48–66

Lays out the title and cogwheel, and in the Options window the language label, its button with a turning chevron, and the list of languages, where the current one is marked with a bar and bold.

### 2.1.4 Panes

lines 68–83

Two panes side by side from 720 px wide; below that they share a phone's height, and the footer is hidden. Each pane's head stays in place while its list scrolls.

### 2.1.5 Controls

lines 84–101

Buttons and the Find field are at least 44 by 44 px (WCAG 2.5.5), with visible borders and a hover colour. Find's message keeps its column, so the box never changes size while typing (Rule 18). The legend, and the scrolling lists.

### 2.1.6 Tree

lines 103–121

Indented rows with dotted guide lines, 44 px rows, turning arrows, pressed and greyed-out rows, and found rows in yellow with the light colours (Req. 20) whatever the colour scheme.

### 2.1.7 Check boxes

lines 122–140

Marks drawn with borders (Req. 14, 23, 26): a green tick, a red cross, and a white dash on a dark box, in system colours when colours are forced. A setting that is on in the Options window shows the green tick.

### 2.1.8 Names and views

lines 141–145

Hides Civ view and Tooltips, which only the pictures use; makes top-level and group names bold; keeps counts on one line at the right; and draws focus rings inside rows.

### 2.1.9 Filters and list

lines 147–163

Filters as a wrapping row of rounded chips under the pane's head, the civilisation list in columns that never split a name, the empty hint and the footer.

### 2.1.10 Dialogs

lines 165–188

Size, padding, border and backdrop of the Options and report windows and of the popup, which is as wide as its message. The Options window is a grid of fixed height, so opening the language list never resizes it (Rule 18); only the report's list scrolls.

### 2.1.11 Small screens

lines 190–201

On short screens everything scrolls as one, as at 400% zoom (WCAG 1.4.10); on narrow ones the padding and the tree's indent shrink.

### 2.1.12 Test helper

lines 203–204

The wider text spacing the tests apply (WCAG 1.4.12). Used only while testing.

# 2.2 pictures.css

**Folder:** css/ 262 lines 16 parts

**Loaded by:** index.html

How the page looks without Simple UI. Every rule here that shows a picture starts with `.pictures` or `html.pictures`, so Simple UI turns them all off (Rules 9, 22), and picture addresses go from this folder, as `../assets/`. The `[data-assets="4k"]` and `[data-background]` rules pick the pictures for the screen.

### 2.2.1 Parchment and cursors

lines 1–24

The parchment behind the page, fitted so the content lies inside its frame on any window, in the four sizes; the ink, gold and serif font the other rules name; the sword and the pointing hand as cursors, the `_x2` ones drawn at two thirds of their size, at `1.5x` in `image-set`, so they are as large as the `_x1` ones.

### 2.2.2 Header and footer

lines 25–47

The footer's colours, the title in its frame with the short title, the load error, and the Options button from its three-state sheet.

### 2.2.3 Fonts

lines 48–57

Georgia, Times New Roman Bold and Trajan Pro Bold from `assets/fonts/`, under their own names, trying the installed copy first. Only rules for the pictures use them, so Simple UI never downloads a font.

### 2.2.4 Text buttons

lines 58–67

Every button with words, but those of the report and the language box, shows the button picture, normal, under the mouse or pressed, with its ends drawn apart so it can be any width.

### 2.2.5 Loading screen

lines 69–72

The screen that covers the page while the pictures load, and its turning picture, the only animation, which stops with reduced motion (Rule 11).

### 2.2.6 Boxes and radio buttons

lines 74–90

Check boxes from their sheet, a row for each state and a column for normal, under the mouse and pressed, with a black dash; Tooltips and the Civ view radio buttons, shown only here.

### 2.2.7 Frames

lines 91–98

The 9-slice frames of the two panes, and their fonts and colours.

### 2.2.8 Options window

lines 99–114

The window's frame, measured to fit around the same window as with Simple UI, with its contents 60 pixels from the frame's sides, or 120 of the screen's own pixels on a 4K screen (`--pixel`), and the settings lit under the mouse.

### 2.2.9 Language box

lines 115–146

The language menu as the game's combo box: its picture, 219 by 35 pixels at the right of its row, with the language's name and, at its right end, the arrow from its atlas, pointing down while the list is closed and up while it is open, normal, under the mouse or pressed, and 35 by 28 of the screen's own pixels on a 4K screen. The list lies over the settings below the box, as wide as it and seven languages high, in the box's colours, and snaps to whole languages; the chosen language is red and the one under the mouse teal. Its scroll bar lies over it.

### 2.2.10 Options on tall windows

lines 147–160

On windows at least 588 pixels high, the parchment behind the window, OK and Cancel in its corner, and the title on its ribbon.

### 2.2.11 Tree

lines 161–182

The tree's colours, found rows in amber, the rows' icons from the four atlases with an outline by kind and reach as `colour_key.csv` says, and greyed-out icons.

### 2.2.12 Tools

lines 183–191

Filters chips, no legend, Find first, and Clear and Expand all as two buttons that share the pane's width; the buttons of the Options window and of the popup have the button picture's shape.

### 2.2.13 Find box

lines 192–209

Find's field in its 9-slice frame, 325 px wide, with the search picture while empty and the clearing cross while not; the hidden label and Include's hidden description.

### 2.2.14 Tooltip

lines 210–214

The tooltip's place, set by the script as numbers, 32 pixels from the pointer as both cursors are 32 pixels large, and its parchment look.

### 2.2.15 Civilisations

lines 215–230

The civilisations' icons from their atlas, and the tiles, six to a row, each with its icon and its name in Trajan Pro.

### 2.2.16 Scroll bars

lines 231–262

The game's scroll bars: the browser's are hidden, room is kept for the bar, and the bar is placed, sized and drawn from its arrows, body and knob, each from its sheet, as `fit()` sets them. The language list's bar has the combo box's arrows instead, centred on it, and its arrows and knob always show their first picture.

# 3 js/

1 file 62 lines

Where the scripts start: script.js, which index.html loads as a module. Every other script is in one of the four folders below it, or in a folder of theirs: js/model/, js/view/ and js/controller/ for the page, and js/tests/ for the tests.

**Subchapters:** 3.1 script.js

## 3.1 script.js

**Folder:** js/ 62 lines 6 functions, 1 object, 0 exported

**Imports:** js/model/model.js, js/model/data/datahandler.js, js/view/page.js, js/view/pictures/pictures.js, js/controller/treehandler.js, js/controller/optionshandler.js, js/controller/updatehandler.js, and js/tests/run_test.js (only with `?test`)

**Imported by:** index.html

Where the page starts: its top-level code runs once, `RUNS` says what the page does in each of its modes, and `start()` starts the parts of the page in their order. It never touches the page itself.

### 3.1.1 `Start-up` TOP-LEVEL CODE lines 23–25

Sets the screen's pictures once (`startScreen()`), then Simple UI from the saved setting, so pictures load at once or never. Loads data.json, then has `useAndLoadLanguages()` load the languages, `run()` start the page, and `failLoad()` show the load error if loading fails. What each step does depends on the mode, which line 11 reads once from the address (`MODE`): `?test`, `?update` or neither.

Called by *no function of the code*
Calls `useAndLoadLanguages()`, `run()`, `failLoad()`; in js/model/data/datahandler.js: `load()`; in js/view/pictures/pictures.js: `startScreen()`, `applySimple()`; in js/tests/run_test.js: `loadFailed()`, `runTests()`

### 3.1.2 `RUNS` OBJECT, INTERNAL lines 13–21

What each mode does, `?test`, `?update` or the page alone: which languages to load, what to run once the page is up, and what to do if data.json cannot load. `?test` loads every language, runs the tests (`runTests()`) and, if loading fails, marks them failed (`loadFailed()`), so CI never waits for a result that cannot come; `?update` loads the visitor's language and runs the updater (`runUpdate()`), which loads the others itself.

Called by `useAndLoadLanguages()`, `run()`, `failLoad()`
Calls `testRunner()`, `loadPreferred()`; in js/model/data/datahandler.js: `loadLanguages()`; in js/controller/updatehandler.js: `runUpdate()`

### 3.1.3 `useAndLoadLanguages(raw)` INTERNAL lines 27–30

Reads data.json (`useData()`) and loads the languages the mode needs: every one for `?test`, else the visitor's. Runs once, when data.json has loaded.

Called by `Start-up`
Calls `RUNS()`; in js/model/data/datahandler.js: `useData()`

### 3.1.4 `loadPreferred()` INTERNAL lines 32–34

Loads the visitor's language, `preferredLanguage()`, and goes on without it if it cannot be loaded: the page then starts in English, and `start()` opens the popup that says so. A visitor downloads only the language shown.

Called by `RUNS()`
Calls in js/model/data/datahandler.js: `loadLanguage()`, `preferredLanguage()`

### 3.1.5 `run()` INTERNAL lines 36–41

Starts the page (`start()`) and its pictures (`startPictures()`), lets the loading screen go once every picture has loaded (`hideLoadingWhenReady()`), and runs what the mode runs: the tests, the updater or nothing. Runs once, when data.json and the languages have loaded.

Called by `Start-up`
Calls `start()`, `RUNS()`; in js/view/pictures/pictures.js: `startPictures()`, `hideLoadingWhenReady()`

### 3.1.6 `failLoad(error)` INTERNAL lines 43–48

Shows that the page could not load: hides the loading screen, shows `#loadError` and, with `?test`, marks the tests failed. It sees only a failure to load data.json or, with `?test`, a language, not an error of the page or the tests, and throws the error again, so that it shows in the console.

Called by `Start-up`
Calls `RUNS()`; in js/view/page.js: `setLoadError()`; in js/view/pictures/pictures.js: `setLoading()`

### 3.1.7 `start()` INTERNAL

lines 50–60

Starts the page: gives the model the start language (`startLanguage()`), the visitor's or, when it could not be loaded, the first, and the model reads its saved Include settings itself; starts the controllers of the tree and of the Options window, writes the fixed texts, has the page observe the model, draws it once with `redraw("start")` and reveals it. If the visitor's language could not be loaded, it then opens the popup `#languageError` (`showLanguageError()`). Runs once, from `run()`.

Called by `run()`
Calls in js/model/model.js: `startModel()`; in js/model/data/datahandler.js: `startLanguage()`, `preferredLanguage()`; in js/view/page.js: `writeFixed()`, `startPage()`, `redraw()`, `showPage()`, `showLanguageError()`; in js/controller/treehandler.js: `startTree()`; in js/controller/optionshandler.js: `startOptions()`

### 3.1.8 `testRunner()` INTERNAL

line 62

Loads run_test.js, the tests, only when `RUNS` asks for them, so the page alone never downloads them.

Called by `RUNS()`
Calls *no function of the code*

# 4 js/model/

6 files 776 lines

The data and the tree's rules: js/model/data/ loads and stores the data, civsets.js holds the sets of civilisations, texts.js the chosen language and icons.js the icons' file names, itemtree.js builds the tree's items and answers questions about its shape, updater.js fetches the game's newest data for `?update`, and model.js keeps the options, the marks, the filters and the counts with the rules of marking, saves its own choices, and tells the page when they change. None of them touches the page, and none imports a view or a controller.

**Subchapters:** 4.1 civsets.js, 4.2 texts.js, 4.3 icons.js, 4.4 itemtree.js, 4.5 updater.js, 4.6 model.js

## 4.1 civsets.js

**Folder:** js/model/ 30 lines 5 functions, 4 exported

**Imports:** nothing
**Imported by:** js/model/itemtree.js, js/model/model.js, js/model/data/datahandler.js, js/tests/helpers_test.js, js/tests/behaviour/marking_test.js, js/tests/behaviour/tree_test.js, js/tests/behaviour/lines_test.js, js/tests/interface/options_test.js, js/tests/interface/keys_test.js, js/tests/pictures/pictures_test.js, js/tests/static/updates_test.js

Sets of civilisations: each civilisation is one bit, so a set of civilisations is a single number that `&` and `|` combine quickly, and `allCivs` is the set of every one. It imports nothing: `useData()` hands it the civilisations.

### 4.1.1 `useCivs(list)` EXPORTED lines 5–9

Takes the civilisations of data.json and gives each one bit, in their order, and makes `allCivs` from them. Runs once, when data.json has loaded.

Called by in js/model/data/datahandler.js: `useData()`
Calls *no function of the code*

### 4.1.2 `size(set)` EXPORTED lines 11–17

Counts the civilisations in a set by counting its bits. Behind every number on the page.

Called by in js/model/itemtree.js: `ownerCount()`; in js/model/model.js: `countOf()`; in js/tests/helpers_test.js: `fixtures()`; in js/tests/behaviour/marking_test.js: `marking()`; in js/tests/behaviour/tree_test.js: `crowding()`, `wanted()`, `ownCount()`; in js/tests/behaviour/lines_test.js: `beyond()`; in js/tests/interface/options_test.js: `leftOut()`; in js/tests/pictures/pictures_test.js: `simpleUiTests()`
Calls *no function of the code*

### 4.1.3 `setOf(rule)` EXPORTED lines 19–23

Turns an availability rule from data.json (`"all"`, a `has` list or a `lacks` list) into a set. Rules stay short in the file while the code works with sets.

Called by in js/model/itemtree.js: `ownerCount()`, `newItem()`; in js/tests/static/updates_test.js: `shortest()`
Calls `bit()`

### 4.1.4 `bit(civ)` INTERNAL lines 25–28

Returns a civilisation's bit, or stops with an error for an unknown name, so a typo in data.json cannot silently drop a civilisation.

Called by `setOf()`
Calls *no function of the code*

### 4.1.5 `civsOf(set)` EXPORTED line 30

Turns a set back into a list of civilisations, in data.json order. Used to list the civilisations that fit.

Called by in js/model/model.js: `matchingCivs()`; in js/tests/helpers_test.js: `expected()`; in js/tests/static/updates_test.js: `shortest()`
Calls *no function of the code*

# 4.2 texts.js

**Folder:** js/model/ 34 lines 6 functions, 6 exported

**Imports:** js/model/data/datahandler.js

**Imported by:** js/model/updater.js, js/model/model.js, js/view/toolkit.js, js/view/page.js, js/view/civsview.js, js/view/treeview.js, js/view/optionsview.js, js/view/report.js, js/view/pictures/tooltips.js, js/controller/treehandler.js, js/controller/optionshandler.js, js/tests/run_test.js, js/tests/helpers_test.js, js/tests/audit_test.js, js/tests/behaviour/marking_test.js, js/tests/behaviour/samples_test.js, js/tests/behaviour/branches_test.js, js/tests/behaviour/tree_test.js, js/tests/behaviour/lines_test.js, js/tests/behaviour/greying_test.js, js/tests/behaviour/finding_test.js, js/tests/interface/menu_test.js, js/tests/interface/languages_test.js, js/tests/interface/options_test.js, js/tests/interface/keys_test.js, js/tests/interface/uniques_test.js, js/tests/pictures/findbox_test.js, js/tests/pictures/treebuttons_test.js

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The chosen language: its text, its language tag, and how it sorts, counts and writes numbers, with the functions that put names and counts into words in it. Everything that shows, sorts or counts text reads it. `PLACEHOLDER`, the pattern of a placeholder such as `{n}`, is exported too, so that `fill()` and `compose()` in toolkit.js find placeholders alike.

### 4.2.1 `useLanguage(code)` EXPORTED lines 10–17

Switches the active language: its text, language tag, sorting, plural rules and number format. Everything that shows, sorts or counts text reads these, so one call changes them all.

Called by in js/model/model.js: `startModel()`, `setLanguage()`
Calls *no function of the code*

### 4.2.2 `nameOf(item)` EXPORTED line 19

Returns a tree item's or group's name in the current language, or its key if missing. Used wherever an item is shown or searched.

Called by in js/view/civsview.js: `sharedNames()`, `chipName()`; in js/view/treeview.js: `label()`; in js/controller/treehandler.js: `highlightMatches()`; in js/tests/helpers_test.js: `fixtures()`; in js/tests/audit_test.js: `controlChecks()`; in js/tests/behaviour/branches_test.js: `buildingMarks()`, `lineFilters()`, `singleLines()`, `closedInside()`; in js/tests/behaviour/tree_test.js: `countKinds()`, `tree()`; in js/tests/behaviour/lines_test.js: `lineTops()`, `parents()`, `lines()`, `evenLine()`; in js/tests/behaviour/greying_test.js: `excludedBy()`, `sharedRow()`, `shuts()`, `variants()`; in js/tests/behaviour/finding_test.js: `finding()`, `dropsGreyed()`; in js/tests/interface/languages_test.js: `languageTests()`; in js/tests/interface/options_test.js: `leftOut()`; in js/tests/interface/keys_test.js: `findKey()`; in js/tests/interface/uniques_test.js: `uniqueFilterTests()`, `inLine()`, `inGroup()`; in js/tests/pictures/findbox_test.js: `findBoxTests()`
Calls *no function of the code*

### 4.2.3 `civName(civ)` EXPORTED line 21

Returns a civilisation's name in the current language, or its key if missing. Keeps the civilisation list translated.

Called by in js/view/civsview.js: `listCivs()`; in js/tests/helpers_test.js: `expected()`; in js/tests/interface/languages_test.js: `civList()`
Calls *no function of the code*

### 4.2.4 `plural(key, n)` EXPORTED lines 23–26

Picks the right form of a counted text, such as "1 civilisation" or "5 civilisations", by the language's plural rules, and fills in the formatted number.

Called by in js/view/civsview.js: `listCivs()`; in js/view/treeview.js: `showCounts()`, `foundText()`; in js/tests/behaviour/marking_test.js: `marking()`; in js/tests/behaviour/tree_test.js: `wanted()`; in js/tests/behaviour/finding_test.js: `finding()`, `dropsGreyed()`
Calls `fill()`

### 4.2.5 `fill(template, values)` EXPORTED lines 30–31

Replaces `{name}` placeholders in a text with values, leaving unknown ones alone. All text lives in the data with placeholders, so the code never builds sentences itself.

Called by `plural()`; in js/model/updater.js: `refresh()`; in js/view/page.js: `writeTexts()`; in js/view/civsview.js: `chipEntry()`, `chipName()`; in js/view/treeview.js: `showCounts()`; in js/view/report.js: `showUpdating()`, `showUpdate()`, `downloadItem()`, `showUpdateFailed()`; in js/tests/run_test.js: `reportLine()`, `publish()`; in js/tests/behaviour/marking_test.js: `marking()`; in js/tests/behaviour/branches_test.js: `buildingMarks()`; in js/tests/behaviour/tree_test.js: `wanted()`; in js/tests/interface/languages_test.js: `footerTexts()`
Calls *no function of the code*

### 4.2.6 `plain(text)` EXPORTED lines 33–34

Lower-cases a text and strips accents, spaces and punctuation, so Find matches names however they are typed, in any language.

Called by in js/controller/treehandler.js: `highlightMatches()`; in js/tests/behaviour/finding_test.js: `finding()`; in js/tests/interface/keys_test.js: `findKey()`
Calls *no function of the code*

# 4.3 icons.js

**Folder:** js/model/ 24 lines 7 functions, 6 exported
**Imports:** js/model/data/datahandler.js
**Imported by:** js/model/updater.js, js/view/civsview.js, js/view/treeview.js, js/tests/pictures/icons_test.js

The icons' file names, the order of the pictures in an atlas, and where each item's icon is: facts about the data, which the views and the updater both need.

### 4.3.1 `fileName(picture)` INTERNAL line 5

Names a picture's PNG file from its number or name, such as `8.png`: the one place that knows the pattern, for `iconFile()` and `byFile()`.

Called by `iconFile()`, `byFile()`
Calls *no function of the code*

### 4.3.2 `iconFile({ kind, picture })` EXPORTED line 7

Returns the path of a tree item's icon PNG from its kind and picture number, such as `assets/images/icons/nodes/unit/8.png`, for the updater's report of new icons.

Called by in js/model/updater.js: `iconsFor()`
Calls `fileName()`

### 4.3.3 `civFile(civ)` EXPORTED line 9

Returns the name of a civilisation's icon file: its name in lower case, unless `icons.names` in data.json gives another, such as `inca` for the Incas.

Called by in js/view/civsview.js: `civIcon()`; in js/tests/pictures/icons_test.js: `civIcons()`
Calls *no function of the code*

### 4.3.4 `byFile(a, b)` EXPORTED line 11

Sorts picture numbers or names as their PNG files sort by name, `1.png`, `10.png`, `100.png` and on to `11.png`, which is the order of the cells in an atlas: a fact about the data, which `cellOf()` and `iconsFor()` in updater.js both use.

Called by `cellOf()`; in js/model/updater.js: `iconsFor()`
Calls `fileName()`

### 4.3.5 `spriteOf(item)` EXPORTED line 13

Returns an item's sprite, its kind and its picture: a group's from `icons.groups` in data.json, any other item's from `pictures`. The tree view finds an icon through it and `atlasOf()`, so it never reads data.json itself.

Called by in js/view/treeview.js: `dress()`
Calls *no function of the code*

### 4.3.6 `atlasOf(kind)` EXPORTED line 15

Returns the pictures of a kind's atlas, from `icons.atlas` in data.json, in which `cellOf()` finds a cell.

Called by in js/view/treeview.js: `dress()`
Calls *no function of the code*

### 4.3.7 `cellOf(names, name, file = String)` EXPORTED lines 17–24

Finds a picture's cell in its atlas, which holds the pictures in file-name order, row by row, with as many columns as the square root of their number, rounded up. The order is worked out once per list. It stops with an error for a picture that is not in its atlas, as the page would otherwise quietly show a cell outside it. The civilisations' atlas is in the order of their icons' files, so for a civilisation it is given `civFile()`.

Called by in js/view/civsview.js: `civIcon()`; in js/view/treeview.js: `dress()`
Calls `byFile()`

# 4.4 itemtree.js

**Folder:** js/model/ 202 lines 38 functions, 11 exported
**Imports:** js/model/civsets.js, js/model/data/datahandler.js
**Imported by:** js/model/updater.js, js/model/model.js, js/tests/helpers_test.js, js/tests/static/data_test.js

Builds the tree's items from data.json: what Include and Include Regionals hide, the rows a variant borrows from the items it stands in for, each item's branch set, the civilisations that have it and everything under it, and the copies, rivals, opponents and alternatives. It owns `items` and `roots`, which model.js passes on, so the imports go one way and every other script imports them from model.js.

It also answers the questions about the tree's shape, which need no marks: an item's line of upgrades, the items under it in a chain, the items above and below it, and whether it counts as a unit or technology.

### 4.4.1 `buildItems({ include, regional })` EXPORTED lines 8–16

Builds the items afresh from data.json and the tree's settings, `{ include, regional }`: lends the variants their rows, leaves out what Options leaves out, turns the rest into items, stores their branch sets, and links copies, rivals and alternatives. `rebuild()` in model.js runs it and keeps the marks.

Called by in js/model/model.js: `rebuild()`
Calls `collect()`, `prune()`, `withBorrowedKids()`, `placesIn()`, `storeBranchSets()`, `linkOriginals()`, `linkRivals()`, `linkAlternatives()`

### 4.4.2 `reachOf(key)` EXPORTED line 18

Returns an item's reach in data.json, `"common"`, `"regional"` or `"unique"`, or nothing for a group. The one place that reads it, for pruning, for lending and for `isUnique()` in model.js.

Called by `isLendable()`, `prune()`, `isHidden()`; in js/model/model.js: `isUnique()`
Calls *no function of the code*

### 4.4.3 `withBorrowedKids(nodes, places)` INTERNAL lines 20–23

Adds under every item that names items it stands in for the rows those items lend, in the order named, so their technologies are written only once in data.json.

Called by `buildItems()`
Calls `keysIn()`, `borrowedKids()`

### 4.4.4 `placesIn(nodes)` INTERNAL line 25

Lists every node of data.json's tree once, top to bottom, each with whether it lies inside an item or a group of lines. `borrowedKids()` looks an item up in it.

Called by `buildItems()`
Calls `placesBelow()`

### 4.4.5 `placesBelow(nodes, inLine)` INTERNAL lines 27–29

Lists the nodes under a place in the tree for `placesIn()`, top to bottom, each with whether an item or a group of lines lies above it.

Called by `placesIn()`
Calls *no function of the code*

### 4.4.6 `treeKeys(nodes, key)` EXPORTED lines 31–33

Lists every item (or group) name in a tree, top to bottom. Used to check the tree, to find groups that need text, and by the updater to know which items to look up.

Called by in js/model/updater.js: `refresh()`, `missingText()`; in js/tests/helpers_test.js: `keys()`; in js/tests/static/data_test.js: `textTests()`
Calls *no function of the code*

### 4.4.7 `borrowedKids(places, key)` INTERNAL lines 35–38

Finds the item that a `"replaces"` names and returns the rows it lends; none if it is not in the tree or counts as a unit or technology, as its rows are then its own upgrades.

Called by `withBorrowedKids()`
Calls `lentCopy()`

### 4.4.8 `lentCopy(node)` INTERNAL line 40

Copies a row and the rows under it that can be lent, marked as copies, for an item that stands in for it.

Called by `borrowedKids()`
Calls `isLendable()`

### 4.4.9 `isLendable(kid)` INTERNAL line 42

Tells whether a row can be lent to an item that stands in for its own: neither unique nor marked `"own"`.

Called by `lentCopy()`
Calls `reachOf()`

### 4.4.10 `keysIn(link)` INTERNAL line 44

Turns a `"replaces"` or `"excludes"` value, one name, a list or nothing, into a list.

Called by `withBorrowedKids()`, `linkBothWays()`, `addSiblingAlternatives()`
Calls *no function of the code*

### 4.4.11 `prune(nodes, options)` INTERNAL lines 46–54

Leaves out what Options leaves out: the unique groups and unique items (keeping their rows), regional items without Include Regionals, and items that one, all or all but one civilisation have, unless something under them is shown.

Called by `buildItems()`
Calls `isHidden()`, `reachOf()`, `liftedKids()`, `isBareLeaf()`

### 4.4.12 `isHidden(node, { include, regional })` INTERNAL lines 56–57

Tells whether Options hides a node and everything under it: a unique group without Include Unique and Universal, or a regional item without Include Regionals.

Called by `prune()`
Calls `reachOf()`

### 4.4.13 `ownerCount(node)` INTERNAL lines 59–62

Counts the civilisations that have an item, from its rule in data.json; none for a group.

Called by `tellsNothing()`
Calls in js/model/civsets.js: `size()`, `setOf()`

### 4.4.14 `tellsNothing(node)` INTERNAL lines 64–67

Tells whether an item tells nothing about who fits, as one civilisation, all or all but one have it.

Called by `isBareLeaf()`
Calls `ownerCount()`

### 4.4.15 `isBareLeaf(node, kids, include)` INTERNAL line 69

Tells whether a node is left out because nothing under it is shown: always for a group, and for an item that tells nothing (`tellsNothing()`) without Include Unique and Universal.

Called by `prune()`
Calls `tellsNothing()`

### 4.4.16 `liftedKids(node, kids)` INTERNAL lines 71–72

Hands the rows of a unique item that is left out to its place, noting the item in their path, so that without the Camel Scout the Camel Rider is a row of the Stable. Lent rows are dropped.

Called by `prune()`
Calls *no function of the code*

### 4.4.17 `collect(nodes, parent)` INTERNAL lines 74–83

Turns the shown part of data.json's tree into the items the page uses (`newItem()`), top to bottom, and notes on each whether it is a unit or technology. Stops with an error for an item without availability.

Called by `buildItems()`
Calls `newItem()`, `inUnitOrTechLine()`

### 4.4.18 `newItem(node, parent)` INTERNAL lines 85–95

Makes one item: its key, a path that tells copies apart, its set of civilisations (all, for a group), and room for its mark, its branch set and its links, which the steps after `collect()` fill in.

Called by `collect()`
Calls in js/model/civsets.js: `setOf()`

### 4.4.19 `inUnitOrTechLine(item)` INTERNAL line 97

Tells whether an item is a unit or technology: an item with another item, or a `"lines"` group, above it. Worked out once, as the item is made, for `isUnitOrTech()`.

Called by `collect()`
Calls `ancestors()`

### 4.4.20 `storeBranchSets()` INTERNAL lines 99–103

Stores each item's branch set in `item.all`, walking the items from the last to the first, so the rows' sets are ready before the item above them needs them. A row's count of matching civilisations uses it.

Called by `buildItems()`
Calls `branchSetOf()`

### 4.4.21 `branchSetOf(item)` INTERNAL line 105

An item's branch set: the civilisations that have it and, by their branch sets, everything under it.

Called by `storeBranchSets()`
Calls *no function of the code*

### 4.4.22 `linkOriginals()` INTERNAL lines 107–111

Links every copied item to its original, the item of the same key that is not a copy, so that model.js can share their marks (`shareMarksWithOriginals()`).

Called by `buildItems()`
Calls *no function of the code*

### 4.4.23 `linkRivals()` INTERNAL lines 113–116

Links the items that `"excludes"` names as rivals, both ways, and gives every item the opponents its rival's branch holds (`addOpponents()`).

Called by `buildItems()`
Calls `linkBothWays()`, `addOpponents()`

### 4.4.24 `linkAlternatives()` INTERNAL lines 118–121

Links the items that `"replaces"` names as alternatives, both ways, and adds the other items that stand in for the same (`addSiblingAlternatives()`).

Called by `buildItems()`
Calls `linkBothWays()`, `addSiblingAlternatives()`

### 4.4.25 `linkBothWays(link, list)` INTERNAL lines 123–132

Links items both ways by `"excludes"` or `"replaces"`, as rivals or as alternatives.

Called by `linkRivals()`, `linkAlternatives()`
Calls `keysIn()`

### 4.4.26 `addOpponents()` INTERNAL lines 134–140

Gives every item the opponents that its must have greys out: everything on its rival's branch, and the other way round.

Called by `linkRivals()`
Calls `opposeBranches()`

### 4.4.27 `opposeBranches(item, rival)` INTERNAL lines 142–147

Gives everything on an item's branch the whole of its rival's branch as opponents, each branch without the other. The rival's branch is worked out once, not once for every item on the item's.

Called by `addOpponents()`
Calls `branchWithout()`

### 4.4.28 `branchWithout(item, rival)` INTERNAL lines 149–151

An item and everything under it, but its rival and what is under that, so the Stable's branch leaves out the Xolotl Warrior inside it.

Called by `opposeBranches()`
Calls *no function of the code*

### 4.4.29 `addSiblingAlternatives()` INTERNAL lines 153–161

Adds to the alternatives of an item that stands in for others the other items that stand in for the same, so marking the Folwark greys out the Settlement too.

Called by `linkAlternatives()`
Calls `keysIn()`

### 4.4.30 `lineOf(item)` EXPORTED lines 163–174

The whole line of upgrades an item is on, from its top to its end, or just the item if it is on none. Filters names and clears lines with it.

Called by in js/model/model.js: `the top-level code`, `unmarkLine()`, `loneFilters()`, `markedFilters()`, `entryOf()`, `isLoneUnique()`
Calls `endsInLine()`

### 4.4.31 `endsInLine(item)` EXPORTED line 176

Tells whether an item's chain (`itemChain()`) ends a line of upgrades: the item is no group, and the chain's last item has nothing under it, so the line neither forks nor goes on into a group.

Called by `lineOf()`; in js/model/model.js: `the top-level code`
Calls `lineEnd()`

### 4.4.32 `lineEnd(item)` EXPORTED line 178

The last item of an item's chain (`itemChain()`): the end of its line of upgrades, unless the line forks or goes on into a group first. `endsInLine()` tells which.

Called by `endsInLine()`; in js/model/model.js: `the top-level code`
Calls `itemChain()`

### 4.4.33 `chainOf(item)` EXPORTED line 180

An item and the rows below it for as long as each has exactly one row under it. Marking opens these.

Called by in js/model/model.js: `the top-level code`, `spreadOf()`
Calls `chainWhile()`

### 4.4.34 `itemChain(item)` EXPORTED line 182

An item and the items below it for as long as each has exactly one row under it and that row is not a group. `passedTo()` and `lineEnd()` follow it.

Called by `lineEnd()`; in js/model/model.js: `passedTo()`
Calls `chainWhile()`

### 4.4.35 `chainWhile(item, takes)` INTERNAL lines 184–190

An item and the rows below it for as long as each has exactly one row under it and that row is taken by the test it is given. `chainOf()` and `itemChain()` are made with it.

Called by `chainOf()`, `itemChain()`
Calls *no function of the code*

### 4.4.36 `isUnitOrTech(item)` EXPORTED line 192

Tells whether an item is a unit or technology, as `collect()` noted on it (`inUnitOrTechLine()`). Buildings and groups mark everything in them; units and technologies pass marks along lines.

Called by in js/model/model.js: `the top-level code`, `toggleState()`, `spreadOf()`, `applyMark()`, `prerequisites()`, `checkboxState()`, `entryOf()`
Calls *no function of the code*

### 4.4.37 `ancestors(item)` EXPORTED lines 194–200

The items above an item, from its parent up to the top. Find opens them to bring a found row into sight, and `inUnitOrTechLine()` reads them.

Called by `inUnitOrTechLine()`; in js/model/model.js: `the top-level code`
Calls *no function of the code*

### 4.4.38 `descendants(item)` EXPORTED line 202

Everything under an item, at every depth.

Called by in js/model/model.js: `spreadOf()`, `applyMark()`, `checkboxState()`
Calls *no function of the code*

# 4.5 updater.js

**Folder:** js/model/ 272 lines 31 functions, 7 exported

**Imports:** js/model/texts.js, js/model/icons.js, js/model/itemtree.js, js/model/data/datahandler.js
**Imported by:** js/controller/updatehandler.js, js/tests/static/data_test.js, js/tests/static/updates_test.js

The updater, which `?update` runs: it fetches aoe2techtree.net's data, trees and texts and its page, works out the civilisations, who has what, the icons, every name and the game update, compares them with data.json and writes the new files. It reads files only through `load()` in datahandler.js. The game's nodes and the `extras` of data.json are read into one shape (`fromGame()`, `fromExtra()`), an Adapter, and the trees and texts are kept in maps keyed by civilisation and language.

### 4.5.1 `fetchUpdate()` EXPORTED lines 11–16

Loads every language of the page, has `fetchUpstream()` download aoe2techtree.net's data, trees, texts and page, runs `refresh()`, and returns what report.js shows: the report's lines, whether the tree blocks the update, the numbers of civilisations, items and languages, and the files that changed, data.json always. So a game update needs only a browser.

Called by in js/controller/updatehandler.js: `runUpdate()`
Calls `refresh()`, `fetchUpstream()`, `countsOf()`, `changedFiles()`; in js/model/data/datahandler.js: `loadLanguages()`

### 4.5.2 `fetchUpstream()` INTERNAL lines 18–29

Downloads aoe2techtree.net's data first, as it names the civilisations, sorted here once, then the tree of every civilisation, the game's texts in every language of the page and its page, all at once.

Called by `fetchUpdate()`
Calls `keyedBy()`; in js/model/data/datahandler.js: `load()`

### 4.5.3 `keyedBy(keys, fetchOne)` INTERNAL lines 31–33

Fetches one file for each civilisation or language, all at once, and keeps them in a map by its key.

Called by `fetchUpstream()`
Calls *no function of the code*

### 4.5.4 `countsOf(fresh)` INTERNAL lines 35–39

Counts the civilisations, items and languages of the new data, for the report's last line.

Called by `fetchUpdate()`
Calls *no function of the code*

### 4.5.5 `changedFiles(current, fresh)` INTERNAL lines 41–45

Splits the current and the new data into their files (`pack()`), writes each as data.json is laid out (`toJson()`), and keeps the new files that differ from the current ones, and data.json always.

Called by `fetchUpdate()`
Calls `toJson()`; in js/model/data/datahandler.js: `pack()`

### 4.5.6 `refresh(current, upstream)` INTERNAL lines 47–69

The updater's core: from the current data and what `fetchUpstream()` fetched, it works out the civilisations, who has what, the icons and every name from aoe2techtree.net's data and the extras, takes the game update from its page (`gameUpdate()`), and writes the report. Stops if `treeProblems()` finds an unknown item, or one named twice.

Called by `fetchUpdate()`
Calls `gather()`, `treeProblems()`, `freshData()`, `iconsFor()`, `translated()`, `gameUpdate()`, `civChanges()`, `availabilityChanges()`, `unowned()`, `renames()`, `unplaced()`, `missingText()`; in js/model/texts.js: `fill()`; in js/model/itemtree.js: `treeKeys()`

### 4.5.7 `gather({ civs, trees }, extras)` INTERNAL lines 71–81

The updater's first step: reads aoe2techtree.net's trees into three tables (`fromTrees()`) and adds the items of `extras` in data.json to them (`fromExtra()`). Lists any extra that aoe2techtree.net now has itself, for the report, so it can be taken out of `extras`.

Called by `refresh()`
Calls `fromTrees()`, `fromExtra()`

### 4.5.8 `fromTrees(civs, trees)` INTERNAL lines 83–100

Reads aoe2techtree.net's tree of every civilisation into three tables: each item's node, in the updater's shape (`fromGame()`), the civilisations that have it, and how often each of its pictures is used.

Called by `gather()`
Calls `fromGame()`

### 4.5.9 `fromGame(node)` INTERNAL lines 102–110

Turns a node of aoe2techtree.net's trees into the updater's shape: its name, the numbers of its name and help texts, its kind, its reach (common, regional or unique), its picture, and whether the civilisation has it. With `fromExtra()`, the rest of the updater reads one shape, wherever an item comes from.

Called by `fromTrees()`
Calls *no function of the code*

### 4.5.10 `fromExtra(name, extra)` EXPORTED lines 112–119

Turns an entry of `extras` in data.json, an item aoe2techtree.net gives to no civilisation, into the node, owners and picture counts the updater gathers for every other item, the node in the shape `fromGame()` gives. A test checks the extras the same way.

Called by `gather()`; in js/tests/static/data_test.js: `kept()`
Calls *no function of the code*

### 4.5.11 `capitalise(text)` INTERNAL line 121

Upper-cases the first letter of a text, as some names in the game start in lower case.

Called by `translated()`, `localName()`
Calls *no function of the code*

### 4.5.12 `treeProblems(keys, nodes)` INTERNAL lines 123–127

Lists what stops the updater, for its report: the items the tree names twice, in alphabetical order, then those that aoe2techtree.net does not have.

Called by `refresh()`
Calls *no function of the code*

### 4.5.13 `freshData(current, civs, { keys, nodes, owners, pictures })` INTERNAL lines 129–135

Makes the new data: a copy of the current one with aoe2techtree.net's civilisations, and who has each item of the tree (`availabilityOf()`) and its icon (`pictureOf()`).

Called by `refresh()`
Calls `availabilityOf()`, `pictureOf()`

### 4.5.14 `availabilityOf(owners, civs)` EXPORTED lines 137–142

Writes who has an item in its shortest form: `"all"`, a `has` list or a `lacks` list. Keeps data.json small; a test checks that every rule is stored this way.

Called by `freshData()`; in js/tests/static/data_test.js: `kept()`; in js/tests/static/updates_test.js: `shortest()`
Calls *no function of the code*

### 4.5.15 `pictureOf(node, counts)` EXPORTED lines 144–147

Picks an item's icon for data.json from aoe2techtree.net's data: its kind, the picture most civilisations show, and whether it is common, regional or unique, which the outlines and Include Regionals use.

Called by `freshData()`; in js/tests/static/data_test.js: `kept()`
Calls *no function of the code*

### 4.5.16 `iconsFor(known, pictures)` INTERNAL lines 149–160

Adds every new picture to a copy of the list of its atlas in `icons.atlas`, in file-name order, and lists it for the report with its PNG, so the atlas can be built again.

Called by `refresh()`
Calls in js/model/icons.js: `byFile()`, `iconFile()`

### 4.5.17 `translated(current, fresh, { keys, nodes, techtree, strings })` INTERNAL lines 162–176

Works out every civilisation's and item's name in every language from the game's texts, unless an override says otherwise, writing out abbreviated names from the help text, and returns the new data's texts with them.

Called by `refresh()`
Calls `capitalise()`, `joinLines()`, `localName()`

### 4.5.18 `joinLines(text, settings)` EXPORTED lines 178–189

Joins a game text broken with <br> into one line, by the language's rules for spaces and hyphens, and tidies spaces. The game breaks names to fit its buttons.

Called by `translated()`, `localName()`; in js/tests/static/updates_test.js: `updaterTests()`
Calls `isLower()`

### 4.5.19 `isLower(char)` INTERNAL line 191

Tells whether a character is a lower-case letter. `joinLines()` uses it to tell a hyphenated word from a real hyphen.

Called by `joinLines()`
Calls *no function of the code*

### 4.5.20 `localName(node, { strings, settings, offset })` INTERNAL lines 193–198

Finds an item's name in a language's game texts. If that name looks abbreviated, it takes the bold name from the item's help text instead, which is written out in full.

Called by `translated()`
Calls `joinLines()`, `capitalise()`

### 4.5.21 `gameUpdate(current, page)` INTERNAL lines 200–206

Reads the game update from aoe2techtree.net's page: the number after "Update" and the address of its notes, which its link leads to and the footer links too. A new number goes into the report; if the page has none, the report says so and data.json keeps the update it had.

Called by `refresh()`
Calls *no function of the code*

### 4.5.22 `civChanges(current, fresh)` INTERNAL lines 208–211

Lists the civilisations added or gone, for the report.

Called by `refresh()`
Calls *no function of the code*

### 4.5.23 `availabilityChanges(current, fresh, keys)` INTERNAL lines 213–215

Lists the items of the tree whose civilisations changed, for the report.

Called by `refresh()`
Calls *no function of the code*

### 4.5.24 `unowned(keys, owners)` INTERNAL line 217

Lists the items of the tree that no civilisation has, for the report.

Called by `refresh()`
Calls *no function of the code*

### 4.5.25 `renames(current, fresh)` INTERNAL lines 219–225

Lists the civilisations and items renamed in each language, for the report. A name that was not there before is new, not renamed.

Called by `refresh()`
Calls *no function of the code*

### 4.5.26 `unplaced(current, keys, nodes)` INTERNAL lines 227–229

Lists the items aoe2techtree.net has that are not yet in the tree, in alphabetical order, for the report, leaving out those that `ignore` in data.json names.

Called by `refresh()`
Calls *no function of the code*

### 4.5.27 `missingText(siteData)` EXPORTED lines 231–242

Lists, per language, the page texts and group names that are missing. A test and the updater's report use it, so no language shows a blank.

Called by `refresh()`
Calls in js/model/itemtree.js: `treeKeys()`; in js/model/data/datahandler.js: `firstLanguage()`

### 4.5.28 `toJson(value, indent = "")` EXPORTED lines 244–262

Writes JSON the way data.json is laid out: short objects and lists on one line, the rest indented, and a list that ends an object of scalars opened on the object's line. Keeps the updater's files readable and their changes easy to review.

Called by `changedFiles()`; in js/tests/static/updates_test.js: `updaterTests()`
Calls `isScalar()`, `isFlatList()`, `oneLine()`

### 4.5.29 `isScalar(value)` INTERNAL line 264

Tells whether a value is text, a number, true or false, or null. Helps `toJson()` decide what fits on one line.

Called by `toJson()`, `isFlatList()`, `oneLine()`
Calls *no function of the code*

### 4.5.30 `isFlatList(value)` INTERNAL

line 266

Tells whether a value is a list of scalars only, which `toJson()` writes on one line.

Called by `toJson()`
Calls `isScalar()`

### 4.5.31 `oneLine(value)` INTERNAL

lines 268–272

Writes a value as one line of JSON, with a space after each comma and colon. The building block of `toJson()`.

Called by `toJson()`
Calls `isScalar()`

# 4.6 model.js

**Folder:** js/model/ 214 lines 47 functions, 19 exported

**Imports:** js/model/civsets.js, js/model/texts.js, js/model/itemtree.js, js/model/data/datahandler.js
**Imported by:** js/script.js, js/view/page.js, js/view/civsview.js, js/view/treeview.js, js/view/pictures/tooltips.js, js/controller/treehandler.js, js/controller/optionshandler.js, js/tests/run_test.js, js/tests/helpers_test.js, js/tests/audit_test.js, js/tests/behaviour/marking_test.js, js/tests/behaviour/branches_test.js, js/tests/behaviour/tree_test.js, js/tests/behaviour/lines_test.js, js/tests/behaviour/greying_test.js, js/tests/behaviour/finding_test.js, js/tests/interface/languages_test.js, js/tests/interface/options_test.js, js/tests/interface/keys_test.js, js/tests/interface/uniques_test.js, js/tests/interface/headings_test.js, js/tests/pictures/pictures_test.js, js/tests/pictures/icons_test.js, js/tests/pictures/findbox_test.js, js/tests/pictures/tooltips_test.js, js/tests/pictures/treebuttons_test.js, js/tests/pictures/scrollbars_test.js, js/tests/static/layers_test.js

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The marks and their rules: how marks pass on, what is greyed out, which civilisations fit and how many, and what Filters shows, with Include, Include Regionals and the language, which it saves itself. itemtree.js builds the items; model.js passes them on with the questions about the tree's shape, so every other script imports them from here.

It is the subject of the Observer pattern: only its own functions change it, and each then tells its observer, page.js, what changed. It never touches the page.

### 4.6.1 `subscribe(observer)` EXPORTED lines 16–18

Adds an observer, which `notify()` then calls after every change. Only page.js subscribes (Test 255), and it never leaves, so there is no `unsubscribe`.

Called by in js/view/page.js: `startPage()`
Calls *no function of the code*

### 4.6.2 `notify(change)` INTERNAL line 20

Tells every observer what kind of change was made, once and as soon as `matching` is up to date: `"marks"`, `"tree"` or `"language"`. The observers read the rest from the model themselves. It is the model's own (Test 256), so nothing else can announce a change.

Called by `saveAndRebuild()`, `setLanguage()`, `marksChanged()`
Calls *no function of the code*

### 4.6.3 `startModel(code)` EXPORTED lines 22–27

Takes the language, reads the saved Include settings (`recallSetting()`), and builds the tree once, saving nothing and telling no one: the page is drawn afterwards, all at once, with `redraw("start")`. Runs once, from `start()`.

Called by in js/script.js: `start()`
Calls `rebuild()`; in js/model/texts.js: `useLanguage()`

### 4.6.4 `setInclude(on)` EXPORTED lines 29–32

Turns Include Unique and Universal on or off, and has `saveAndRebuild()` save the choice, build the tree again, keeping marks, and tell the page that the tree changed.

Called by in js/controller/optionshandler.js: `chooseInclude()`
Calls `saveAndRebuild()`

### 4.6.5 `setRegional(on)` EXPORTED lines 34–37

Turns Include Regionals on or off, and has `saveAndRebuild()` save the choice, build the tree again, keeping marks, and tell the page that the tree changed.

Called by in js/controller/optionshandler.js: `chooseRegional()`
Calls `saveAndRebuild()`

### 4.6.6 `saveAndRebuild(key, on)` INTERNAL lines 39–43

Saves a setting of the tree (`rememberSetting()`), builds the tree again, keeping marks, and tells the page that the tree changed. What `setInclude()` and `setRegional()` share.

Called by `setInclude()`, `setRegional()`
Calls `rebuild()`, `notify()`

### 4.6.7 `setLanguage(code)` EXPORTED lines 45–50

Makes a language the one the tree's names are in, saves the choice, has `keepInAddress()` put it in the address (`?lng=`), so a shared link keeps it, and tells the page that the language changed.

Called by in js/controller/optionshandler.js: `chooseLanguage()`; in js/tests/run_test.js: `restore()`
Calls `notify()`; in js/model/texts.js: `useLanguage()`

### 4.6.8 `rebuild()` INTERNAL lines 52–60

Has `buildItems()` build the items afresh from data.json and the Include settings, keeps every mark by its path, shares the marks of copies with their originals, and works out `matching` again (`updateMatching()`). Runs at the start and whenever an Include setting changes.

Called by `startModel()`, `saveAndRebuild()`
Calls `shareMarksWithOriginals()`, `updateMatching()`; in js/model/itemtree.js: `buildItems()`

### 4.6.9 `shareMarksWithOriginals()` INTERNAL lines 62–67

Ties every lent row to the original that `linkOriginals()` found, so both share one mark: marking Two-Man Saw under the Mule Cart marks it under the Lumber Camp too, with one entry in Filters. The copy reads and writes its mark, `self`, through its original, a Proxy for one property.

Called by `rebuild()`
Calls *no function of the code*

### 4.6.10 `updateMatching()` INTERNAL lines 69–72

Works out the civilisations that fit every mark, as `matching`.

Called by `rebuild()`, `marksChanged()`
Calls *no function of the code*

### 4.6.11 `toReveal(item, mark)` EXPORTED line 74

The rows a mark will reach, or none if it clears the item or the item is greyed out. The controller opens the way to them before it marks, so they come into sight before the greyed-out branches close (Req. 28, 31).

Called by in js/controller/treehandler.js: `toggle()`
Calls `barred()`, `toggleState()`, `spreadOf()`

### 4.6.12 `toggleMark(item, mark)` EXPORTED lines 76–82

Marks an item must have or must lack, or clears it if it already has that mark, passes must have up to its prerequisites, works out `matching` again and tells the page that the marks changed. A greyed-out item is left as it is, and no one is told (Req. 37).

Called by in js/controller/treehandler.js: `toggle()`; in js/tests/behaviour/greying_test.js: `excludedBy()`
Calls `barred()`, `toggleState()`, `applyMark()`, `prerequisites()`, `marksChanged()`

### 4.6.13 `clear()` EXPORTED lines 84–87

Clears every mark (Req. 17), works out `matching` again and tells the page that the marks changed.

Called by in js/controller/treehandler.js: `startTree()`
Calls `setMark()`, `marksChanged()`

### 4.6.14 `unmarkLone(item)` EXPORTED lines 89–92

Clears the must have of a unique item whose lone filter entry is cleared, and only that, and tells the page that the marks changed.

Called by in js/controller/treehandler.js: `unchip()`
Calls `marksChanged()`

### 4.6.15 `unmarkLine(item, mark)` EXPORTED lines 94–101

Clears what an entry of Filters stands for: the marks of its kind along its line, or the item and everything in it, keeping the lone filter entries (`keepingLoneFilters()`), and tells the page that the marks changed.

Called by in js/controller/treehandler.js: `unchip()`
Calls `keepingLoneFilters()`, `setMark()`, `marksChanged()`; in js/model/itemtree.js: `lineOf()`

### 4.6.16 `keepingLoneFilters(change)` INTERNAL lines 103–107

Makes a change, then gives back its must have to every unique item that had a lone filter entry before it, so clearing a line or a building keeps those entries (Req. 36).

Called by `unmarkLine()`
Calls `keepsOwnFilter()`

### 4.6.17 `marksChanged()` INTERNAL lines 109–112

Works out `matching` again with `updateMatching()` and tells the page that the marks changed. Every change of the marks ends with it.

Called by `toggleMark()`, `clear()`, `unmarkLone()`, `unmarkLine()`
Calls `updateMatching()`, `notify()`

### 4.6.18 `barred(item)` EXPORTED lines 114–115

Tells whether an item is greyed out: not marked must have itself, while one of its alternatives or opponents is (Req. 37).

Called by `toReveal()`, `toggleMark()`, `locked()`; in js/view/treeview.js: `paintBar()`
Calls *no function of the code*

### 4.6.19 `toggleState(item)` INTERNAL line 117

The mark an item has now, which `toggleMark()` sets or clears: its own for a unit or technology, otherwise what all of it shows (`branchState()`).

Called by `toReveal()`, `toggleMark()`
Calls `branchState()`; in js/model/itemtree.js: `isUnitOrTech()`

### 4.6.20 `spreadOf(item)` INTERNAL line 119

The rows a mark reached, which marking opens: down to the first fork, and the rows it passed to or everything inside.

Called by `toReveal()`
Calls `passedTo()`; in js/model/itemtree.js: `chainOf()`, `isUnitOrTech()`, `descendants()`

### 4.6.21 `applyMark(item, mark)` INTERNAL lines 121–128

Sets or clears a mark: on everything in a building or group, or along a unit's or technology's line. Must lack also clears must have below that depended on it.

Called by `toggleMark()`
Calls `setMark()`, `prerequisites()`, `passedTo()`; in js/model/itemtree.js: `isUnitOrTech()`, `descendants()`

### 4.6.22 `setMark(item, mark)` INTERNAL line 130

Marks an item and everything under it, groups only through their rows, skipping the rivals of the item marked. How a building or group is marked or cleared.

Called by `clear()`, `unmarkLine()`, `applyMark()`
Calls `setMarkUnder()`

### 4.6.23 `setMarkUnder(item, mark, top)` INTERNAL lines 132–137

Does the work of `setMark()`: marks an item and, row by row, everything under it but the rivals of `top`, the item that `setMark()` was given.

Called by `setMark()`
Calls *no function of the code*

### 4.6.24 `passedTo(item)` INTERNAL lines 139–141

The items a mark passes down to: along the item's line until it forks, and only those the same civilisations have (Req. 29, 30).

Called by `spreadOf()`, `applyMark()`
Calls in js/model/itemtree.js: `itemChain()`

### 4.6.25 `prerequisites(item)` INTERNAL lines 143–149

The units and technologies above an item that its must have passes up to, stopping at a building or at an item that some civilisations with it lack (Req. 34).

Called by `toggleMark()`, `applyMark()`
Calls `covers()`; in js/model/itemtree.js: `isUnitOrTech()`

### 4.6.26 `covers(above, item)` INTERNAL line 151

Tells whether every civilisation that has an item also has the one above it, so that must have may pass up to it.

Called by `prerequisites()`
Calls *no function of the code*

### 4.6.27 `behind(item)` EXPORTED line 153

Tells whether an item lies inside a closed greyed-out item, where Find does not look.

Called by in js/controller/treehandler.js: `highlightMatches()`; in js/tests/behaviour/finding_test.js: `dropsGreyed()`
Calls `locked()`

### 4.6.28 `locked(item)` EXPORTED line 155

Tells whether a greyed-out item is also closed: unless its own rival is inside it, as the Xolotl Warrior is inside the Stable, which stays open so the Xolotl Warrior stays in sight.

Called by `behind()`; in js/view/treeview.js: `treeView()`, `paintBar()`; in js/view/pictures/tooltips.js: `branchTip()`; in js/controller/treehandler.js: `expand()`, `expandAll()`
Calls `barred()`, `isInside()`

### 4.6.29 `isInside(item, top)` INTERNAL line 157

Tells whether an item lies anywhere under another.

Called by `locked()`
Calls *no function of the code*

### 4.6.30 `branchState(item)` EXPORTED line 159

The mark a branch shows: the one an item's own mark and all its rows share, or `"mixed"` when they differ, leaving out the rivals of the item asked about.

Called by `toggleState()`, `checkboxState()`, `covered()`, `entryOf()`, `filterMark()`; in js/tests/behaviour/branches_test.js: `buildingMarks()`, `groupMarks()`; in js/tests/behaviour/greying_test.js: `rivals()`
Calls `stateUnder()`

### 4.6.31 `stateUnder(item, top)` INTERNAL lines 161–165

Does the work of `branchState()`: the mark that an item and every row under it share, or `"mixed"`, leaving out the rivals of `top`, the item asked about.

Called by `branchState()`
Calls *no function of the code*

### 4.6.32 `checkboxState(item)` EXPORTED lines 167–170

What a row's box shows: a unit's or technology's own mark, or a dash if a row below has another; otherwise `branchState()`.

Called by in js/view/treeview.js: `paintMark()`
Calls `branchState()`; in js/model/itemtree.js: `isUnitOrTech()`, `descendants()`

### 4.6.33 `countOf(item)` EXPORTED lines 172–175

The two counts beside an item's name, `withItem` and `withBranch`: the civilisations that fit the marks and have the item, or for a group everything in it, and those that fit and have the item and everything under it, its branch set. The tree's view shows one or both.

Called by in js/view/treeview.js: `showCounts()`
Calls in js/model/civsets.js: `size()`

### 4.6.34 `matchingCivs()` EXPORTED line 177

The civilisations that fit every mark, for the list on the right.

Called by in js/view/civsview.js: `listCivs()`
Calls in js/model/civsets.js: `civsOf()`

### 4.6.35 `filtersOf()` EXPORTED lines 179–181

Works out the entries of Filters, in the tree's order: the lone filter entries (`loneFilters()`), and one entry per marked building, group or line (`markedFilters()`), but none for what a marked row above covers, and only one for each line and mark (`topFilters()`).

Called by in js/view/civsview.js: `listFilters()`
Calls `loneFilters()`, `topFilters()`, `markedFilters()`, `byIndex()`

### 4.6.36 `loneFilters()` INTERNAL lines 183–184

The lone filter entries: one for every unique item that is not a copy and whose must have keeps its own entry (`keepsOwnFilter()`), so it stays named in Filters while its line or group is marked too (Req. 36).

Called by `filtersOf()`
Calls `keepsOwnFilter()`; in js/model/itemtree.js: `lineOf()`

### 4.6.37 `topFilters(marked)` INTERNAL line 186

Keeps of the marked entries those that no other entry outranks (`outranks()`), so that each line has one entry for each of its marks.

Called by `filtersOf()`
Calls `outranks()`

### 4.6.38 `outranks(other, filter)` INTERNAL lines 188–189

Tells whether one entry stands in for another: the same mark on the same line, and deeper for must have or higher for must lack, so that a line's entry is its deepest must have and its highest must lack.

Called by `topFilters()`
Calls `depth()`

### 4.6.39 `depth({ item, line })` INTERNAL line 191

How deep an entry's item lies in its line, counted from the line's top.

Called by `outranks()`
Calls *no function of the code*

### 4.6.40 `markedFilters()` INTERNAL lines 193–195

An entry for every marked item that is not a copy, with its mark (`filterMark()`) and its line, unless a marked row above covers it (`covered()`) or it keeps a lone filter entry instead.

Called by `filtersOf()`
Calls `filterMark()`, `covered()`, `keepsOwnFilter()`; in js/model/itemtree.js: `lineOf()`

### 4.6.41 `covered(item, mark)` INTERNAL lines 197–198

Tells whether an item's mark is already in the entry of the row above it, or above a copy of it: that row's whole branch has the mark, and the row does not keep a lone filter entry of its own.

Called by `markedFilters()`
Calls `branchState()`, `keepsOwnFilter()`

### 4.6.42 `byIndex(a, b)` INTERNAL line 200

Sorts entries in the tree's order, by their items' places in `items`.

Called by `filtersOf()`
Calls *no function of the code*

### 4.6.43 `entryOf({ item, mark, lone })` EXPORTED lines 202–206

What an entry of Filters, as `filtersOf()` gives it, names: a line's deepest item with must have, unless the entry is lone, and whether it reads "Full": a building or group marked in full, with more than one row (Req. 24). `chipName()` puts it into words.

Called by in js/view/civsview.js: `chipName()`
Calls `isLoneUnique()`, `branchState()`; in js/model/itemtree.js: `lineOf()`, `isUnitOrTech()`

### 4.6.44 `filterMark(item)` INTERNAL line 208

The mark an item stands for in Filters: the one all of it shows, else its own.

Called by `markedFilters()`
Calls `branchState()`

### 4.6.45 `keepsOwnFilter(item, mark)` INTERNAL line 210

Tells whether a mark keeps an item's lone filter entry: must have on an item that `isLoneUnique()` picks.

Called by `keepingLoneFilters()`, `loneFilters()`, `markedFilters()`, `covered()`
Calls `isLoneUnique()`

### 4.6.46 `isLoneUnique(item)` INTERNAL line 212

Tells whether an item is unique in a line that is not all unique, so it keeps its own entry in Filters (Req. 36).

Called by `entryOf()`, `keepsOwnFilter()`
Calls `isUnique()`; in js/model/itemtree.js: `lineOf()`

### 4.6.47 `isUnique(item)` INTERNAL line 214

Tells whether data.json counts an item as unique, by its reach (`reachOf()`).

Called by `isLoneUnique()`
Calls in js/model/itemtree.js: `reachOf()`

# 5 js/model/data/

4 files 108 lines

The only scripts that load or store data or read the address. datahandler.js, the Facade, loads data.json and the language files, holds the rule of which language to start in, and passes on what the other three hold: storage.js the saved settings, address.js the address's parameters, and guard.js `attempt()`, which turns what a browser refuses into nothing. Every other script imports from datahandler.js alone.

**Subchapters:** 5.1 datahandler.js, 5.2 storage.js, 5.3 address.js, 5.4 guard.js

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## 5.1 datahandler.js

**Folder:** js/model/data/ 69 lines 12 functions, 12 exported

**Imports:** js/model/civsets.js, js/model/data/storage.js, js/model/data/address.js
**Imported by:** js/script.js, js/model/texts.js, js/model/icons.js, js/model/itemtree.js, js/model/updater.js, js/model/model.js, js/view/page.js, js/view/civsview.js, js/view/optionsview.js, js/view/report.js, js/view/pictures/pictures.js, js/view/pictures/scrollbars.js, js/controller/optionshandler.js, js/tests/run_test.js, js/tests/helpers_test.js, js/tests/audit_test.js, js/tests/behaviour/greying_test.js, js/tests/interface/menu_test.js, js/tests/interface/languages_test.js, js/tests/interface/options_test.js, js/tests/interface/uniques_test.js, js/tests/pictures/pictures_test.js, js/tests/pictures/icons_test.js, js/tests/pictures/tooltips_test.js, js/tests/static/scanner_test.js, js/tests/static/rules_test.js, js/tests/static/data_test.js, js/tests/static/updates_test.js

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The Facade over the data: it loads data.json and the language files, unpacks and packs them, and holds the language rule, which language the page starts in. It passes on what the other three scripts of js/model/data/ hold, the saved settings of storage.js, the address of address.js and `attempt()` of guard.js, so every script outside js/model/data/ imports from it alone, and none of them reads a file, the browser's storage or the address (Tests 250, 251).

### 5.1.1 `useData(raw)` EXPORTED lines 10–13

Reads data.json and hands its list of civilisations to `useCivs()`, which gives each civilisation one bit.

Called by in js/script.js: `useAndLoadLanguages()`
Calls `unpack()`; in js/model/civsets.js: `useCivs()`

### 5.1.2 `unpack(file)` EXPORTED lines 15–18

Turns data.json as stored into the shape used in memory, where overrides and text are kept per language, so other languages can be added beside English.

Called by `useData()`; in js/tests/static/updates_test.js: `readBack()`
Calls `firstLanguage()`

### 5.1.3 `loadLanguages()` EXPORTED line 20

Loads every language's file. The tests and the updater need them all.

Called by in js/script.js: `RUNS()`; in js/model/updater.js: `fetchUpdate()`
Calls `loadLanguage()`

### 5.1.4 `loadLanguage(code)` EXPORTED lines 22–24

Loads a language's file unless it is loaded already. A visitor downloads only the language shown, not all of them.

Called by `loadLanguages()`; in js/script.js: `loadPreferred()`; in js/controller/optionshandler.js: `chooseLanguage()`
Calls `hasLanguage()`, `load()`, `localeFile()`, `addLanguage()`

### 5.1.5 `load(url)` EXPORTED lines 26–34

Fetches a file as text and returns a promise. It uses XMLHttpRequest because, unlike `fetch()`, it also reads files when the page is opened from disk, as the tests in CI do.

Called by `loadLanguage()`; in js/script.js: `Start-up`; in js/model/updater.js: `fetchUpstream()`; in js/tests/run_test.js: `runAll()`; in js/tests/pictures/icons_test.js: `edges()`; in js/tests/static/scanner_test.js: `scripts()`, `styles()`
Calls *no function of the code*

### 5.1.6 `localeFile(code)` EXPORTED line 36

Returns the path of a language's file, such as `locales/de.json`. Keeps the pattern in one place for loading and for the updater's downloads.

Called by `loadLanguage()`, `pack()`; in js/tests/interface/options_test.js: `failing()`; in js/tests/static/updates_test.js: `updaterTests()`
Calls *no function of the code*

### 5.1.7 `addLanguage(siteData, code, file)` EXPORTED lines 38–41

Adds one language file's overrides and text to the data in memory. Used when a language loads, and when a test reads back the files the updater would write.

Called by `loadLanguage()`; in js/tests/static/updates_test.js: `readBack()`
Calls *no function of the code*

### 5.1.8 `firstLanguage(from = data)` EXPORTED line 43

The first language of data.json, English, or of the data it is given. The language rule rests on it: it is the default (Req. 10), the one whose texts data.json itself holds, and the one the page falls back on.

Called by `unpack()`, `preferredLanguage()`, `startLanguage()`, `pack()`; in js/model/updater.js: `missingText()`
Calls *no function of the code*

### 5.1.9 `hasLanguage(code)` EXPORTED line 45

Tells whether a language's file has loaded, so that its texts are in the data.

Called by `loadLanguage()`, `startLanguage()`; in js/controller/optionshandler.js: `chooseLanguage()`
Calls *no function of the code*

### 5.1.10 `preferredLanguage()` EXPORTED lines 47–50

The language the visitor prefers: the one in the address (`?lng=`), else the visitor's saved choice, else the first, English (Req. 10). Shared links and return visits keep their language; first visits are English.

Called by `startLanguage()`; in js/script.js: `loadPreferred()`, `start()`; in js/tests/interface/menu_test.js: `englishFirst()`
Calls `firstLanguage()`; in js/model/data/storage.js: `recall()`

### 5.1.11 `startLanguage()` EXPORTED lines 52–55

The start language: the preferred language if its file has loaded, else the first. `start()` gives it to the model, and opens the popup `#languageError` when it is not the preferred one.

Called by in js/script.js: `start()`
Calls `preferredLanguage()`, `hasLanguage()`, `firstLanguage()`

### 5.1.12 `pack(siteData)` EXPORTED lines 57–69

Splits the data in memory back into files: data.json with English, and one file per other language. The updater writes these, and a test checks that nothing is lost.

Called by in js/model/updater.js: `changedFiles()`; in js/tests/static/updates_test.js: `updaterTests()`
Calls `firstLanguage()`, `localeFile()`

# 5.2 storage.js

**Folder:** js/model/data/ 22 lines 4 functions, 4 exported
**Imports:** js/model/data/guard.js
**Imported by:** js/model/data/datahandler.js

The saved choices, in the browser's storage. `SETTINGS` lists each setting of the Options window with its values, the first being its default, so that each is read and stored the same way everywhere. Every read and write goes through `attempt()`, so a browser that refuses its storage, as in a private window, only forgets the choices. datahandler.js passes it all on.

### 5.2.1 `recall(key)` EXPORTED

line 11

Reads a saved choice from the browser's storage, or null. Remembers the visitor's language and options between visits.

Called by `recallSetting()`; in js/model/data/datahandler.js: `the top-level code`, `preferredLanguage()`
Calls in js/model/data/guard.js: `attempt()`

### 5.2.2 `remember(key, value)` EXPORTED

lines 13–15

Saves a choice in the browser's storage, or removes it when the value is null. Failures are ignored, as the page works without storage.

Called by `rememberSetting()`
Calls in js/model/data/guard.js: `attempt()`

### 5.2.3 `recallSetting(key)` EXPORTED

lines 17–20

Reads a saved setting back as one of its values in `SETTINGS`, or its default when none is saved or the saved one is none of them. The model reads Include and Include Regionals with it, script.js Simple UI, and optionshandler.js Civ view and Tooltips.

Called by *no function of the code*
Calls `recall()`

### 5.2.4 `rememberSetting(key, value)` EXPORTED

line 22

Saves a setting's value as text, which `recallSetting()` turns back into the value. The model saves Include and Include Regionals with it, and optionshandler.js Simple UI, Civ view and Tooltips.

Called by *no function of the code*
Calls `remember()`

# 5.3 address.js

**Folder:** js/model/data/ 9 lines 1 function, 1 exported
**Imports:** js/model/data/guard.js
**Imported by:** js/model/data/datahandler.js

The address: `params`, the parameters the page was opened with, such as `?test`, `?update` and `?lng=`, read once, and `keepInAddress()`, the one function that writes it. datahandler.js passes both on.

## 5.3.1 `keepInAddress(key, value)` EXPORTED

lines 5–9

Puts a choice into the address, as `?lng=` for the language, so that a shared link keeps it, without a new step in the browser's history. The only function that writes the address, and only model.js calls it; failures are ignored.

Called by *no function of the code*
Calls in js/model/data/guard.js: `attempt()`

# 5.4 guard.js

**Folder:** js/model/data/ 8 lines 1 function, 1 exported

**Imports:** nothing

**Imported by:** js/model/data/storage.js, js/model/data/address.js

`attempt()`, which runs what the browser may refuse. storage.js and address.js use it, and so do the views, through datahandler.js, to read the style sheets, try a selector and capture the pointer.

## 5.4.1 `attempt(action)` EXPORTED

lines 1–8

Runs a function and returns null instead of a `DOMException`, the error with which the browser refuses, such as storage in a private window, so a refusal never breaks the page. Any other error is thrown again, so that a mistake in the code is not hidden.

Called by in js/model/data/storage.js: `recall()`, `remember()`; in js/model/data/address.js: `keepInAddress()`

Calls *no function of the code*

# 6 js/view/

6 files 722 lines

What the page shows: toolkit.js holds the small tools every view uses, page.js holds the views as one view of the whole page and writes its fixed texts, civsview.js draws the civilisations and Filters, treeview.js the tree, optionsview.js the Options window, and report.js the report window; js/view/pictures/ holds what only the pictures need. No view imports a controller.

**Subchapters:** 6.1 toolkit.js, 6.2 page.js, 6.3 civsview.js, 6.4 treeview.js, 6.5 optionsview.js, 6.6 report.js

## 6.1 toolkit.js

**Folder:** js/view/ 32 lines 8 functions, 8 exported

**Imports:** js/model/texts.js

**Imported by:** js/view/page.js, js/view/civsview.js, js/view/treeview.js, js/view/optionsview.js, js/view/report.js, js/view/pictures/pictures.js, js/view/pictures/tooltips.js, js/view/pictures/scrollbars.js, js/controller/optionshandler.js, js/tests/run_test.js, js/tests/helpers_test.js, js/tests/audit_test.js, js/tests/behaviour/marking_test.js, js/tests/behaviour/samples_test.js, js/tests/behaviour/branches_test.js, js/tests/behaviour/tree_test.js, js/tests/behaviour/lines_test.js, js/tests/behaviour/greying_test.js, js/tests/behaviour/finding_test.js, js/tests/interface/menu_test.js, js/tests/interface/languages_test.js, js/tests/interface/options_test.js, js/tests/interface/keys_test.js, js/tests/interface/uniques_test.js, js/tests/interface/reporting_test.js, js/tests/pictures/pictures_test.js, js/tests/pictures/icons_test.js, js/tests/pictures/findbox_test.js, js/tests/pictures/tooltips_test.js, js/tests/pictures/treebuttons_test.js, js/tests/pictures/scrollbars_test.js

What every view uses: finding and copying the page's elements, telling whether Simple UI is on, what a change means for a view, writing a text with elements in it, placing an icon in its atlas, and checking a controller's actions. It imports only the placeholder pattern of texts.js.

### 6.1.1 `$(id)` EXPORTED

line 3

Short for `document.getElementById`, used wherever the page is touched.

Called by `clone()`; in js/view/page.js: `writeTexts()`, `writeFixed()`, `setLink()`, `showPage()`, `setLoadError()`, `showLanguageError()`; in js/view/civsview.js: `startCivsView()`, `refocusChip()`, `chipsShown()`, `listCivs()`, `setView()`, `listFilters()`; in js/view/treeview.js: `startTreeView()`, `wireTools()`, `handOffFocus()`, `findText()`, `replant()`, `highlight()`; in js/view/optionsview.js: `wireLanguages()`, `setMenu()`, `coverSettings()`, `menuOpen()`, `showLanguage()`, `languageOptions()`, `onMenuKey()`, `wireOptions()`, `setTicked()`, `checked()`, `buildLanguages()`, `readers()`, `showView()`; in js/view/report.js: `showUpdate()`, `setStatus()`, `openReport()`, `report()`; in js/view/pictures/pictures.js: `hideLoadingWhenReady()`, `onPage()`, `setLoading()`; in js/view/pictures/tooltips.js: `startTips()`; in js/tests/run_test.js: `publish()`; in js/tests/helpers_test.js: `ticked()`, `nameFor()`, `type()`, `pick()`, `reset()`, `beside()`, `listed()`; in js/tests/audit_test.js: `controlChecks()`; in js/tests/behaviour/marking_test.js: `marking()`, `chipMarks()`; in js/tests/behaviour/samples_test.js: `chipNames()`; in js/tests/behaviour/branches_test.js: `lineFilters()`; in js/tests/behaviour/tree_test.js: `crowding()`, `counts()`, `tree()`; in js/tests/behaviour/lines_test.js: `parents()`, `lines()`, `evenLine()`, `unevenLine()`, `beyond()`; in js/tests/behaviour/greying_test.js: `rivals()`, `excludedBy()`, `sharedRow()`, `variants()`; in js/tests/behaviour/finding_test.js: `finding()`, `dropsGreyed()`; in js/tests/interface/menu_test.js: `menuKeys()`, `menuOpening()`, `menuClosing()`, `menuChoosing()`, `windowSize()`; in js/tests/interface/languages_test.js: `footerTexts()`, `footerOnce()`; in js/tests/interface/options_test.js: `optionKeys()`, `openingOptions()`, `cancellingOptions()`, `closedAndFocused()`, `languagePopupTests()`, `failing()`, `confirmBox()`; in js/tests/interface/keys_test.js: `filterFocus()`, `treeMoves()`, `findKey()`, `menuAudit()`, `languageAudits()`, `optionAudits()`; in js/tests/interface/uniques_test.js: `uniqueFilterTests()`, `inLine()`, `names()`, `unmark()`; in js/tests/interface/reporting_test.js: `reportTests()`; in js/tests/pictures/pictures_test.js: `wrongSizes()`, `offParchment()`, `optionsButtonTests()`, `simpleUiTests()`, `picturesOn()`, `simpleSpot()`, `civLinks()`, `civViewTests()`, `pickView()`, `sideBySide()`; in js/tests/pictures/icons_test.js: `boxTests()`, `civIcons()`; in js/tests/pictures/findbox_test.js: `findBoxTests()`, `lens()`, `crossTests()`; in js/tests/pictures/tooltips_test.js: `treeTipTests()`, `includeTipTests()`, `silentTipTests()`, `everyTipTests()`, `tipsOver()`; in js/tests/pictures/treebuttons_test.js: `treeButtonTests()`, `sideBySide()`, `everyLanguage()`, `shown()`; in js/tests/pictures/scrollbars_test.js: `scrollBarTests()`, `partsOf()`, `comboArrows()`, `languageWheel()`

Calls *no function of the code*

### 6.1.2 `clone(id)` EXPORTED

line 5

Copies the element inside a template in index.html. Repeated markup lives in index.html (Rule 2); the code only fills it in.

Called by in js/view/civsview.js: `civEntry()`, `civIcon()`, `chipEntry()`; in js/view/treeview.js: `build()`, `dress()`; in js/view/optionsview.js: `buildLanguages()`; in js/view/report.js: `downloadItem()`, `report()`; in js/view/pictures/pictures.js: `setLoading()`; in js/view/pictures/scrollbars.js: `startBars()`

Calls `$()`

### 6.1.3 `simple()` EXPORTED line 7

Tells whether Simple UI is on, from the `pictures` class on the page, which is what shows the pictures. One source of truth, readable from every script.

Called by in js/view/civsview.js: `civEntry()`; in js/view/treeview.js: `dress()`; in js/view/optionsview.js: `coverSettings()`; in js/view/pictures/pictures.js: `applySimple()`, `hideLoadingWhenReady()`; in js/view/pictures/tooltips.js: `tipsWanted()`; in js/view/pictures/scrollbars.js: `wire()`, `fit()`; in js/controller/optionshandler.js: `choices()`; in js/tests/run_test.js: `snapshot()`, `toStart()`; in js/tests/pictures/scrollbars_test.js: `barStates()`

Calls *no function of the code*

### 6.1.4 `changesNames(change)` EXPORTED line 9

Tells whether a change can change the names the page shows: `"start"`, `"tree"` and `"language"`. The tree's view then writes its rows' names again, and the view of Filters notes the shared names again.

Called by in js/view/civsview.js: `filtersView()`; in js/view/treeview.js: `treeView()`

Calls *no function of the code*

### 6.1.5 `rebuilds(change)` EXPORTED line 11

Tells whether a change brings new items, at the start and after `"tree"`, so that the tree's view draws its rows again.

Called by in js/view/treeview.js: `treeView()`

Calls *no function of the code*

### 6.1.6 `compose(node, template, values)` EXPORTED lines 13–18

Writes a text with placeholders into an element, putting elements such as the source link where their placeholder is. Lets a translated sentence contain a link. It splits the text at the placeholder pattern that `fill()` uses too, and leaves a placeholder without a value as it is.

Called by in js/view/page.js: `writeTexts()`, `writeFixed()`; in js/view/treeview.js: `showCounts()`

Calls *no function of the code*

### 6.1.7 `place(icon, { column, row, columns, rows })` EXPORTED lines 20–26

Tells an icon which cell of its atlas to show, as four numbers for CSS: its column and row, and the atlas's columns and rows. Scripts give CSS only numbers.

Called by in js/view/civsview.js: `civIcon()`; in js/view/treeview.js: `dress()`

Calls *no function of the code*

### 6.1.8 `demand(actions, names)` EXPORTED lines 28–32

Checks that a controller has every action a view needs, and stops with an error that names the missing ones. treeview.js, civsview.js and optionsview.js call it as they start, so a missing action shows at once.

Called by in js/view/civsview.js: `startCivsView()`; in js/view/treeview.js: `startTreeView()`; in js/view/optionsview.js: `startOptionsView()`

Calls *no function of the code*

# 6.2 page.js

**Folder:** js/view/ 57 lines 8 functions, 1 object, 6 exported

**Imports:** js/model/texts.js, js/model/model.js, js/model/data/datahandler.js, js/view/toolkit.js, js/view/civsview.js, js/view/treeview.js, js/view/optionsview.js

**Imported by:** js/script.js, js/controller/optionshandler.js

The whole page as one view (Composite), and the model's only observer (Observer). It holds the views of the texts, Options, the tree, the civilisations and Filters, and `redraw()` passes each change to all of them, always in that order: the model's notices, and the start and Simple UI. It also writes the page's fixed texts and those of the chosen language, shows the page once it is drawn, and shows the messages that the data or a language could not be loaded.

### 6.2.1 `textsView` OBJECT, INTERNAL lines 9–13

The view of the texts: writes them again at the start and when the language changes.

Called by `the top-level code`
Calls `writeTexts()`

### 6.2.2 `startPage()` EXPORTED line 17

Has the model tell the page of its changes, by subscribing `{ update: redraw }`. Runs once, from `start()`.

Called by in js/script.js: `start()`
Calls `redraw()`; in js/model/model.js: `subscribe()`

### 6.2.3 `redraw(change)` EXPORTED line 19

Passes a change to every view, in order: the model's `"marks"`, `"tree"` and `"language"`, `"start"` once at the start, from `start()`, and `"look"` after Simple UI changes, from `chooseSimple()`.

Called by `startPage()`; in js/script.js: `start()`; in js/controller/optionshandler.js: `chooseSimple()`
Calls *no function of the code*

### 6.2.4 `writeTexts()` INTERNAL lines 21–28

Writes every text of the chosen language: the page's `lang`, each element marked `data-text`, and the footer's source and game update, whose link reads "Update" and its number in the language (Req. 5, 11).

Called by `textsView()`
Calls in js/model/texts.js: `fill()`; in js/view/toolkit.js: `$()`, `compose()`

### 6.2.5 `writeFixed()` EXPORTED lines 30–39

Writes the texts that never change: the report window's English words, the source's, Microsoft's and the project's links (`setLink()`), the address of the game update's notes, and Microsoft's notice around its links. Runs once at the start.

Called by in js/script.js: `start()`
Calls `setLink()`; in js/view/toolkit.js: `$()`, `compose()`

### 6.2.6 `setLink(id, { url, name })` INTERNAL lines 41–44

Gives a link its address and its name, both from `data.json`.

Called by `writeFixed()`
Calls in js/view/toolkit.js: `$()`

### 6.2.7 `showPage()` EXPORTED lines 46–49

Shows the header and the content, which stay hidden until the page has started.

Called by in js/script.js: `start()`
Calls in js/view/toolkit.js: `$()`

### 6.2.8 `setLoadError(on)` EXPORTED lines 51–53

Shows or hides the message that the data could not be loaded, `#loadError`, which is in English, as no language has loaded then.

Called by in js/script.js: `failLoad()`
Calls in js/view/toolkit.js: `$()`

### 6.2.9 `showLanguageError()` EXPORTED lines 55–57

Opens the popup `#languageError`, "The language could not be loaded.", unless it is open already: when the file of the language chosen in Options cannot be fetched, and at the start when the visitor's own language cannot. It is a modal dialog over the page, so the panes neither move nor change size (Req. 35); its OK closes it, and the focus goes back to where it was.

Called by in js/script.js: `start()`; in js/controller/optionshandler.js: `languageFailed()`
Calls in js/view/toolkit.js: `$()`

# 6.3 civsview.js

**Folder:** js/view/ 116 lines 14 functions, 2 objects, 6 exported

**Imports:** js/model/texts.js, js/model/icons.js, js/model/model.js, js/model/data/datahandler.js, js/view/toolkit.js
**Imported by:** js/view/page.js, js/controller/treehandler.js, js/controller/optionshandler.js, js/tests/run_test.js, js/tests/helpers_test.js, js/tests/behaviour/marking_test.js

The civilisations' pane: the count, the list or the tiles with the Civ view setting, and Filters, whose buttons it makes and names. It is a part of the page in page.js, and hands a click on a Filters button to the actions of treehandler.js, its controller (Strategy). Each button carries its filter in a `WeakMap`, not in its attributes, so that nothing has to read the filter back from the markup.

### 6.3.1 `startCivsView(controller)` EXPORTED lines 21–24

Starts the civilisations' view with its controller's actions, checked with `demand()`, and connects the Filters buttons. Runs once, from `startTree()`.

Called by in js/controller/treehandler.js: `startTree()`
Calls `onChip()`; in js/view/toolkit.js: `demand()`, `$()`

### 6.3.2 `onChip(e)` INTERNAL lines 26–31

Has the controller clear the marks behind a Filters button, handing it the filter the button carries, and notes the button's place, so that the focus can go on from there once Filters is drawn again (`refocusChip()`).

Called by `startCivsView()`
Calls `chipsShown()`

### 6.3.3 `refocusChip(at)` INTERNAL lines 33–36

Moves focus, once Filters is drawn again, to the button now in the place of the one clicked, or to the one before it when the last was cleared, or to the civilisations' heading when no button is left (WCAG 2.4.3).

Called by `listFilters()`
Calls `chipsShown()`; in js/view/toolkit.js: `$()`

### 6.3.4 `chipsShown()` INTERNAL line 38

Lists the Filters buttons on the page, in their order.

Called by `onChip()`, `refocusChip()`
Calls in js/view/toolkit.js: `$()`

### 6.3.5 `civsView` OBJECT, EXPORTED line 40

The view of the civilisations: lists them again after every change.

Called by in js/view/page.js: `the top-level code`
Calls `listCivs()`

### 6.3.6 `listCivs()` INTERNAL lines 42–48

Lists the civilisations that fit the marks (`matchingCivs()`), translated and sorted (Req. 4), each made by `civEntry()`; writes the count, and the hint when the list is empty.

Called by `civsView()`
Calls `byName()`, `civEntry()`; in js/model/texts.js: `civName()`, `plural()`; in js/model/model.js: `matchingCivs()`; in js/view/toolkit.js: `$()`

### 6.3.7 `byName([a], [b])` INTERNAL line 50

Sorts civilisations by their translated names, in the order of the language's collator.

Called by `listCivs()`
Calls *no function of the code*

### 6.3.8 `civEntry([name, civ])` INTERNAL lines 52–61

Makes a civilisation's entry from its template, with its name, and without Simple UI its icon and its link to its page on aoe2techtree.net.

Called by `listCivs()`
Calls `civIcon()`; in js/view/toolkit.js: `clone()`, `simple()`

### 6.3.9 `civIcon(civ)` INTERNAL line 63

Makes a civilisation's icon from its template, showing its cell of the civilisations' atlas. Only used without Simple UI.

Called by `civEntry()`
Calls in js/model/icons.js: `cellOf()`, `civFile()`; in js/view/toolkit.js: `place()`, `clone()`

### 6.3.10 `setView(choice)` EXPORTED lines 65–68

Shows the civilisations as a list or as tiles, and keeps the choice as `view`, the Civ view setting. Tiles are pictures, so CSS shows them only without Simple UI.

Called by in js/controller/optionshandler.js: `startOptions()`, `chooseView()`; in js/tests/run_test.js: `toStart()`, `restore()`
Calls in js/view/toolkit.js: `$()`

### 6.3.11 `filtersView` OBJECT, EXPORTED lines 70–75

The view of Filters: notes the shared names again at the start and when the tree or the language changed (`changesNames()`), then makes the buttons again.

Called by in js/view/page.js: `the top-level code`
Calls `sharedNames()`, `listFilters()`; in js/view/toolkit.js: `changesNames()`

### 6.3.12 `sharedNames()` INTERNAL lines 77–84

Lists the names that more than one row uses, so that Filters can say which one is meant.

Called by `filtersView()`
Calls in js/model/texts.js: `nameOf()`

### 6.3.13 `listFilters()` INTERNAL lines 86–92

Makes one Filters button per filter entry (Req. 22, 23), with `chipEntry()`, and hides Filters when nothing is marked. After a click on a button, it moves the focus on (`refocusChip()`).

Called by `filtersView()`
Calls `chipEntry()`, `refocusChip()`; in js/model/model.js: `filtersOf()`; in js/view/toolkit.js: `$()`

### 6.3.14 `chipEntry(filter)` INTERNAL lines 94–105

Makes a Filters button from its template, with its mark and name, the filter it stands for, kept for `onChip()`, and a label telling screen readers what a click clears.

Called by `listFilters()`
Calls `chipName()`; in js/model/texts.js: `fill()`; in js/view/toolkit.js: `clone()`

### 6.3.15 `chipName(filter)` EXPORTED lines 107–114

Names an entry of Filters, as `filtersOf()` gives it, from what `entryOf()` makes of it: a shared name "in" its parent, and "Full" before a building or group marked in full (Req. 24).

Called by `chipEntry()`; in js/tests/behaviour/marking_test.js: `marking()`
Calls in js/model/texts.js: `nameOf()`, `fill()`; in js/model/model.js: `entryOf()`

### 6.3.16 `nameIsShared(name)` EXPORTED line 116

Tells whether more than one row uses a name. Only the tests ask it.

Called by in js/tests/helpers_test.js: `fixtures()`
Calls *no function of the code*

# 6.4 treeview.js

**Folder:** js/view/ 326 lines 41 functions, 1 object, 13 exported

**Imports:** js/model/texts.js, js/model/icons.js, js/model/model.js, js/view/toolkit.js
**Imported by:** js/view/page.js, js/view/pictures/tooltips.js, js/controller/treehandler.js, js/tests/helpers_test.js, js/tests/audit_test.js, js/tests/behaviour/marking_test.js, js/tests/behaviour/branches_test.js, js/tests/behaviour/tree_test.js, js/tests/behaviour/lines_test.js, js/tests/behaviour/greying_test.js, js/tests/behaviour/finding_test.js, js/tests/interface/keys_test.js, js/tests/interface/headings_test.js, js/tests/pictures/treebuttons_test.js

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The tree on the left as the visitor sees it: its rows with their names, boxes, icons and counts, which branches are open, what Find found, and the focus, with the buttons above the tree and the Find box. It is a part of the page in page.js, and hands every click, touch and key to the actions of treehandler.js, its controller (Strategy). The tooltip asks it what is under the mouse.

### 6.4.1 `startTreeView(controller)` EXPORTED lines 24–35

Starts the tree's view with its controller's actions, checked with `demand()`, and connects keys, presses, moves that cancel a long press, the menu and clicks, then the tools around the tree. Runs once, from `startTree()`.

Called by in js/controller/treehandler.js: `startTree()`
Calls `onKey()`, `onPress()`, `onMove()`, `onRelease()`, `endPress()`, `onMenu()`, `onClick()`, `wireTools()`; in js/view/toolkit.js: `$()`, `demand()`

### 6.4.2 `onKey(e)` INTERNAL lines 37–43

Hands a key in the tree to the controller's `key()`, with Shift, the other modifier keys, whether an arrow has focus, and a way to stop the browser's own action, which the controller takes when it uses the key (WCAG 2.1.1).

Called by `startTreeView()`
Calls *no function of the code*

### 6.4.3 `onPress(e)` INTERNAL lines 45–59

Starts a right-click, or a touch that becomes a long press after half a second, which marks must lack (Req. 3). A press is one object: its item, its row, its pointer, where it began and whether it is a touch. The row shows as pressed when the long press is ready.

Called by `startTreeView()`
Calls `endPress()`, `itemAt()`

### 6.4.4 `onMove(e)` INTERNAL lines 61–63

Cancels a press that has drifted (`drifted()`), as when a finger scrolls the tree.

Called by `startTreeView()`
Calls `drifted()`, `endPress()`

### 6.4.5 `drifted(e)` INTERNAL line 65

Tells whether the pointer has moved more than `PRESS_DRIFT_PX`, 10 pixels, from where the press began.

Called by `onMove()`
Calls *no function of the code*

### 6.4.6 `endPress()` INTERNAL lines 67–73

Cancels a long press in progress and its highlight, when the finger moves or lifts, or the tree is drawn again, and notes when a touch ended.

Called by `startTreeView()`, `onPress()`, `onMove()`, `onRelease()`, `replant()`
Calls *no function of the code*

### 6.4.7 `itemAt(target)` EXPORTED lines 75–78

Returns the tree item an element belongs to, but not a heading, so one listener on the tree serves every row, and the tooltip knows when it is over the tree.

Called by `onPress()`, `onRelease()`, `onMenu()`, `onClick()`; in js/view/pictures/tooltips.js: `tipFor()`
Calls `heading()`

### 6.4.8 `heading(item)` INTERNAL line 80

Tells whether a group is only a heading inside an item, which has no box and cannot be marked.

Called by `itemAt()`, `branchAt()`, `build()`, `paintMark()`, `focusRow()`
Calls *no function of the code*

### 6.4.9 `branchAt(target)` EXPORTED lines 82–86

Returns the item whose branch an element opens or closes: that of a branch's arrow, or a heading. `onClick()` and the tooltip's `branchTip()` ask it, so that neither reads the row's markup.

Called by `onClick()`; in js/view/pictures/tooltips.js: `branchTip()`
Calls `heading()`

### 6.4.10 `onRelease(e)` INTERNAL lines 88–95

Ends a press: reports a hold on the item to the controller if the press was ready and ends on the same item, and swallows the click that follows.

Called by `startTreeView()`
Calls `endPress()`, `itemAt()`

### 6.4.11 `onMenu(e)` INTERNAL lines 97–103

Stops the browser's own menu over the tree, and reports the menu key to the controller. Right-clicks, long presses and the menu event that follows a touch are left to `onPress()` and `onRelease()` (`menuHandledElsewhere()`).

Called by `startTreeView()`
Calls `itemAt()`, `menuHandledElsewhere()`

### 6.4.12 `menuHandledElsewhere(e)` INTERNAL line 105

Tells whether a menu event comes from a right-click, a press in progress or a touch that has just ended, which `onPress()` and `onRelease()` handle.

Called by `onMenu()`
Calls `justTouched()`

### 6.4.13 `justTouched()` INTERNAL line 107

Tells whether a touch ended less than `MENU_AFTER_TOUCH_MS`, 600 milliseconds, ago, so that the menu event of a long touch is not taken for the menu key.

Called by `menuHandledElsewhere()`
Calls *no function of the code*

### 6.4.14 `onClick(e)` INTERNAL lines 109–117

Opens or closes a branch from its arrow, or a heading from its name (`branchAt()`), or reports a click on an item to the controller, unless the click ends a long press. What a gesture marks is the controller's to decide.

Called by `startTreeView()`
Calls `branchAt()`, `itemAt()`

### 6.4.15 `wireTools()` INTERNAL lines 119–132

Connects Expand all and Collapse all (`handOffFocus()`), Clear (Req. 17 to 19), and Find with its clearing cross and Enter, each to its action. Runs once at the start.

Called by `startTreeView()`
Calls `handOffFocus()`, `findText()`; in js/view/toolkit.js: `$()`

### 6.4.16 `handOffFocus(from, to, act)` INTERNAL lines 134–138

Connects Expand all or Collapse all to its action, and moves focus to the other if the one clicked had it and no longer shows, as without Simple UI, where the two take turns in one place.

Called by `wireTools()`
Calls in js/view/toolkit.js: `$()`

### 6.4.17 `findText()` EXPORTED line 140

What is typed in the Find box.

Called by `wireTools()`; in js/controller/treehandler.js: `redrawn()`
Calls in js/view/toolkit.js: `$()`

### 6.4.18 `treeView` OBJECT, EXPORTED lines 142–152

The view of the tree: draws it again at the start and when the tree was built again (`rebuilds()`), dresses its rows when Simple UI changes, names them when the names may have changed (`changesNames()`), then closes greyed-out items, paints the boxes and writes every count. Then it hands the change to its controller's `redrawn()`, so that Find follows the tree as it is drawn now.

Called by in js/view/page.js: `the top-level code`
Calls `replant()`, `dress()`, `label()`, `setOpen()`, `paint()`, `showCounts()`; in js/model/model.js: `locked()`; in js/view/toolkit.js: `rebuilds()`, `changesNames()`

### 6.4.19 `replant()` INTERNAL lines 154–164

Draws the tree from the model again, keeping the branches that were open open by their paths, and dresses each row. Runs at the start and whenever the tree is built again.

Called by `treeView()`
Calls `opened()`, `endPress()`, `build()`, `dress()`, `setOpen()`; in js/view/toolkit.js: `$()`

### 6.4.20 `opened(row)` INTERNAL line 166

Tells whether a row's branch is open, read from its arrow's `aria-expanded`, which is the one place this is kept.

Called by `replant()`, `isOpen()`
Calls *no function of the code*

### 6.4.21 `build(list, ul)` INTERNAL lines 168–176

Makes a row for each item from the template (`newRow()`), takes a heading's box away (`makeHeading()`), and gives it its branch (`buildBranch()`), which calls `build()` again for the items below.

Called by `replant()`, `buildBranch()`
Calls `newRow()`, `heading()`, `makeHeading()`, `buildBranch()`; in js/view/toolkit.js: `clone()`

### 6.4.22 `newRow(li, item)` INTERNAL lines 178–191

Finds the parts of a new row by their class: its entry, row, name, "must lack" words and counts, and gives the row its item's number and its name an id.

Called by `build()`
Calls `labelId()`

### 6.4.23 `labelId(item)` INTERNAL line 193

The id of an item's name, which labels its arrow for screen readers.

Called by `newRow()`, `buildBranch()`
Calls *no function of the code*

### 6.4.24 `kidsId(item)` INTERNAL line 195

The id of the list of the items below an item, which its arrow controls.

Called by `buildBranch()`
Calls *no function of the code*

### 6.4.25 `makeHeading(row)` INTERNAL lines 197–200

Makes a row a heading, which cannot be marked: it loses its box, its role, its place in the tab order and `aria-checked`.

Called by `build()`
Calls *no function of the code*

### 6.4.26 `buildBranch(row, item)` INTERNAL lines 202–218

Gives a row with items below it its arrow and its list, with ids that tie the arrow and the list to the name for screen readers, and builds the items below; a row without any loses both.

Called by `build()`
Calls `kidsId()`, `labelId()`, `build()`

### 6.4.27 `dress(item)` INTERNAL lines 220–230

Gives a tree row its icon between its box and name, from its sprite (`spriteOf()`) and its cell of its atlas (`atlasOf()`), with its kind and reach for the outline, or takes it away with Simple UI on (Rules 9, 22).

Called by `treeView()`, `replant()`
Calls in js/model/icons.js: `spriteOf()`, `cellOf()`, `atlasOf()`; in js/view/toolkit.js: `simple()`, `place()`, `clone()`

### 6.4.28 `setOpen(item, open)` EXPORTED lines 232–238

Opens or closes an item's branch, and rewrites its count, since a closed item shows two numbers.

Called by `treeView()`, `replant()`; in js/controller/treehandler.js: `toggle()`, `branch()`, `expand()`, `leaveOrCollapse()`, `showMatches()`, `enter()`, `expandAll()`, `collapseAll()`; in js/tests/helpers_test.js: `openTo()`; in js/tests/behaviour/greying_test.js: `standIns()`, `shuts()`; in js/tests/pictures/treebuttons_test.js: `treeButtonTests()`
Calls `showCounts()`

### 6.4.29 `showCounts(item)` INTERNAL lines 240–257

Writes an item's count of civilisations that fit the marks, from the model's `countOf()`: its own number, or "own–all" when `showsRange()` says so, and a spoken version for screen readers. A root without a civilisation shows none.

Called by `treeView()`, `setOpen()`
Calls `showsRange()`; in js/model/texts.js: `plural()`, `fill()`; in js/model/model.js: `countOf()`; in js/view/toolkit.js: `compose()`

### 6.4.30 `showsRange(item)` INTERNAL line 259

Tells whether an item's count shows "own–all": when it is no group, has items below it and is closed.

Called by `showCounts()`
Calls `isOpen()`

### 6.4.31 `isOpen(item)` EXPORTED line 261

Tells whether an item's branch is open.

Called by `showsRange()`, `shown()`; in js/view/pictures/tooltips.js: `branchTip()`; in js/controller/treehandler.js: `branch()`, `enterOrExpand()`, `leaveOrCollapse()`; in js/tests/behaviour/tree_test.js: `countKinds()`, `wanted()`, `tree()`; in js/tests/behaviour/lines_test.js: `lines()`; in js/tests/behaviour/greying_test.js: `rivals()`, `sharedRow()`, `shuts()`; in js/tests/interface/keys_test.js: `treeMoves()`, `branchKeys()`; in js/tests/interface/headings_test.js: `headingTests()`
Calls `opened()`

### 6.4.32 `label()` INTERNAL lines 263–267

Writes every row's name in the current language.

Called by `treeView()`
Calls in js/model/texts.js: `nameOf()`

### 6.4.33 `paint()` INTERNAL lines 269–275

Updates every row: its mark (`paintMark()`) and whether it is greyed out (`paintBar()`).

Called by `treeView()`
Calls `paintMark()`, `paintBar()`

### 6.4.34 `paintMark(row, item)` INTERNAL lines 277–284

Shows a row's mark: its box's state for the eye, `aria-checked` for screen readers, from `ARIA_CHECKED`, the hidden "must lack" words, and the bold of a row's own mark. A heading has no box.

Called by `paint()`
Calls `heading()`; in js/model/model.js: `checkboxState()`

### 6.4.35 `paintBar(row, item)` INTERNAL lines 286–291

Shows a row as greyed out or not, for the eye and as `aria-disabled`, and disables a branch's arrow while the branch is locked.

Called by `paint()`
Calls in js/model/model.js: `barred()`, `locked()`

### 6.4.36 `focusRow(item)` EXPORTED lines 293–296

Moves focus to an item's row: a heading's arrow, or any other row's name.

Called by `focusTwisty()`; in js/controller/treehandler.js: `key()`, `enter()`
Calls `heading()`

### 6.4.37 `focusTwisty(item)` EXPORTED lines 298–302

Moves focus to an item's arrow, or to its row when it has none. The controller uses it when an arrow had focus, so that the keys go on from arrow to arrow.

Called by in js/controller/treehandler.js: `key()`
Calls `focusRow()`

### 6.4.38 `highlight(found, searched)` EXPORTED lines 304–309

Highlights the items Find found, and announces how many there are, or that there are none (Req. 20, 21), as `foundText()` words it.

Called by in js/controller/treehandler.js: `highlightMatches()`
Calls `foundText()`; in js/view/toolkit.js: `$()`

### 6.4.39 `foundText(found, searched)` INTERNAL lines 311–314

What Find announces: how many items it found, or that it found none, and nothing while its box is empty.

Called by `highlight()`
Calls in js/model/texts.js: `plural()`

### 6.4.40 `reveal(item)` EXPORTED lines 316–322

Scrolls the tree's own list just enough to show a row, and nothing around it, so the Find box being typed in never moves (Rule 18).

Called by in js/controller/treehandler.js: `showMatches()`
Calls *no function of the code*

### 6.4.41 `rowOf(item)` EXPORTED line 324

The elements of an item's row: its entry, row, name, "must lack" words, counts, arrow, list and icon. They are kept here, by item, so that the model's items hold no element (Test 258).

Called by in js/tests/helpers_test.js: `nodeOf()`, `twistyOf()`, `labelOf()`; in js/tests/audit_test.js: `controlChecks()`; in js/tests/behaviour/marking_test.js: `marking()`; in js/tests/behaviour/branches_test.js: `groupMarks()`; in js/tests/behaviour/tree_test.js: `miscounted()`, `tree()`

Calls *no function of the code*

### 6.4.42 `shown(item)` EXPORTED line 326

Tells whether an item's row can be seen, that is, whether every branch above it is open.

Called by in js/controller/treehandler.js: `moveFocus()`; in js/tests/behaviour/branches_test.js: `buildingMarks()`, `groupMarks()`, `closedInside()`; in js/tests/behaviour/lines_test.js: `lines()`; in js/tests/behaviour/greying_test.js: `rivals()`; in js/tests/behaviour/finding_test.js: `finding()`; in js/tests/interface/headings_test.js: `headingTests()`

Calls `isOpen()`

# 6.5 optionsview.js

**Folder:** js/view/ 141 lines 20 functions, 3 objects, 7 exported

**Imports:** js/model/texts.js, js/model/data/datahandler.js, js/view/toolkit.js

**Imported by:** js/view/page.js, js/controller/optionshandler.js, js/tests/helpers_test.js, js/tests/audit_test.js, js/tests/interface/menu_test.js, js/tests/interface/options_test.js, js/tests/pictures/tooltips_test.js, js/tests/pictures/scrollbars_test.js

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The Options window as the visitor sees it: the language menu, the boxes and Civ view, as they are ticked and picked before OK. It is a part of the page in page.js, and hands opening the window and OK to the actions of optionshandler.js, its controller (Strategy).

### 6.5.1 `startOptionsView(controller)` EXPORTED lines 9–13

Starts the Options window's view with its controller's actions, checked with `demand()`, and connects the language menu and the window. Runs once, from `startOptions()`.

Called by in js/controller/optionshandler.js: `startOptions()`
Calls `wireLanguages()`, `wireOptions()`; in js/view/toolkit.js: `demand()`

### 6.5.2 `wireLanguages()` INTERNAL lines 15–32

Connects the language menu: its button, a click on a language or anywhere else in the list (Req. 11, 12), its keys, and closing when focus or a click goes elsewhere. Runs once at the start.

Called by `startOptionsView()`
Calls `setMenu()`, `menuOpen()`, `onScrollbar()`, `showLanguage()`, `onMenuKey()`; in js/view/toolkit.js: `$()`

### 6.5.3 `setMenu(open, returnTo)` EXPORTED lines 34–39

Opens or closes the language menu, has `coverSettings()` hide or show the settings, and moves focus back to an element if asked.

Called by `wireLanguages()`, `onMenuKey()`, `wireOptions()`; in js/tests/helpers_test.js: `reset()`; in js/tests/pictures/tooltips_test.js: `everyTipTests()`; in js/tests/pictures/scrollbars_test.js: `scrollBarTests()`
Calls `coverSettings()`; in js/view/toolkit.js: `$()`

### 6.5.4 `coverSettings(open)` INTERNAL lines 41–43

With Simple UI, hides the settings while the language list is open, so that the window keeps its size (Rule 18); without Simple UI the list lies over them, as the game's combo box does.

Called by `setMenu()`
Calls in js/view/toolkit.js: `$()`, `simple()`

### 6.5.5 `menuOpen()` INTERNAL line 45

Tells whether the language list is open.

Called by `wireLanguages()`, `onMenuKey()`
Calls in js/view/toolkit.js: `$()`

### 6.5.6 `onScrollbar(e)` INTERNAL lines 47–48

Tells whether a click landed on the language list's own scroll bar, which Chrome reports as a click, so that scrolling the list does not close it.

Called by `wireLanguages()`
Calls *no function of the code*

### 6.5.7 `showLanguage(code)` INTERNAL lines 50–55

Shows a language as chosen in the window: its name on the menu's button and its mark in the list. OK makes it the page's language.

Called by `the top-level code`, `wireLanguages()`, `optionsView()`
Calls `languageOptions()`; in js/view/toolkit.js: `$()`

### 6.5.8 `languageOptions()` EXPORTED line 57

Lists the language menu's buttons. Used by the menu's keys and by the tests.

Called by `showLanguage()`, `onMenuKey()`, `picked()`; in js/tests/helpers_test.js: `pick()`; in js/tests/audit_test.js: `structureChecks()`; in js/tests/interface/menu_test.js: `menuKeys()`, `menuOpening()`, `menuClosing()`, `menuChoosing()`; in js/tests/interface/options_test.js: `cancellingOptions()`; in js/tests/pictures/scrollbars_test.js: `languageWheel()`
Calls in js/view/toolkit.js: `$()`

### 6.5.9 `onMenuKey(e)` INTERNAL lines 59–72

Handles the language menu's keys: Down and Up open it at the current language and move through it, Home and End jump, as `MENU_MOVES` says, and Escape closes it and returns focus (WCAG 2.1.1).

Called by `wireLanguages()`
Calls `menuOpen()`, `setMenu()`, `languageOptions()`, `MENU_MOVES()`; in js/view/toolkit.js: `$()`

### 6.5.10 `MENU_MOVES` OBJECT, INTERNAL lines 74–79

Where each key of the language menu goes: Down and Up one language on or back (`stepFrom()`), and Home and End to the first and the last while the menu is open.

Called by `onMenuKey()`
Calls `stepFrom()`

### 6.5.11 `stepFrom({ list, at }, by)` INTERNAL line 81

Steps one language on or back from the focused one, round from the last to the first and back, or goes to the chosen language when none has focus.

Called by `MENU_MOVES()`
Calls `isPicked()`

### 6.5.12 `isPicked(option)` INTERNAL line 83

Tells whether a language is the one marked as chosen in the list; the one test of the current language.

Called by `stepFrom()`, `picked()`
Calls *no function of the code*

### 6.5.13 `wireOptions()` INTERNAL lines 85–100

Connects the cogwheel button, which has the controller show the page's values in the window before it opens, OK, which has the controller apply the window, and the boxes, by click and by Space. Closing the window closes the language menu. Runs once at the start.

Called by `startOptionsView()`
Calls `setMenu()`, `toggle()`; in js/view/toolkit.js: `$()`

### 6.5.14 `toggle(id)` INTERNAL line 102

Ticks or unticks an Options box in the window only; OK applies it.

Called by `wireOptions()`
Calls `setTicked()`, `checked()`

### 6.5.15 `setTicked(id, on)` INTERNAL lines 104–106

Shows an Options box as ticked or not, in its `aria-checked`, which screen readers read and from which CSS draws the switch.

Called by `toggle()`, `showChoice()`
Calls in js/view/toolkit.js: `$()`

### 6.5.16 `checked(id)` INTERNAL line 108

Tells whether an Options box is ticked.

Called by `toggle()`, `readChoice()`
Calls in js/view/toolkit.js: `$()`

### 6.5.17 `buildLanguages()` EXPORTED lines 110–119

Makes one button per language, named and tagged in its own language, so screen readers say each name right (WCAG 3.1.2). Runs once at the start.

Called by in js/controller/optionshandler.js: `startOptions()`
Calls in js/view/toolkit.js: `$()`, `clone()`

### 6.5.18 `optionsView` OBJECT, EXPORTED lines 121–125

The view of the Options window: marks the page's language in it at the start and when the language changes.

Called by in js/view/page.js: `the top-level code`
Calls `showLanguage()`

### 6.5.19 `readChoice(id)` EXPORTED line 127

What the window shows for a setting: its reading in `readers`, or its box. `apply()` reads every setting through it.

Called by in js/controller/optionshandler.js: `apply()`
Calls `readers()`, `checked()`

### 6.5.20 `picked()` INTERNAL line 129

The language marked in the window's list, which OK makes the page's language.

Called by `readers()`
Calls `languageOptions()`, `isPicked()`

### 6.5.21 `readers` OBJECT, INTERNAL line 131

How the window reads the two settings that are no boxes: the language and Civ view.

Called by `readChoice()`
Calls `picked()`; in js/view/toolkit.js: `$()`

### 6.5.22 `showChoice(id, value)` EXPORTED line 133

Shows a value of a setting in the window: the language, Civ view, or a box. `showChoices()` shows every setting through it.

Called by in js/controller/optionshandler.js: `showChoices()`, `languageFailed()`
Calls `setTicked()`

### 6.5.23 `showView(choice)` INTERNAL lines 135–139

Marks the chosen view among the Civ view radio buttons.

Called by `the top-level code`
Calls in js/view/toolkit.js: `$()`

# 6.6 report.js

**Folder:** js/view/ 50 lines 7 functions, 5 exported

**Imports:** js/model/texts.js, js/model/data/datahandler.js, js/view/toolkit.js
**Imported by:** js/controller/updatehandler.js, js/tests/run_test.js, js/tests/interface/reporting_test.js

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The report window of the updater and the tests. It only shows what they found.

### 6.6.1 `showUpdating()` EXPORTED lines 5–6

Opens the report window for the updater, saying that it is loading the source's data.

Called by in js/controller/updatehandler.js: `runUpdate()`
Calls `openReport()`; in js/model/texts.js: `fill()`

### 6.6.2 `showUpdate({ messages, blocked, counts, files })` EXPORTED lines 8–18

Shows what the updater found: each change, the counts of civilisations, items and languages, and each changed file to download (`downloadItem()`); or that the update is blocked, when the new data cannot be used.

Called by in js/controller/updatehandler.js: `runUpdate()`
Calls `report()`, `setStatus()`, `downloadItem()`; in js/model/texts.js: `fill()`; in js/view/toolkit.js: `$()`

### 6.6.3 `setStatus(text)` INTERNAL lines 20–22

Writes the report window's status line.

Called by `showUpdate()`, `showUpdateFailed()`, `openReport()`
Calls in js/view/toolkit.js: `$()`

### 6.6.4 `downloadItem({ path, json })` INTERNAL lines 24–31

Makes the link that downloads a changed file, such as `data.json`: its new text, ending in a line break, under the file's own name.

Called by `showUpdate()`
Calls in js/model/texts.js: `fill()`; in js/view/toolkit.js: `clone()`

### 6.6.5 `showUpdateFailed(error)` EXPORTED lines 33–35

Shows why the update failed.

Called by in js/controller/updatehandler.js: `runUpdate()`
Calls `setStatus()`; in js/model/texts.js: `fill()`

### 6.6.6 `openReport(title, status)` EXPORTED lines 37–44

Opens the report window, used by the tests and the updater, with a title and a status line, and empties it.

Called by `showUpdating()`; in js/tests/run_test.js: `publish()`; in js/tests/interface/reporting_test.js: `reportTests()`
Calls `setStatus()`; in js/view/toolkit.js: `$()`

### 6.6.7 `report(line)` EXPORTED lines 46–50

Adds a line to the report window.

Called by `showUpdate()`; in js/tests/run_test.js: `publish()`; in js/tests/interface/reporting_test.js: `reportTests()`
Calls in js/view/toolkit.js: `clone()`, `$()`

# 7 js/view/pictures/

3 files 316 lines

The picture look, which Simple UI turns off: pictures.js picks and preloads the screen's pictures, runs the loading screen and starts the other two, tooltips.js shows the tooltip, and scrollbars.js draws the game's scroll bars.

**Subchapters:** 7.1 pictures.js, 7.2 tooltips.js, 7.3 scrollbars.js

## 7.1 pictures.js

**Folder:** js/view/pictures/ 115 lines 19 functions, 6 exported

**Imports:** js/model/data/datahandler.js, js/view/toolkit.js, js/view/pictures/tooltips.js, js/view/pictures/scrollbars.js
**Imported by:** js/script.js, js/controller/optionshandler.js, js/tests/pictures/picturetools_test.js

The picture look, what Simple UI off adds: the pictures the screen uses, their preloading and the loading screen. It also starts the tooltip of tooltips.js and the scroll bars of scrollbars.js.

### 7.1.1 `startPictures()` EXPORTED lines 28–31

Starts the tooltip (`startTips()`) and the scroll bars (`startBars()`). Runs once, after `start()`.

Called by in js/script.js: `run()`
Calls in js/view/pictures/tooltips.js: `startTips()`; in js/view/pictures/scrollbars.js: `startBars()`

### 7.1.2 `startScreen()` EXPORTED lines 33–36

Works out the screen's pictures once, as the page opens, into `data-background` and `data-assets` on `<html>` (`screenPictures()`), and sets `--pixel`, the size of one screen pixel. Resizing the window or zooming changes neither, so the page uses one parchment and one size of each picture for the whole visit.

Called by in js/script.js: `Start-up`
Calls `screenPictures()`

### 7.1.3 `applySimple(on)` EXPORTED lines 38–43

Switches the look (Req. 32, 33): shows the loading screen when leaving Simple UI and hides it when turning it on, and adds or removes the `pictures` class. Leaving Simple UI, it puts the loading picture on the page before the class, so that the browser asks for it first, and then preloads the pictures of `<html>` and `<body>` (`appliesToRoot()`). Redrawing the page and ending the loading screen are left to its callers: the top-level code of script.js at the start, and `chooseSimple()`.

Called by in js/script.js: `Start-up`; in js/controller/optionshandler.js: `chooseSimple()`
Calls `setLoading()`, `pictureFiles()`, `appliesToRoot()`, `preload()`; in js/view/toolkit.js: `simple()`

### 7.1.4 `screenPictures(width = screen.width, height = screen.height, ratio = devicePixelRatio)` EXPORTED lines 45–48

Works out the pictures for the screen: the parchment from `backgroundFor()` (`fhd`, `qhd`, `uhd` or `ultrawide`) and the size of every other picture, `4k` with `uhd` or `ultrawide` and `sd` otherwise.

Called by `startScreen()`; in js/tests/pictures/picturetools_test.js: `misjudged()`
Calls `backgroundFor()`

### 7.1.5 `backgroundFor(width, height, ratio)` INTERNAL lines 50–56

Picks the parchment for a screen from its size in device pixels: `fhd` on a phone, whose shorter side is under `PHONE_SIDE`, 720 pixels; `ultrawide` if it is more than twice as wide as high; then `uhd` beyond `QHD`, `qhd` beyond `FHD`, and `fhd` otherwise.

Called by `screenPictures()`
Calls `beyond()`

### 7.1.6 `beyond(wide, high, limit)` INTERNAL line 58

Tells whether a screen is wider or higher in device pixels than a size, such as `QHD`, 2560 by 1440.

Called by `backgroundFor()`
Calls *no function of the code*

### 7.1.7 `pictureFiles(use = () => true)` INTERNAL lines 60–61

Lists the pictures and fonts in the style sheets that the screen uses, so the loading screen can wait for them all.

Called by `applySimple()`, `hideLoadingWhenReady()`
Calls `filesOf()`, `fits()`

### 7.1.8 `filesOf(sheet, use)` INTERNAL lines 63–64

Lists the addresses in a style sheet's rules whose selectors pass a test.

Called by `pictureFiles()`
Calls `rulesIn()`, `linksIn()`

### 7.1.9 `rulesIn(rules)` INTERNAL line 66

Lists every rule of a style sheet, with the rules inside media queries.

Called by `filesOf()`
Calls *no function of the code*

### 7.1.10 `linksIn(text, base)` INTERNAL line 68

Lists the picture and font addresses in a rule's text, made whole from the style sheet's own address.

Called by `filesOf()`
Calls *no function of the code*

### 7.1.11 `fits(src)` INTERNAL lines 70–74

Tells whether a picture is the one for the screen: the parchment of `data-background`, and the size of `data-assets`. Pictures without a size always fit.

Called by `pictureFiles()`
Calls *no function of the code*

### 7.1.12 `appliesToRoot(selector)` INTERNAL lines 76–77

Tells whether a selector already holds for `<html>` or `<body>`, as for the parchment and the cursors, whose pictures show at once.

Called by `applySimple()`
Calls *no function of the code*

### 7.1.13 `preload(src)` INTERNAL lines 79–82

Loads a picture once into a kept image, so that the browser has it before the page shows it.

Called by `applySimple()`, `hideLoadingWhenReady()`
Calls *no function of the code*

### 7.1.14 `hideLoadingWhenReady()` EXPORTED lines 84–97

Hides the loading screen once every picture and font has loaded, at once if they have already. Without Simple UI only; a later call takes over from an earlier one.

Called by in js/script.js: `run()`; in js/controller/optionshandler.js: `chooseSimple()`
Calls `onPage()`, `pictureFiles()`, `preload()`, `ready()`, `setLoading()`, `done()`; in js/view/toolkit.js: `simple()`, `$()`

### 7.1.15 `onPage()` INTERNAL line 99

The addresses of the images already in the page, but for the loading screen's own.

Called by `hideLoadingWhenReady()`
Calls in js/view/toolkit.js: `$()`

### 7.1.16 `ready(pictures, fonts)` INTERNAL lines 101–102

Tells whether every picture and font has loaded already.

Called by `hideLoadingWhenReady()`
Calls *no function of the code*

### 7.1.17 `done(img)` INTERNAL line 104

Waits for a kept image to be decoded, or to fail, so that waiting never stops.

Called by `hideLoadingWhenReady()`
Calls *no function of the code*

### 7.1.18 `setLoading(on)` EXPORTED lines 106–113

Shows or hides the loading screen, with its turning picture, made once, and makes the page behind it unreachable while it shows.

Called by `applySimple()`, `hideLoadingWhenReady()`; in js/script.js: `failLoad()`
Calls `loadSpinner()`; in js/view/toolkit.js: `$()`, `clone()`

### 7.1.19 `loadSpinner(img)` INTERNAL line 115

Gives the loading picture its address, kept in `data-src` so that it loads only when the loading screen shows it.

Called by `setLoading()`
Calls *no function of the code*

# 7.2 tooltips.js

**Folder:** js/view/pictures/ 65 lines 9 functions, 2 exported
**Imports:** js/model/texts.js, js/model/model.js, js/view/toolkit.js, js/view/treeview.js
**Imported by:** js/view/pictures/pictures.js, js/controller/optionshandler.js, js/tests/run_test.js, js/tests/pictures/tooltips_test.js

The tooltip of the picture look: what it says over each control, and where it shows. It asks the tree's view what is under the mouse and the model whether a branch is locked, so it reads no markup of the tree. It keeps the Tooltips setting, as `tips`.

### 7.2.1 `setTips(on)` EXPORTED lines 12–14

Switches the tooltips on or off, as the Tooltips setting says.

Called by in js/controller/optionshandler.js: `startOptions()`, `chooseTips()`; in js/tests/run_test.js: `toStart()`, `restore()`
Calls *no function of the code*

### 7.2.2 `startTips()` EXPORTED lines 16–30

Has the tooltip follow the mouse (`showTip()`), and hides it when the mouse leaves the page. After every click it works the tooltip out again for what is under the mouse (`elementFromPoint`), so that it follows a button that changes, such as Expand all turning into Collapse all. Runs once, from `startPictures()`.

Called by in js/view/pictures/pictures.js: `startPictures()`
Calls `showTip()`; in js/view/toolkit.js: `$()`

### 7.2.3 `showTip(tip, target, { clientX: x, clientY: y, pointerType })` INTERNAL lines 32–39

Writes the tooltip for what the mouse is over (`tipFor()`), or hides it when there is none or none is wanted (`tipsWanted()`), puts it into the Options window while that is open (`hostFor()`), and places it (`placeTip()`).

Called by `startTips()`
Calls `tipsWanted()`, `tipFor()`, `hostFor()`, `placeTip()`

### 7.2.4 `tipsWanted(pointerType)` INTERNAL line 41

Tells whether the tooltip may show: only for a mouse, without Simple UI and with Tooltips on.

Called by `showTip()`
Calls in js/view/toolkit.js: `simple()`

### 7.2.5 `hostFor(target)` INTERNAL line 43

Where the tooltip goes: into the open window under the mouse, such as Options, so that it shows over the window, or else into the page's body.

Called by `showTip()`
Calls *no function of the code*

### 7.2.6 `placeTip(tip, x, y)` INTERNAL lines 45–50

Places the tooltip with its corner at the cursor picture's corner, below the mouse where it fits and above it otherwise, and inside the window by `TIP_MARGIN`.

Called by `showTip()`
Calls `clamp()`

### 7.2.7 `clamp(low, value, high)` INTERNAL line 52

Keeps a number between a lowest and a highest value.

Called by `placeTip()`
Calls *no function of the code*

### 7.2.8 `tipFor(target)` INTERNAL lines 54–59

Returns the tooltip's text for what the mouse is over: the marking hint over the tree's items (`itemAt()`), the text that an element's `data-tip` names over every control, the hint of a branch's arrow (`branchTip()`), or the address of a civilisation's or the footer's link.

Called by `showTip()`
Calls `branchTip()`; in js/view/treeview.js: `itemAt()`

### 7.2.9 `branchTip(target)` INTERNAL lines 61–65

The tooltip of a branch's arrow, or of the name of a group inside a building, which has no box and opens on a click (`branchAt()`): to collapse the branch while it is open, to expand it while it is closed, and none on a locked branch, whose arrow is disabled.

Called by `tipFor()`
Calls in js/model/model.js: `locked()`; in js/view/treeview.js: `branchAt()`, `isOpen()`

# 7.3 scrollbars.js

**Folder:** js/view/pictures/ 136 lines 11 functions, 2 exported
**Imports:** js/model/data/datahandler.js, js/view/toolkit.js
**Imported by:** js/view/pictures/pictures.js, js/tests/pictures/scrollbars_test.js

The game's scroll bars, which replace the browser's without Simple UI. A bar is placed after each list that scrolls, so it does not scroll with it, and CSS draws it from the numbers `fit()` gives it.

### 7.3.1 `startBars()` EXPORTED lines 16–27

Puts a bar from its template after every list that scrolls, connects it, and fits the bars again whenever anything about them changes. Runs once, from `startPictures()`.

Called by in js/view/pictures/pictures.js: `startPictures()`
Calls `wire()`, `watch()`, `fitAll()`; in js/view/toolkit.js: `clone()`

### 7.3.2 `wire(bar, box)` INTERNAL lines 29–54

Connects a bar, whose parts it finds by their class: the arrows scroll a line, a press on the body above or below the knob scrolls a page, less a line, towards it, and not at all once the knob reaches the pointer, the knob drags and the wheel over the bar scrolls the list; the list's scrolling fits the bar again. A line is `LINE_PX`, 40 pixels, but in the language list without Simple UI it is one language, and the wheel there, over the list or its bar, scrolls a language at a time, so that the list always shows whole languages.

Called by `startBars()`
Calls `wheelUnit()`, `hold()`, `drag()`, `fit()`; in js/view/toolkit.js: `simple()`

### 7.3.3 `wheelUnit(e, box)` INTERNAL lines 56–60

How far one step of the wheel scrolls, by the step's kind: a pixel, a line of `LINE_PX`, or the list's height.

Called by `wire()`
Calls *no function of the code*

### 7.3.4 `hold(part, step)` INTERNAL lines 62–83

Makes an arrow or the bar's body act on a press, and again every 50 milliseconds after 400 while held (`REPEAT_EVERY_MS`, `REPEAT_AFTER_MS`), keeping hold of the pointer so the press stays while it leaves.

Called by `wire()`
Calls *no function of the code*

### 7.3.5 `drag(knob, track, box)` INTERNAL lines 85–102

Lets the knob be dragged: the list scrolls as far as the knob moves, scaled from the body's room to the list's.

Called by `wire()`
Calls *no function of the code*

### 7.3.6 `fit(bar, box)` INTERNAL lines 104–112

Gives a list's bar its place and height (`placeBar()`) and the knob's position, as numbers for CSS, and hides the bar with Simple UI or when the list holds no more than it shows, and the knob when the bar's body is too short for it (`trackTooShort()`).

Called by `wire()`, `fitAll()`
Calls `placeBar()`, `trackTooShort()`; in js/view/toolkit.js: `simple()`

### 7.3.7 `placeBar(bar, box, frame)` INTERNAL lines 114–121

Places a bar along its list's right edge, inside the list's borders and as high as the list's inside.

Called by `fit()`
Calls *no function of the code*

### 7.3.8 `trackTooShort(track, bar)` INTERNAL line 123

Tells whether a bar's body is shorter than the bar is wide, too short for the knob.

Called by `fit()`
Calls *no function of the code*

### 7.3.9 `fitAll()` INTERNAL line 125

Fits every bar again. Runs when a list or what is in it changes size, when anything is added or taken out, when Simple UI changes and when the window does.

Called by `the top-level code`, `startBars()`
Calls `fit()`

### 7.3.10 `watch(box)` INTERNAL line 129

Has the size observer watch a list and everything directly in it, so a bar is fitted again when they change size.

Called by `the top-level code`, `startBars()`
Calls *no function of the code*

### 7.3.11 `barsByBox()` EXPORTED line 136

A copy of the bars, by their lists, for the tests; the bars themselves stay inside the module.

Called by in js/tests/pictures/scrollbars_test.js: `scrollBarTests()`, `ownBars()`, `barStates()`, `comboArrows()`
Calls *no function of the code*

# 8 js/controller/

3 files 212 lines

What the visitor's clicks, touches and keys mean: treehandler.js for the tree and Filters, optionshandler.js for the Options window, and updatehandler.js for the updater. They change the model or ask a view to change, and never touch the page.

**Subchapters:** 8.1 treehandler.js, 8.2 optionshandler.js, 8.3 updatehandler.js

## 8.1 treehandler.js

**Folder:** js/controller/ 116 lines 18 functions, 1 object, 1 exported
**Imports:** js/model/texts.js, js/model/model.js, js/view/civsview.js, js/view/treeview.js
**Imported by:** js/script.js

What the visitor's clicks, touches and keys in the tree mean: marking, opening and closing, the arrow keys, Find, the buttons above the tree and the buttons of Filters. These are the actions treeview.js and civsview.js are started with (Strategy). They change the model, or ask the tree's view to open a branch or move the focus, and never touch the page. The view hands them each change of the model once it has drawn it, so they never subscribe to the model.

### 8.1.1 `startTree()` EXPORTED lines 16–19

Starts the tree's view with these actions, and with the model's `clear()` for Clear, and the civilisations' view with `unchip()` for the Filters buttons. Runs once, from `start()`.

Called by in js/script.js: `start()`
Calls `gesture()`, `branch()`, `key()`, `find()`, `enter()`, `expandAll()`, `collapseAll()`, `redrawn()`, `unchip()`; in js/model/model.js: `clear()`; in js/view/civsview.js: `startCivsView()`; in js/view/treeview.js: `startTreeView()`

### 8.1.2 `gesture(item, how)` INTERNAL line 21

Turns what the tree's view reports into a mark: a click marks must have (Req. 3), and a long press, a right-click and the menu key must lack, as `keyAction()` decides for Space and X.

Called by `startTree()`
Calls `toggle()`

### 8.1.3 `toggle(item, mark)` INTERNAL lines 23–30

Handles a mark from a gesture or a key: opens the way to every row the mark will reach (`toReveal()`, Req. 28, 31), then calls `toggleMark()`. The model refuses a greyed-out item, which then reaches nothing.

Called by `gesture()`, `keyAction()`
Calls in js/model/model.js: `toReveal()`, `toggleMark()`; in js/view/treeview.js: `setOpen()`

### 8.1.4 `branch(item)` INTERNAL line 32

Opens or closes a branch, from its arrow or a heading's name.

Called by `startTree()`
Calls `expand()`; in js/view/treeview.js: `isOpen()`, `setOpen()`

### 8.1.5 `expand(item)` INTERNAL lines 34–36

Opens an item and each only child below it, so opening Militia shows its whole line in one step, unless the item is greyed out and closed. Used by the arrows and the Right key.

Called by `branch()`, `enterOrExpand()`
Calls in js/model/model.js: `locked()`; in js/view/treeview.js: `setOpen()`

### 8.1.6 `key(item, keyPress)` INTERNAL lines 38–43

Acts on a key in the tree as `keyAction()` finds it, moving the focus from arrow to arrow when an arrow had focus and from row to row otherwise, and has the view stop the browser's own action for a key it used (WCAG 2.1.1).

Called by `startTree()`
Calls `keyAction()`; in js/view/treeview.js: `focusTwisty()`, `focusRow()`

### 8.1.7 `keyAction({ key: name, shift, modified, onTwisty })` INTERNAL lines 45–52

Finds what a key in the tree means, without acting on it: Space marks must have and X must lack, as `MARK_KEYS` says, but not on an arrow (Req. 3); Up, Down, Home and End move as `MOVES` says, and Right and Left open, close and move as `BRANCH_KEYS` says. A key with Alt, Ctrl or Meta, a move with Shift and any other key mean nothing.

Called by `key()`
Calls `toggle()`, `MOVES()`, `moveFocus()`

### 8.1.8 `moveFocus(item, move, focusOn)` INTERNAL lines 54–57

Moves the focus where a key of `MOVES` goes, among the rows in sight (`shown()`).

Called by `keyAction()`
Calls in js/view/treeview.js: `shown()`

### 8.1.9 `MOVES` OBJECT, INTERNAL lines 59–64

Where each key that moves the focus goes from an item, among the rows in sight: Down and Up to the next and the previous row, and Home and End to the first and the last.

Called by `keyAction()`
Calls *no function of the code*

### 8.1.10 `enterOrExpand(item, focusOn)` INTERNAL lines 68–71

The Right key: goes into an open branch, to its first row, or opens a closed one and stays.

Called by `the top-level code`
Calls `expand()`; in js/view/treeview.js: `isOpen()`

### 8.1.11 `leaveOrCollapse(item, focusOn)` INTERNAL lines 73–76

The Left key: closes an open branch and stays, or goes out of a closed one, to the item above.

Called by `the top-level code`
Calls in js/view/treeview.js: `isOpen()`, `setOpen()`

### 8.1.12 `find(text)` INTERNAL lines 78–81

Runs Find: highlights the matches (`highlightMatches()`) and opens the tree to them (`showMatches()`) (Req. 20, 21).

Called by `startTree()`, `redrawn()`
Calls `highlightMatches()`, `showMatches()`

### 8.1.13 `highlightMatches(text)` INTERNAL lines 83–87

Notes the items whose names hold Find's text, but those inside closed greyed-out items (`behind()`), and has the view highlight them and say how many there are (Req. 20, 21).

Called by `find()`, `redrawn()`
Calls in js/model/texts.js: `plain()`, `nameOf()`; in js/model/model.js: `behind()`; in js/view/treeview.js: `highlight()`

### 8.1.14 `showMatches()` INTERNAL lines 89–96

Opens the branches down to the matches, their `ancestors()`, and closes every other, then scrolls the first into view (`reveal()`).

Called by `find()`
Calls in js/view/treeview.js: `setOpen()`, `reveal()`

### 8.1.15 `enter()` INTERNAL lines 98–102

Opens the branches down to the first item Find found and moves focus to it, on Enter in its box.

Called by `startTree()`
Calls in js/view/treeview.js: `setOpen()`, `focusRow()`

### 8.1.16 `expandAll()` INTERNAL line 104

Opens every branch but those of closed greyed-out items.

Called by `startTree()`
Calls in js/model/model.js: `locked()`; in js/view/treeview.js: `setOpen()`

### 8.1.17 `collapseAll()` INTERNAL line 106

Closes every branch.

Called by `startTree()`
Calls in js/view/treeview.js: `setOpen()`

### 8.1.18 `unchip({ item, mark, lone })` INTERNAL lines 108–111

Clears what a Filters button stands for: a lone filter entry, a unique item's own, with `unmarkLone()`, and any other with `unmarkLine()`. The civilisations' view hands it the button's filter, `{ item, mark, lone }`.

Called by `startTree()`
Calls in js/model/model.js: `unmarkLone()`, `unmarkLine()`

### 8.1.19 `redrawn(change)` INTERNAL

lines 113–116

Keeps Find up to date once the tree's view has drawn a change: after marks it only highlights again (`highlightMatches()`), so that no branch opens or closes, and after a new tree, a new language or the start it runs the whole of Find again, since the names or the rows changed. Nothing changes for `"look"`.

Called by `startTree()`
Calls `highlightMatches()`, `find()`; in js/view/treeview.js: `findText()`

# 8.2 optionshandler.js

**Folder:** js/controller/ 85 lines 10 functions, 1 object, 4 exported

**Imports:** js/model/texts.js, js/model/model.js, js/model/data/datahandler.js, js/view/toolkit.js, js/view/page.js, js/view/civsview.js, js/view/optionsview.js, js/view/pictures/pictures.js, js/view/pictures/tooltips.js

**Imported by:** js/script.js, js/tests/run_test.js, js/tests/behaviour/lines_test.js, js/tests/behaviour/greying_test.js, js/tests/interface/options_test.js, js/tests/interface/keys_test.js, js/tests/interface/uniques_test.js, js/tests/interface/headings_test.js, js/tests/interface/reporting_test.js, js/tests/pictures/pictures_test.js, js/tests/pictures/icons_test.js, js/tests/pictures/scrollbars_test.js

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What the settings of the Options window do: the language, the two Include boxes, Simple UI, Civ view and Tooltips, each applied on OK. Its `showChoices()` and `apply()` are the actions optionsview.js is started with (Strategy). The model saves Include, Include Regionals and the language itself; this file saves only the look settings: Simple UI, Civ view and Tooltips. The language in use is the model's alone: only the one last chosen is kept here, as `wanted`, while its file loads.

### 8.2.1 `choices` OBJECT, INTERNAL lines 11–18

Every setting of the Options window as an entry: how to read its value now and how to choose it. `showChoices()` and `apply()` go through it.

Called by `showChoices()`, `apply()`
Calls `chooseLanguage()`, `chooseInclude()`, `chooseRegional()`, `chooseSimple()`, `chooseView()`, `chooseTips()`; in js/view/toolkit.js: `simple()`

### 8.2.2 `startOptions()` EXPORTED lines 22–28

Starts the Options window: starts the window's view with `showChoices()` and `apply()`, builds the language menu, restores Civ view and Tooltips as they were stored (`recallSetting()`), and shows every setting in the window. Runs once, from `start()`.

Called by in js/script.js: `start()`
Calls `showChoices()`, `apply()`; in js/view/civsview.js: `setView()`; in js/view/optionsview.js: `startOptionsView()`, `buildLanguages()`; in js/view/pictures/tooltips.js: `setTips()`

### 8.2.3 `showChoices()` INTERNAL lines 30–34

Shows the page's values in every control of the window, dropping what was changed: at the start, each time the window opens, and after OK.

Called by `startOptions()`, `apply()`
Calls `choices()`; in js/view/optionsview.js: `showChoice()`

### 8.2.4 `apply()` INTERNAL lines 36–42

Applies every setting whose window value differs from the page's, when OK is pressed. It reads every control before it chooses any, then shows the page's values again (`showChoices()`).

Called by `startOptions()`
Calls `choices()`, `showChoices()`; in js/view/optionsview.js: `readChoice()`

### 8.2.5 `chooseLanguage(code)` INTERNAL lines 44–55

Chooses a language: has the model use it with `setLanguage()`, which saves it and puts it in the address, after loading its file if needed, as `wanted`, so that only the last one chosen is used. If the file cannot be loaded, it calls `languageFailed()`.

Called by `choices()`
Calls `languageFailed()`; in js/model/model.js: `setLanguage()`; in js/model/data/datahandler.js: `hasLanguage()`, `loadLanguage()`

### 8.2.6 `languageFailed()` INTERNAL lines 57–60

Puts the window back on the language in use and opens the popup that the language could not be loaded (`showLanguageError()`). `apply()` shows the language in use after OK too, but the window may be open again by the time the language fails.

Called by `chooseLanguage()`
Calls in js/view/page.js: `showLanguageError()`; in js/view/optionsview.js: `showChoice()`

### 8.2.7 `chooseInclude(on)` EXPORTED lines 62–64

Switches Include Unique and Universal (Req. 13 to 15): has the model build the tree again, keeping marks, which saves the choice (`setInclude()`).

Called by `choices()`; in js/tests/run_test.js: `toStart()`, `restore()`; in js/tests/behaviour/lines_test.js: `beyond()`; in js/tests/behaviour/greying_test.js: `standIns()`; in js/tests/interface/options_test.js: `optionKeys()`; in js/tests/interface/keys_test.js: `optionAudits()`; in js/tests/interface/uniques_test.js: `uniqueFilterTests()`; in js/tests/interface/headings_test.js: `headingTests()`; in js/tests/pictures/icons_test.js: `iconTests()`
Calls in js/model/model.js: `setInclude()`

### 8.2.8 `chooseRegional(on)` EXPORTED lines 66–68

Switches Include Regionals: has the model build the tree again, keeping marks, which saves the choice (`setRegional()`).

Called by `choices()`; in js/tests/run_test.js: `toStart()`, `restore()`
Calls in js/model/model.js: `setRegional()`

### 8.2.9 `chooseSimple(on)` EXPORTED lines 70–75

Switches Simple UI (Req. 32, 33): saves the choice, has `applySimple()` switch the look, `redraw()` draw the page's views again with `"look"`, and `hideLoadingWhenReady()` end the loading screen. Simple UI is only how the page looks, so it does not go through the model.

Called by `choices()`; in js/tests/run_test.js: `toStart()`, `restore()`; in js/tests/interface/reporting_test.js: `reportTests()`; in js/tests/pictures/pictures_test.js: `pictureTests()`; in js/tests/pictures/scrollbars_test.js: `scrollBarTests()`
Calls in js/view/page.js: `redraw()`; in js/view/pictures/pictures.js: `applySimple()`, `hideLoadingWhenReady()`

### 8.2.10 `chooseView(choice)` INTERNAL lines 77–80

Chooses Civ view, list or tiles: saves the choice and shows it (`setView()`).

Called by `choices()`
Calls in js/view/civsview.js: `setView()`

### 8.2.11 `chooseTips(on)` INTERNAL lines 82–85

Switches Tooltips on or off: saves the choice and passes it to the tooltip (`setTips()`).

Called by `choices()`
Calls in js/view/pictures/tooltips.js: `setTips()`

# 8.3 updatehandler.js

**Folder:** js/controller/ 11 lines 1 function, 1 exported

**Imports:** js/model/updater.js, js/view/report.js
**Imported by:** js/script.js

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Runs the updater, which `?update` opens.

### 8.3.1 `runUpdate()` EXPORTED lines 4–11

Runs the updater: shows that it is loading, has updater.js fetch the new data and work out what changed, and has report.js show it, or why it failed. It changes nothing by itself.

Called by in js/script.js: `RUNS()`
Calls in js/model/updater.js: `fetchUpdate()`; in js/view/report.js: `showUpdating()`, `showUpdate()`, `showUpdateFailed()`

# 9 js/tests/

3 files 573 lines

The tests, loaded only when the address ends in `?test`. run_test.js runs them, helpers_test.js holds the tools they share and audit_test.js the accessibility audit; the tests themselves are in the four folders below, one for each kind, and TESTS.md describes them one by one, each with the one thing it tests (Rule 17).

**Subchapters:** 9.1 run_test.js, 9.2 helpers_test.js, 9.3 audit_test.js

## 9.1 run_test.js

**Folder:** js/tests/ 144 lines 16 functions, 2 exported

**Imports:** js/model/texts.js, js/model/model.js, js/model/data/datahandler.js, js/view/toolkit.js, js/view/civsview.js, js/view/report.js, js/view/pictures/tooltips.js, js/controller/optionshandler.js, js/tests/helpers_test.js, js/tests/behaviour/marking_test.js, js/tests/interface/menu_test.js, js/tests/interface/languages_test.js, js/tests/interface/options_test.js, js/tests/interface/keys_test.js, js/tests/interface/uniques_test.js, js/tests/interface/headings_test.js, js/tests/interface/reporting_test.js, js/tests/pictures/pictures_test.js, js/tests/static/scanner_test.js, js/tests/static/rules_test.js, js/tests/static/data_test.js, js/tests/static/updates_test.js

**Imported by:** js/script.js (only with `?test`)

Runs the tests in groups and shows the report. Loaded only when the address ends in `?test`.

### 9.1.1 `merge(where)` INTERNAL lines 28–39

Combines the results into one line per test, failed if any of its checks failed, with the first details and the window's size. A test may check in several states and languages.

Called by `runAndPublish()`
Calls *no function of the code*

### 9.1.2 `reportLine(result, tools)` INTERNAL line 41

Writes a test's report line, such as "Test 15 failed: ...", from its number, name and details, in the form `outcome()` picks.

Called by `publish()`
Calls `outcome()`; in js/model/texts.js: `fill()`

### 9.1.3 `outcome(result, tools)` INTERNAL lines 43–46

Picks the form of a report line: passed or failed for a numbered test, and a plain "Failed" for the line "the tests ran to the end", which has no number.

Called by `reportLine()`
Calls *no function of the code*

### 9.1.4 `announce(lines)` INTERNAL line 48

Writes lines of the report to the console, one by one and each starting with `tests:`, where the workflow on GitHub reads them from Chrome's log.

Called by `loadFailed()`, `publish()`, `stopped()`
Calls *no function of the code*

### 9.1.5 `loadFailed()` EXPORTED lines 50–53

Marks the tests as failed when the page could not load, and says so in the console, so CI never waits for a result that cannot come.

Called by in js/script.js: `Start-up`
Calls `announce()`

### 9.1.6 `snapshot()` INTERNAL lines 55–58

Notes the visitor's language, Include, Include Regionals, Simple UI, Civ view, Tooltips and address, and the stored language and settings, for `restore()`.

Called by `runAndPublish()`
Calls in js/view/toolkit.js: `simple()`

### 9.1.7 `toStart()` INTERNAL lines 60–68

Sets the state every group of tests starts from: the page as `reset()` leaves it, Include unticked, Include Regionals ticked, Simple UI and Tooltips on, the civilisations as a list and the first language. So no group depends on what the one before it left.

Called by `runGroup()`
Calls in js/view/toolkit.js: `simple()`; in js/view/civsview.js: `setView()`; in js/view/pictures/tooltips.js: `setTips()`; in js/controller/optionshandler.js: `chooseInclude()`, `chooseRegional()`, `chooseSimple()`; in js/tests/helpers_test.js: `reset()`, `codes()`, `pick()`

### 9.1.8 `restore(before)` INTERNAL lines 70–80

Puts back the visitor's language, options, view, address and saved choices after the tests.

Called by `runAndPublish()`
Calls in js/model/model.js: `setLanguage()`; in js/view/civsview.js: `setView()`; in js/view/pictures/tooltips.js: `setTips()`; in js/controller/optionshandler.js: `chooseInclude()`, `chooseRegional()`, `chooseSimple()`; in js/tests/helpers_test.js: `reset()`

### 9.1.9 `runAll()` INTERNAL lines 82–94

Reads index.html, the style sheets and the scripts as text, then runs the 15 groups of tests one after another through `runGroup()`, in the order TESTS.md lists them; the language tests run once for each language.

Called by `runAndPublish()`
Calls `runGroup()`; in js/model/data/datahandler.js: `load()`; in js/tests/helpers_test.js: `codes()`; in js/tests/behaviour/marking_test.js: `behaviourTests()`; in js/tests/interface/menu_test.js: `menuTests()`; in js/tests/interface/languages_test.js: `languageTests()`; in js/tests/interface/options_test.js: `optionTests()`, `languagePopupTests()`; in js/tests/interface/keys_test.js: `keyboardTests()`, `accessibilityTests()`; in js/tests/interface/uniques_test.js: `uniqueFilterTests()`; in js/tests/interface/headings_test.js: `headingTests()`; in js/tests/interface/reporting_test.js: `reportTests()`; in js/tests/pictures/pictures_test.js: `pictureTests()`; in js/tests/static/scanner_test.js: `styles()`, `scripts()`; in js/tests/static/rules_test.js: `sourceTests()`; in js/tests/static/data_test.js: `textTests()`, `dataTests()`; in js/tests/static/updates_test.js: `updaterTests()`

### 9.1.10 `runGroup(group)` INTERNAL lines 96–103

Runs one group of tests from the state `toStart()` sets. If the group throws, as `fixtures()` does when data.json lacks a sample, it fails "the tests ran to the end" with the group's name and the error, and the next group runs.

Called by `runAll()`
Calls `toStart()`; in js/tests/helpers_test.js: `check()`

### 9.1.11 `runTests()` EXPORTED lines 105–111

Runs the tests and shows their report through `runAndPublish()`. If the runner itself throws, `stopped()` reports it, so that the run still ends in a result.

Called by in js/script.js: `Start-up`
Calls `runAndPublish()`, `stopped()`

### 9.1.12 `runAndPublish()` INTERNAL lines 113–118

Keeps the visitor's settings (`snapshot()`), runs every group, puts the settings back, and publishes one numbered line per test.

Called by `runTests()`
Calls `snapshot()`, `runAll()`, `restore()`, `publish()`, `numbered()`, `merge()`, `windowSize()`

### 9.1.13 `windowSize()` INTERNAL line 120

Names the window's size, such as "1920 by 1080", for the details of a failure and the report's summary.

Called by `runAndPublish()`, `publish()`
Calls *no function of the code*

### 9.1.14 `numbered(results)` INTERNAL lines 122–125

Numbers the tests in the order they first ran, the numbers TESTS.md gives them; the line "the tests ran to the end" gets none.

Called by `runAndPublish()`
Calls *no function of the code*

### 9.1.15 `publish(results)` INTERNAL lines 127–139

Marks the page passed or failed, and shows the report, failures first, under a summary with the count, the colours and the window's size. For CI it then has `announce()` write the failed tests, the count and whether they passed. Its Close leaves the tests: it opens the page's own address, without `?test`.

Called by `runAndPublish()`
Calls `windowSize()`, `reportLine()`, `announce()`; in js/model/texts.js: `fill()`; in js/view/toolkit.js: `$()`; in js/view/report.js: `openReport()`, `report()`

### 9.1.16 `stopped(error)` INTERNAL

lines 141–144

Marks the tests as failed when the runner itself stops with an error, and says so in the console, so CI never waits for a result that cannot come.

Called by `runTests()`
Calls `announce()`

# 9.2 helpers_test.js

**Folder:** js/tests/ 229 lines 48 functions, 40 exported

**Imports:** js/model/civsets.js, js/model/texts.js, js/model/itemtree.js, js/model/model.js, js/model/data/datahandler.js, js/view/toolkit.js, js/view/civsview.js, js/view/treeview.js, js/view/optionsview.js

**Imported by:** js/tests/run_test.js, js/tests/audit_test.js, js/tests/behaviour/marking_test.js, js/tests/behaviour/samples_test.js, js/tests/behaviour/branches_test.js, js/tests/behaviour/tree_test.js, js/tests/behaviour/lines_test.js, js/tests/behaviour/greying_test.js, js/tests/behaviour/finding_test.js, js/tests/interface/menu_test.js, js/tests/interface/languages_test.js, js/tests/interface/options_test.js, js/tests/interface/keys_test.js, js/tests/interface/uniques_test.js, js/tests/interface/headings_test.js, js/tests/interface/reporting_test.js, js/tests/pictures/pictures_test.js, js/tests/pictures/picturetools_test.js, js/tests/pictures/icons_test.js, js/tests/pictures/findbox_test.js, js/tests/pictures/tooltips_test.js, js/tests/pictures/treebuttons_test.js, js/tests/pictures/scrollbars_test.js, js/tests/static/rules_test.js, js/tests/static/layers_test.js, js/tests/static/data_test.js, js/tests/static/updates_test.js

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Small tools the test files share: recording results, the names of the tests that run in more than one file (`NAMES`), finding and measuring elements, reading names as screen readers do, acting like a visitor, and picking sample items. Helpers that only one folder's tests use sit in that folder.

### 9.2.1 `check(name, ok, detail = "")` EXPORTED

line 15

Records one test result. Every test reports this way, and each name is one line in the report.

Called by `expect()`; in js/tests/run_test.js: `runGroup()`; in js/tests/audit_test.js: `audit()`; in js/tests/behaviour/marking_test.js: `marking()`, `menuKey()`, `gestures()`; in js/tests/behaviour/branches_test.js: `buildingMarks()`, `lineFilters()`, `singleLines()`, `groupMarks()`; in js/tests/behaviour/tree_test.js: `crowding()`, `tree()`; in js/tests/behaviour/lines_test.js: `lineTops()`, `parents()`, `lines()`, `evenLine()`, `unevenLine()`, `beyond()`; in js/tests/behaviour/greying_test.js: `rivals()`, `excludedBy()`, `standIns()`, `sharedRow()`, `shuts()`, `borrowedRows()`, `variants()`; in js/tests/behaviour/finding_test.js: `finding()`, `dropsGreyed()`; in js/tests/interface/menu_test.js: `menuTests()`, `menuKeys()`, `menuOpening()`, `menuClosing()`, `menuChoosing()`; in js/tests/interface/languages_test.js: `languageTests()`, `footerTexts()`, `footerOnce()`, `civList()`; in js/tests/interface/options_test.js: `optionTests()`, `optionKeys()`, `openingOptions()`, `confirmingOptions()`, `cancellingOptions()`, `closedAndFocused()`, `languagePopupTests()`; in js/tests/interface/keys_test.js: `filterFocus()`, `treeMoves()`, `stepKeys()`, `branchKeys()`, `findKey()`; in js/tests/interface/uniques_test.js: `uniqueFilterTests()`, `inLine()`, `inGroup()`; in js/tests/interface/headings_test.js: `headingTests()`; in js/tests/interface/reporting_test.js: `reportTests()`; in js/tests/pictures/pictures_test.js: `screenTests()`, `optionsButtonTests()`, `simpleUiTests()`, `civViewTests()`; in js/tests/pictures/findbox_test.js: `findBoxTests()`, `crossTests()`; in js/tests/pictures/tooltips_test.js: `treeTipTests()`, `includeTipTests()`, `silentTipTests()`; in js/tests/pictures/treebuttons_test.js: `treeButtonTests()`, `showing()`, `sideBySide()`, `everyLanguage()`; in js/tests/pictures/scrollbars_test.js: `scrollBarTests()`, `scrollTests()`, `languageWheel()`; in js/tests/static/updates_test.js: `updaterTests()`

Calls *no function of the code*

### 9.2.2 `expect(name, offenders)` EXPORTED

lines 17–18

Records a test that passes when its list of offenders is empty, naming the first few, so a failure points at the exact items.

Called by in js/tests/behaviour/tree_test.js: `crowding()`, `countKinds()`, `tree()`; in js/tests/interface/languages_test.js: `footerOnce()`; in js/tests/interface/options_test.js: `optionTests()`, `optionKeys()`, `confirmingOptions()`; in js/tests/interface/keys_test.js: `textSpacing()`; in js/tests/pictures/pictures_test.js: `screenTests()`, `simpleUiTests()`; in js/tests/pictures/icons_test.js: `iconTests()`, `boxTests()`, `treeIconTests()`, `civIconTests()`; in js/tests/pictures/tooltips_test.js: `everyTipTests()`; in js/tests/pictures/treebuttons_test.js: `looks()`; in js/tests/pictures/scrollbars_test.js: `scrollBarTests()`, `barStates()`; in js/tests/static/rules_test.js: `codeTests()`, `loadingTests()`, `sizeTests()`, `assetTests()`; in js/tests/static/layers_test.js: `pageTests()`, `importTests()`, `modelTests()`, `viewTests()`; in js/tests/static/data_test.js: `textTests()`, `treeTests()`, `itemTests()`, `textDataTests()`, `sourceDataTests()`; in js/tests/static/updates_test.js: `updaterTests()`

Calls `check()`

### 9.2.3 `phone()` EXPORTED

line 29

Tells whether the window uses the phone layout: narrower than 720 px and at least 640 px tall.

Called by `footerAsWide()`, `footerHidden()`

Calls *no function of the code*

### 9.2.4 `width(selector)` INTERNAL

line 31

Measures the width of the first element a selector finds.

Called by `footerAsWide()`

Calls *no function of the code*

### 9.2.5 `footerAsWide()` EXPORTED

line 33

Tells whether the footer is as wide as the panes, or the window uses the phone layout, which hides it.

Called by in js/tests/interface/languages_test.js: `footerOnce()`; in js/tests/pictures/pictures_test.js: `screenTests()`

Calls `phone()`, `width()`

### 9.2.6 `footerHidden()` EXPORTED line 35

Tells whether the footer is hidden in the phone layout; in any other layout it holds by itself.

Called by in js/tests/interface/languages_test.js: `footerOnce()`; in js/tests/pictures/pictures_test.js: `screenTests()`
Calls `phone()`

### 9.2.7 `need(ok, problem)` EXPORTED line 37

Returns a problem in a list if a condition fails, and an empty list if it holds, so a test can list exactly what went wrong.

Called by in js/tests/pictures/pictures_test.js: `screenTests()`; in js/tests/pictures/icons_test.js: `misfit()`; in js/tests/pictures/scrollbars_test.js: `unloadedSheets()`, `offEdge()`, `outOfRoom()`, `unequal()`, `misshapen()`, `strayKnob()`; in js/tests/static/rules_test.js: `codeTests()`; in js/tests/static/layers_test.js: `modelTests()`; in js/tests/static/data_test.js: `itemTests()`, `sourceDataTests()`
Calls *no function of the code*

### 9.2.8 `same(a, b)` EXPORTED line 39

Tells whether two lists are equal, item by item. Used to compare lists of civilisations.

Called by in js/tests/behaviour/marking_test.js: `marking()`; in js/tests/behaviour/samples_test.js: `listedAre()`; in js/tests/behaviour/branches_test.js: `buildingMarks()`, `lineFilters()`, `singleLines()`; in js/tests/behaviour/greying_test.js: `rivals()`, `excludedBy()`, `sharedRow()`, `borrowedRows()`, `variants()`; in js/tests/interface/menu_test.js: `menuTests()`; in js/tests/interface/languages_test.js: `civList()`; in js/tests/interface/options_test.js: `optionTests()`, `optionKeys()`, `openingOptions()`, `withoutRegionals()`, `languagePopupTests()`; in js/tests/interface/keys_test.js: `filterFocus()`; in js/tests/interface/uniques_test.js: `uniqueFilterTests()`, `inLine()`, `inGroup()`; in js/tests/pictures/pictures_test.js: `simpleUiTests()`; in js/tests/pictures/treebuttons_test.js: `showing()`; in js/tests/static/updates_test.js: `updaterTests()`
Calls *no function of the code*

### 9.2.9 `codes()` EXPORTED line 41

Lists the language codes from data.json.

Called by in js/tests/run_test.js: `toStart()`, `runAll()`; in js/tests/interface/menu_test.js: `menuKeys()`, `menuOpening()`, `menuChoosing()`, `englishFirst()`; in js/tests/interface/languages_test.js: `languageTests()`; in js/tests/interface/options_test.js: `languagePopupTests()`; in js/tests/interface/keys_test.js: `accessibilityTests()`, `languageAudits()`; in js/tests/pictures/treebuttons_test.js: `everyLanguage()`; in js/tests/static/data_test.js: `textTests()`, `textDataTests()`; in js/tests/static/updates_test.js: `updaterTests()`
Calls *no function of the code*

### 9.2.10 `nodeOf(item)` EXPORTED line 43

The row of an item's box and name, from the tree's view, which keeps the elements away from the items.

Called by `mustHave()`, `mustLack()`; in js/tests/behaviour/marking_test.js: `marking()`, `menuKey()`, `gestures()`; in js/tests/behaviour/branches_test.js: `buildingMarks()`, `lineFilters()`, `singleLines()`, `markedAlone()`, `groupMarks()`; in js/tests/behaviour/tree_test.js: `crowding()`, `counts()`; in js/tests/behaviour/lines_test.js: `lineTops()`, `parents()`, `lines()`, `evenLine()`, `unevenLine()`, `beyond()`; in js/tests/behaviour/greying_test.js: `rivals()`, `excludedBy()`, `standIns()`, `sharedRow()`, `shuts()`, `variants()`, `greyedItems()`, `greyed()`; in js/tests/behaviour/finding_test.js: `finding()`, `dropsGreyed()`, `found()`; in js/tests/interface/menu_test.js: `menuTests()`; in js/tests/interface/options_test.js: `optionTests()`, `optionKeys()`; in js/tests/interface/keys_test.js: `treeMoves()`, `stepKeys()`, `branchKeys()`, `findKey()`, `menuAudit()`, `textSpacing()`; in js/tests/interface/uniques_test.js: `uniqueFilterTests()`, `inLine()`, `inGroup()`; in js/tests/interface/headings_test.js: `headingTests()`; in js/tests/pictures/pictures_test.js: `simpleUiTests()`; in js/tests/pictures/icons_test.js: `iconTests()`, `treeIconTests()`; in js/tests/pictures/tooltips_test.js: `treeTipTests()`, `silentTipTests()`, `everyTipTests()`; in js/tests/pictures/treebuttons_test.js: `treeButtonTests()`; in js/tests/pictures/scrollbars_test.js: `scrollBarTests()`
Calls in js/view/treeview.js: `rowOf()`

### 9.2.11 `twistyOf(item)` EXPORTED line 45

The arrow of an item's row, if it has one.

Called by `chainOf()`; in js/tests/audit_test.js: `controlChecks()`; in js/tests/behaviour/tree_test.js: `tree()`; in js/tests/behaviour/lines_test.js: `lines()`; in js/tests/behaviour/greying_test.js: `rivals()`, `shuts()`; in js/tests/interface/keys_test.js: `treeMoves()`; in js/tests/interface/headings_test.js: `headingTests()`; in js/tests/pictures/treebuttons_test.js: `treeButtonTests()`
Calls in js/view/treeview.js: `rowOf()`

### 9.2.12 `labelOf(item)` EXPORTED line 47

The name in an item's row.

Called by in js/tests/interface/languages_test.js: `languageTests()`; in js/tests/interface/headings_test.js: `headingTests()`; in js/tests/pictures/icons_test.js: `treeIconTests()`
Calls in js/view/treeview.js: `rowOf()`

### 9.2.13 `openTo(item)` EXPORTED line 49

Opens every branch above an item, its `ancestors()`, so that its row is in sight.

Called by in js/tests/behaviour/branches_test.js: `buildingMarks()`; in js/tests/behaviour/lines_test.js: `lines()`, `beyond()`; in js/tests/behaviour/greying_test.js: `rivals()`, `standIns()`; in js/tests/interface/uniques_test.js: `uniqueFilterTests()`; in js/tests/interface/headings_test.js: `headingTests()`

Calls in js/view/treeview.js: `setOpen()`

### 9.2.14 `ticked(id)` EXPORTED line 51

Tells whether an Options box is ticked, by its `aria-checked`, as screen readers hear it.

Called by in js/tests/interface/options_test.js: `optionKeys()`, `openingOptions()`, `cancellingOptions()`

Calls in js/view/toolkit.js: `$()`

### 9.2.15 `ui()` EXPORTED line 53

Returns the page's texts in the current language.

Called by in js/tests/pictures/pictures_test.js: `simpleUiTests()`; in js/tests/pictures/tooltips_test.js: `treeTipTests()`, `includeTipTests()`

Calls *no function of the code*

### 9.2.16 `box(el)` EXPORTED line 55

Short for an element's `getBoundingClientRect()`: its place and size on the screen.

Called by in js/tests/behaviour/marking_test.js: `gestures()`; in js/tests/behaviour/tree_test.js: `crowding()`; in js/tests/interface/menu_test.js: `windowSize()`; in js/tests/interface/keys_test.js: `textSpacing()`; in js/tests/interface/reporting_test.js: `reportTests()`, `tops()`; in js/tests/pictures/findbox_test.js: `findBoxTests()`; in js/tests/pictures/tooltips_test.js: `treeTipTests()`, `includeTipTests()`, `tipsOver()`, `inside()`; in js/tests/pictures/treebuttons_test.js: `sideBySide()`, `spots()`, `cut()`; in js/tests/pictures/scrollbars_test.js: `scrollBarTests()`, `partsOf()`, `unequal()`, `misshapen()`, `strayKnob()`, `comboArrows()`

Calls *no function of the code*

### 9.2.17 `keys()` EXPORTED line 57

Lists every item in the full tree.

Called by in js/tests/interface/options_test.js: `cancellingOptions()`, `missing()`, `regionalKeys()`; in js/tests/static/rules_test.js: `assetTests()`; in js/tests/static/data_test.js: `textTests()`, `dataTests()`; in js/tests/static/updates_test.js: `updaterTests()`

Calls in js/model/itemtree.js: `treeKeys()`

### 9.2.18 `describe(el)` EXPORTED lines 59–62

Names an element briefly, such as `button#clear` or a tree row's text, for failure messages.

Called by in js/tests/audit_test.js: `audit()`, `structureChecks()`, `headingChecks()`, `unmatched()`, `controlChecks()`, `colourChecks()`, `layoutChecks()`, `focusChecks()`; in js/tests/interface/options_test.js: `languagePopupTests()`; in js/tests/interface/keys_test.js: `textSpacing()`; in js/tests/pictures/pictures_test.js: `offParchment()`, `strayCursors()`, `bigCursors()`, `picturesOn()`, `imagesShown()`; in js/tests/pictures/icons_test.js: `boxTests()`; in js/tests/pictures/findbox_test.js: `crossTests()`; in js/tests/pictures/tooltips_test.js: `everyTipTests()`; in js/tests/pictures/treebuttons_test.js: `showing()`, `looks()`; in js/tests/pictures/scrollbars_test.js: `scrollBarTests()`, `ownBars()`, `barStates()`, `partsOf()`

Calls *no function of the code*

### 9.2.19 `rendered(el)` EXPORTED lines 64–65

Tells whether an element can be seen: laid out, not hidden, and not text for screen readers only. The accessibility checks look at what is rendered.

Called by `everything()`, `visibleTexts()`; in js/tests/audit_test.js: `headingChecks()`, `controlChecks()`, `focusables()`; in js/tests/interface/options_test.js: `openingOptions()`; in js/tests/pictures/pictures_test.js: `offParchment()`, `simpleUiTests()`; in js/tests/pictures/findbox_test.js: `findBoxTests()`, `crossTests()`; in js/tests/pictures/tooltips_test.js: `tipsOver()`; in js/tests/pictures/treebuttons_test.js: `shown()`; in js/tests/pictures/scrollbars_test.js: `barStates()`

Calls *no function of the code*

### 9.2.20 `everything()` EXPORTED line 67

Lists every element that can be seen on the page.

Called by in js/tests/audit_test.js: `audit()`; in js/tests/interface/keys_test.js: `textSpacing()`; in js/tests/pictures/pictures_test.js: `wrongSizes()`, `strayCursors()`, `bigCursors()`, `imagesShown()`

Calls `rendered()`

### 9.2.21 `styles(el)` EXPORTED line 69

Gets an element's styles and those of its `::before`, `::after` and `::marker` parts, so decorations are checked too.

Called by `drawn()`

Calls *no function of the code*

### 9.2.22 `rgb(colour)` INTERNAL line 71

Reads the numbers of a CSS colour.

Called by `tone()`
Calls *no function of the code*

### 9.2.23 `tone(colour)` INTERNAL lines 73–79

Names a colour as white, black, green, red or other, loosely enough for light and dark colours.

Called by `drawn()`
Calls `rgb()`

### 9.2.24 `drawn(box)` EXPORTED lines 81–89

Names the mark drawn on a checkbox, such as "green tick", "red cross" or "white dash", from the borders of its `::before` and `::after`, leaving out parts that draw nothing (Req. 3, 14, 23, 26).

Called by in js/tests/behaviour/marking_test.js: `marking()`, `chipMarks()`; in js/tests/behaviour/branches_test.js: `buildingMarks()`; in js/tests/behaviour/lines_test.js: `parents()`; in js/tests/interface/options_test.js: `openingOptions()`; in js/tests/pictures/icons_test.js: `offSprite()`
Calls `styles()`, `tone()`

### 9.2.25 `textOf(node)` INTERNAL lines 93–96

Reads an element's text as a screen reader would, leaving out parts marked `aria-hidden`.

Called by `nameFor()`
Calls *no function of the code*

### 9.2.26 `tidy(text)` EXPORTED line 98

Turns runs of spaces into single spaces and trims the ends.

Called by `nameFor()`; in js/tests/audit_test.js: `layoutChecks()`; in js/tests/interface/languages_test.js: `footerTexts()`
Calls *no function of the code*

### 9.2.27 `nameFor(el)` EXPORTED lines 100–106

Works out an element's accessible name much as a browser does: from `aria-labelledby`, `aria-label`, its label or its text. Lets the tests check that controls are named (WCAG 4.1.2).

Called by in js/tests/audit_test.js: `controlChecks()`; in js/tests/interface/menu_test.js: `menuChoosing()`
Calls `tidy()`, `textOf()`; in js/view/toolkit.js: `$()`

### 9.2.28 `visibleTexts(el)` EXPORTED lines 108–115

Lists the visible pieces of text inside an element. Used to find stray placeholders, and to compare names with what can be seen (WCAG 2.5.3).

Called by in js/tests/audit_test.js: `controlChecks()`; in js/tests/interface/languages_test.js: `languageTests()`
Calls `rendered()`

### 9.2.29 `keyDown(target, key)` EXPORTED lines 117–119

Sends a key press to an element, as a visitor's keyboard would.

Called by `mustLack()`; in js/tests/behaviour/marking_test.js: `marking()`; in js/tests/behaviour/branches_test.js: `lineFilters()`; in js/tests/behaviour/tree_test.js: `counts()`; in js/tests/behaviour/lines_test.js: `lineTops()`, `parents()`, `unevenLine()`, `beyond()`; in js/tests/behaviour/greying_test.js: `rivals()`, `shuts()`; in js/tests/interface/menu_test.js: `menuKeys()`, `menuClosing()`; in js/tests/interface/options_test.js: `optionKeys()`; in js/tests/interface/keys_test.js: `treeMoves()`, `stepKeys()`, `branchKeys()`, `findKey()`; in js/tests/interface/headings_test.js: `headingTests()`
Calls *no function of the code*

### 9.2.30 `tap(el)` EXPORTED lines 121–125

Clicks an element with mouse pointer events, as a visitor would.

Called by `mustHave()`; in js/tests/behaviour/marking_test.js: `marking()`; in js/tests/behaviour/branches_test.js: `buildingMarks()`, `lineFilters()`, `singleLines()`, `markedAlone()`, `groupMarks()`; in js/tests/behaviour/tree_test.js: `crowding()`, `counts()`; in js/tests/behaviour/lines_test.js: `lineTops()`, `parents()`, `lines()`, `evenLine()`, `beyond()`; in js/tests/behaviour/greying_test.js: `rivals()`, `excludedBy()`, `standIns()`, `sharedRow()`, `shuts()`, `variants()`; in js/tests/behaviour/finding_test.js: `dropsGreyed()`; in js/tests/interface/menu_test.js: `menuTests()`; in js/tests/interface/options_test.js: `optionTests()`, `optionKeys()`; in js/tests/interface/keys_test.js: `menuAudit()`, `textSpacing()`; in js/tests/interface/uniques_test.js: `uniqueFilterTests()`, `inLine()`, `inGroup()`; in js/tests/pictures/pictures_test.js: `simpleUiTests()`; in js/tests/pictures/tooltips_test.js: `everyTipTests()`; in js/tests/pictures/treebuttons_test.js: `treeButtonTests()`; in js/tests/pictures/scrollbars_test.js: `scrollBarTests()`
Calls *no function of the code*

### 9.2.31 `mustHave(item)` EXPORTED line 127

Marks an item must have with a click, as a visitor would.

Called by in js/tests/interface/keys_test.js: `languageAudits()`; in js/tests/pictures/icons_test.js: `boxTests()`
Calls `tap()`, `nodeOf()`

### 9.2.32 `mustLack(item)` EXPORTED line 129

Marks an item must lack with X, as a visitor would.

Called by in js/tests/interface/keys_test.js: `filterFocus()`, `languageAudits()`; in js/tests/pictures/icons_test.js: `boxTests()`
Calls `keyDown()`, `nodeOf()`

### 9.2.33 `type(text)` EXPORTED lines 131–134

Types a text into Find, as a visitor would.

Called by `reset()`; in js/tests/behaviour/finding_test.js: `finding()`, `dropsGreyed()`; in js/tests/interface/keys_test.js: `findKey()`; in js/tests/pictures/findbox_test.js: `findBoxTests()`; in js/tests/pictures/tooltips_test.js: `everyTipTests()`
Calls in js/view/toolkit.js: `$()`

### 9.2.34 `pick(code)` EXPORTED lines 136–141

Chooses a language through the Options window, its menu and OK, as a visitor would.

Called by in js/tests/run_test.js: `toStart()`; in js/tests/interface/menu_test.js: `menuTests()`; in js/tests/interface/languages_test.js: `languageTests()`; in js/tests/interface/options_test.js: `languagePopupTests()`; in js/tests/interface/keys_test.js: `languageAudits()`; in js/tests/pictures/treebuttons_test.js: `everyLanguage()`
Calls in js/view/toolkit.js: `$()`; in js/view/optionsview.js: `languageOptions()`

### 9.2.35 `reset()` EXPORTED lines 143–150

Returns the page to its start: Find empty, no marks, every branch closed, the menu and the Options window closed, nothing focused. Tests start from here.

Called by in js/tests/run_test.js: `toStart()`, `restore()`; in js/tests/behaviour/marking_test.js: `marking()`, `gestures()`; in js/tests/behaviour/branches_test.js: `buildingMarks()`, `lineFilters()`, `singleLines()`, `markedAlone()`, `groupMarks()`; in js/tests/behaviour/tree_test.js: `crowding()`, `counts()`, `tree()`; in js/tests/behaviour/lines_test.js: `lineTops()`, `parents()`, `lines()`; in js/tests/behaviour/greying_test.js: `rivals()`, `standIns()`; in js/tests/behaviour/finding_test.js: `finding()`, `dropsGreyed()`; in js/tests/interface/menu_test.js: `menuTests()`, `menuKeys()`; in js/tests/interface/languages_test.js: `languageTests()`; in js/tests/interface/options_test.js: `optionTests()`, `optionKeys()`, `languagePopupTests()`; in js/tests/interface/keys_test.js: `filterFocus()`, `treeMoves()`, `findKey()`, `menuAudit()`, `languageAudits()`, `textSpacing()`, `optionAudits()`; in js/tests/interface/uniques_test.js: `uniqueFilterTests()`; in js/tests/interface/headings_test.js: `headingTests()`; in js/tests/pictures/pictures_test.js: `pictureTests()`, `wrongSizes()`; in js/tests/pictures/icons_test.js: `iconTests()`; in js/tests/pictures/tooltips_test.js: `everyTipTests()`; in js/tests/pictures/scrollbars_test.js: `scrollBarTests()`
Calls `type()`; in js/view/toolkit.js: `$()`; in js/view/optionsview.js: `setMenu()`

### 9.2.36 `beside()` EXPORTED lines 152–157

Tells whether Find's message is on one line, right of the box and level with it (Req. 21).

Called by in js/tests/behaviour/finding_test.js: `finding()`; in js/tests/pictures/findbox_test.js: `findBoxTests()`
Calls in js/view/toolkit.js: `$()`

### 9.2.37 `listed()` EXPORTED line 159

Lists the civilisations the page shows.

Called by in js/tests/behaviour/samples_test.js: `listedAre()`; in js/tests/behaviour/tree_test.js: `crowding()`; in js/tests/interface/menu_test.js: `menuTests()`; in js/tests/interface/languages_test.js: `civList()`; in js/tests/interface/options_test.js: `optionTests()`, `optionKeys()`; in js/tests/interface/keys_test.js: `filterFocus()`
Calls in js/view/toolkit.js: `$()`

### 9.2.38 `expected(set)` EXPORTED line 161

Lists the names a set of civilisations should show, translated and sorted, to compare with `listed()`.

Called by in js/tests/behaviour/samples_test.js: `listedAre()`; in js/tests/interface/menu_test.js: `menuTests()`; in js/tests/interface/keys_test.js: `filterFocus()`
Calls in js/model/civsets.js: `civsOf()`; in js/model/texts.js: `civName()`

### 9.2.39 `itemAbove(item)` INTERNAL lines 163–166

Tells whether an item has another item above it, or sits in a group marked `"lines"`, which makes it a unit or technology. The tests work this out on their own, apart from the page's code.

Called by `onTop()`, `inner()`, `fixtures()`
Calls *no function of the code*

### 9.2.40 `onlyChildren(item, stopAt = () => false)` EXPORTED lines 168–174

Lists an item and those below it while each has one item under it, stopping before an item for which `stopAt` holds: the rows that opening an item opens, or that a mark on it may pass down to.

Called by `chainOf()`; in js/tests/behaviour/lines_test.js: `sameCivs()`
Calls *no function of the code*

### 9.2.41 `chainOf(item)` EXPORTED line 176

The rows that opening an item should open: the item and those below it while each has one row under it (`onlyChildren()`), those with an arrow only. Worked out apart from the page's own `chainOf()`.

Called by in js/tests/behaviour/tree_test.js: `tree()`; in js/tests/interface/keys_test.js: `treeMoves()`
Calls `onlyChildren()`, `twistyOf()`

### 9.2.42 `outside(nodes, regional)` EXPORTED lines 178–180

Lists the items of a tree outside regional items and all under them: what the tree shows with Include ticked and Include Regionals unticked.

Called by in js/tests/interface/options_test.js: `withoutRegionals()`
Calls *no function of the code*

### 9.2.43 `byPath(path)` EXPORTED line 182

Finds the item with a path in the tree as it is now, after it was built again.

Called by in js/tests/behaviour/lines_test.js: `beyond()`; in js/tests/interface/options_test.js: `optionTests()`, `optionKeys()`
Calls *no function of the code*

### 9.2.44 `onTop(item)` INTERNAL line 184

Tells whether an item is the top of its line: a unit or technology right under a building or group. The tests work this out on their own, apart from the page's code.

Called by `inner()`, `fixtures()`
Calls `itemAbove()`

### 9.2.45 `lines()` INTERNAL line 186

Lists the items that start a line.

Called by `fixtures()`
Calls *no function of the code*

### 9.2.46 `inner(item)` INTERNAL line 188

Tells whether an item is a unit or technology below the top of its line with one item under it, where a mark passes down only at the same count (Req. 29, 30).

Called by `fixtures()`
Calls `itemAbove()`, `onTop()`

### 9.2.47 `fixtures()` EXPORTED lines 190–227

Picks sample items from the tree: leaves some civilisations have, buildings, line tops, even and uneven counts, a line in a group of lines, a building some do without, a lone building and an item with a rival. Found from the data, to survive game updates. Each test that uses them picks them itself, after `reset()`, so no test depends on another. It throws when data.json lacks any of the three leaves that most tests mark, so that their group fails "the tests ran to the end" instead of checking nothing.

Called by in js/tests/behaviour/marking_test.js: `marking()`, `gestures()`; in js/tests/behaviour/branches_test.js: `buildingMarks()`, `lineFilters()`, `singleLines()`, `groupMarks()`; in js/tests/behaviour/tree_test.js: `counts()`; in js/tests/behaviour/lines_test.js: `lineTops()`, `parents()`, `lines()`; in js/tests/behaviour/greying_test.js: `rivals()`; in js/tests/behaviour/finding_test.js: `finding()`; in js/tests/interface/menu_test.js: `menuTests()`; in js/tests/interface/keys_test.js: `filterFocus()`, `treeMoves()`, `findKey()`, `menuAudit()`, `languageAudits()`, `textSpacing()`; in js/tests/pictures/icons_test.js: `boxTests()`; in js/tests/pictures/tooltips_test.js: `everyTipTests()`
Calls `itemAbove()`, `lines()`, `onTop()`, `inner()`; in js/model/civsets.js: `size()`; in js/model/texts.js: `nameOf()`; in js/view/civsview.js: `nameIsShared()`

### 9.2.48 `wait(ms)` EXPORTED line 229

Waits a number of milliseconds: to hold a long press, and to give a language that is made to fail time to show its popup.

Called by in js/tests/behaviour/marking_test.js: `gestures()`; in js/tests/interface/options_test.js: `failing()`
Calls *no function of the code*

# 9.3 audit_test.js

**Folder:** js/tests/ 200 lines 17 functions, 1 exported

**Imports:** js/model/texts.js, js/model/model.js, js/model/data/datahandler.js, js/view/toolkit.js, js/view/treeview.js, js/view/optionsview.js, js/tests/helpers_test.js, js/tests/pictures/picturetools_test.js

**Imported by:** js/tests/interface/options_test.js, js/tests/interface/keys_test.js

The accessibility audit: checks the page as it stands against WCAG 2.2 AAA, wherever a test calls it. Its checks are Tests 208 to 227, and it checks too that the page does not scroll (Test 29).

### 9.3.1 `audit(label, { focus = true } = {})` EXPORTED lines 25–38

Checks the page as it stands through the checks below, and the focus checks unless asked not to, with the state named in each failure's detail. It also checks that the page shows no image, media or picture in its styles (Test 226; Rules 9, 22). Called in many states and languages.

Called by in js/tests/interface/options_test.js: `languagePopupTests()`; in js/tests/interface/keys_test.js: `menuAudit()`, `languageAudits()`, `optionAudits()`

Calls `structureChecks()`, `controlChecks()`, `colourChecks()`, `layoutChecks()`, `focusChecks()`; in js/tests/helpers_test.js: `check()`, `everything()`, `describe()`; in js/tests/pictures/picturetools_test.js: `pictured()`

### 9.3.2 `structureChecks(note, list)` INTERNAL lines 40–51

Checks the page's title, its language and every language tag, the language of each language choice, its landmarks, its headings (`headingChecks()`), and that every ARIA reference points to exactly one element (Tests 208 to 216; WCAG 1.3.1, 2.4.2, 3.1.1, 3.1.2, 4.1.2).

Called by `audit()`

Calls `headingChecks()`, `unmatched()`; in js/view/optionsview.js: `languageOptions()`; in js/tests/helpers_test.js: `describe()`

### 9.3.3 `headingChecks(note, list)` INTERNAL lines 53–62

Checks the headings that can be seen: one main heading, none without text, and none that skips a level (Tests 213 to 215; WCAG 1.3.1, 2.4.6).

Called by `structureChecks()`

Calls in js/tests/helpers_test.js: `rendered()`, `describe()`

### 9.3.4 `unmatched()` INTERNAL lines 64–72

Lists every id in an `aria-labelledby`, `aria-controls` or `aria-describedby` that no element or more than one has, with the element that names it and the count (Test 216).

Called by `structureChecks()`

Calls in js/tests/helpers_test.js: `describe()`

### 9.3.5 `controlChecks(list)` INTERNAL lines 74–89

Checks that boxes, branches and the language menu say their state, that every control has a name, and that the name contains its visible text (Tests 217 to 219; WCAG 2.5.3, 4.1.2).

Called by `audit()`

Calls `focusables()`, `squash()`; in js/model/texts.js: `nameOf()`; in js/view/toolkit.js: `$()`; in js/view/treeview.js: `rowOf()`; in js/tests/helpers_test.js: `describe()`, `twistyOf()`, `rendered()`, `nameFor()`, `visibleTexts()`

### 9.3.6 `colourChecks(all, list)` INTERNAL lines 91–108

Checks the contrast of all visible text, 7:1, or 4.5:1 when large, and of control borders, 3:1 (Tests 220, 221; WCAG 1.4.6, 1.4.11).

Called by `audit()`

Calls `contrastOn()`, `backgroundOf()`; in js/tests/helpers_test.js: `describe()`

### 9.3.7 `layoutChecks(all, note, list)` INTERNAL lines 110–131

Checks 44 by 44 targets, but for links inside a sentence, no page scrolling, nothing sticking out sideways, and unjustified text with 1.5 line spacing (Tests 222 to 225 and 29; WCAG 2.5.5, 1.4.10, 1.4.8; Rule 12).

Called by `audit()`

Calls `focusables()`; in js/tests/helpers_test.js: `tidy()`, `describe()`

### 9.3.8 `focusChecks(list)` INTERNAL lines 133–148

Focuses every control in turn, only those of the open window when there is one, and checks that nothing covers it (Test 227; WCAG 2.4.12).

Called by `audit()`

Calls `focusables()`, `visibleCentre()`; in js/tests/helpers_test.js: `describe()`

### 9.3.9 `visibleCentre(el)` INTERNAL lines 150–163

Finds the middle of the visible part of an element's first line, clipped by every scrolling area around it, to see what is drawn there.

Called by `focusChecks()`
Calls *no function of the code*

### 9.3.10 `focusables()` INTERNAL lines 165–166

Lists every element that can be seen and reached with the keyboard.

Called by `controlChecks()`, `layoutChecks()`, `focusChecks()`
Calls in js/tests/helpers_test.js: `rendered()`

### 9.3.11 `squash(text)` INTERNAL line 168

Lower-cases a text and keeps only letters and digits, for loose comparisons.

Called by `controlChecks()`
Calls *no function of the code*

### 9.3.12 `contrastOn(colour, behind)` INTERNAL line 170

Computes the contrast of a colour, which may be see-through, over the colour behind it.

Called by `colourChecks()`
Calls `contrast()`, `blend()`, `rgba()`

### 9.3.13 `contrast(a, b)` INTERNAL lines 172–175

Computes the contrast ratio of two colours, as WCAG defines it, from 1:1 to 21:1.

Called by `contrastOn()`
Calls `luminance()`

### 9.3.14 `luminance(rgb)` INTERNAL lines 177–183

Computes a colour's relative luminance, as WCAG defines it.

Called by `contrast()`
Calls *no function of the code*

### 9.3.15 `backgroundOf(el)` INTERNAL lines 185–193

Works out the colour behind an element by blending its own and its ancestors' backgrounds. Contrast is measured against this.

Called by `colourChecks()`
Calls `rgba()`, `blend()`

### 9.3.16 `blend([r, g, b, a], [R, G, B])` INTERNAL line 195

Blends a see-through colour over the colour below it, to get the colour actually seen.

Called by `contrastOn()`, `backgroundOf()`
Calls *no function of the code*

### 9.3.17 `rgba(value)` INTERNAL lines 197–200

Reads a CSS colour into its red, green, blue and alpha numbers.

Called by `contrastOn()`, `backgroundOf()`
Calls *no function of the code*

# 10 js/tests/behaviour/

7 files 672 lines

The tests of what marking does: the marks and Filters, the counts, opening and closing, greying out and Find. marking_test.js runs them all, and samples_test.js holds the test names and tools the others share.

**Subchapters:** 10.1 marking_test.js, 10.2 samples_test.js, 10.3 branches_test.js, 10.4 tree_test.js, 10.5 lines_test.js, 10.6 greying_test.js, 10.7 finding_test.js

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## 10.1 marking_test.js

**Folder:** js/tests/behaviour/ 100 lines 6 functions, 1 exported
**Imports:** js/model/civsets.js, js/model/texts.js, js/model/model.js, js/view/toolkit.js, js/view/civsview.js, js/view/treeview.js, js/tests/helpers_test.js, js/tests/behaviour/samples_test.js, js/tests/behaviour/branches_test.js, js/tests/behaviour/tree_test.js, js/tests/behaviour/lines_test.js, js/tests/behaviour/greying_test.js, js/tests/behaviour/finding_test.js
**Imported by:** js/tests/run_test.js

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The tests of marking an item: by click, X, Space, the menu key, a long press and a right-click, with the civilisation list, its count and the first filter buttons. Its `behaviourTests()` runs the tests of every file in `js/tests/behaviour/`.

### 10.1.1 `behaviourTests()` EXPORTED lines 23–31

Runs the tests of the tree, in the order TESTS.md lists them, and waits for the long press of `gestures()`.

Called by in js/tests/run_test.js: `runAll()`
Calls `marking()`, `gestures()`; in js/tests/behaviour/branches_test.js: `branches()`; in js/tests/behaviour/tree_test.js: `crowding()`, `counts()`, `tree()`; in js/tests/behaviour/lines_test.js: `lineTops()`, `parents()`, `lines()`; in js/tests/behaviour/greying_test.js: `rivals()`, `standIns()`; in js/tests/behaviour/finding_test.js: `finding()`

### 10.1.2 `marking()` INTERNAL lines 33–62

Tests marking by click, X, the menu key (`menuKey()`) and Space, what screen readers hear, the civilisation list and its count, the first filter buttons with their names and marks, and Clear (Tests 4 to 17; Req. 3, 4, 17, 22, 23).

Called by `behaviourTests()`
Calls `chipMarks()`, `menuKey()`; in js/model/civsets.js: `size()`; in js/model/texts.js: `plural()`, `fill()`; in js/view/toolkit.js: `$()`; in js/view/civsview.js: `chipName()`; in js/view/treeview.js: `rowOf()`; in js/tests/helpers_test.js: `reset()`, `fixtures()`, `tap()`, `nodeOf()`, `check()`, `keyDown()`, `drawn()`, `same()`; in js/tests/behaviour/samples_test.js: `listedAre()`, `boxOf()`

### 10.1.3 `menuKey(item)` INTERNAL lines 64–69

Tests that the menu key marks an item must lack and that the browser's own menu does not open over it (Tests 13, 14).

Called by `marking()`
Calls in js/tests/helpers_test.js: `nodeOf()`, `check()`

### 10.1.4 `gestures()` INTERNAL lines 71–89

Tests that a long press with a finger marks must lack, that a right-click clears it and then marks it again, and that no browser menu opens (Tests 36 to 38 and 14; Req. 3).

Called by `behaviourTests()`
Calls `rightClick()`; in js/tests/helpers_test.js: `reset()`, `fixtures()`, `nodeOf()`, `box()`, `wait()`, `check()`

### 10.1.5 `rightClick(node, at)` INTERNAL lines 91–98

Right-clicks a row as a visitor's mouse would, with pointer events and the `contextmenu` that follows, and returns that event, to see whether the browser's menu was kept shut.

Called by `gestures()`
Calls *no function of the code*

### 10.1.6 `chipMarks()` INTERNAL line 100

Names the mark drawn on each filter button, sorted, such as "green tick" and "red cross".

Called by `marking()`
Calls in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `drawn()`; in js/tests/behaviour/samples_test.js: `boxOf()`

# 10.2 samples_test.js

**Folder:** js/tests/behaviour/ 24 lines 4 functions, 4 exported

**Imports:** js/model/texts.js, js/view/toolkit.js, js/tests/helpers_test.js
**Imported by:** js/tests/behaviour/marking_test.js, js/tests/behaviour/branches_test.js, js/tests/behaviour/lines_test.js, js/tests/behaviour/greying_test.js

What the behaviour tests share: the names of the tests that run in several of their files, the marks as `drawn()` names them, and small questions about the civilisation list, the filter buttons and the tree. It holds no tests itself.

### 10.2.1 `listedAre(set)` EXPORTED line 13

Tells whether the civilisations listed are exactly those of a set, whatever their order (Req. 4).

Called by in js/tests/behaviour/marking_test.js: `marking()`; in js/tests/behaviour/lines_test.js: `lineTops()`, `parents()`, `unevenLine()`, `beyond()`; in js/tests/behaviour/greying_test.js: `rivals()`
Calls in js/tests/helpers_test.js: `same()`, `listed()`, `expected()`

### 10.2.2 `chipNames()` EXPORTED line 15

Lists the names on the filter buttons.

Called by in js/tests/behaviour/branches_test.js: `buildingMarks()`, `lineFilters()`, `singleLines()`, `markedAlone()`; in js/tests/behaviour/lines_test.js: `lineTops()`; in js/tests/behaviour/greying_test.js: `sharedRow()`
Calls in js/view/toolkit.js: `$()`

### 10.2.3 `boxOf(el)` EXPORTED line 17

The box of a row or a filter button, whose mark `drawn()` reads.

Called by in js/tests/behaviour/marking_test.js: `marking()`, `chipMarks()`; in js/tests/behaviour/branches_test.js: `buildingMarks()`; in js/tests/behaviour/lines_test.js: `parents()`
Calls *no function of the code*

### 10.2.4 `below(item, top)` EXPORTED lines 19–24

Tells whether an item sits anywhere under another.

Called by in js/tests/behaviour/branches_test.js: `closedInside()`; in js/tests/behaviour/lines_test.js: `parents()`, `unevenLine()`; in js/tests/behaviour/greying_test.js: `sharedRow()`, `borrowedRows()`
Calls *no function of the code*

# 10.3 branches_test.js

**Folder:** js/tests/behaviour/ 85 lines 7 functions, 1 exported

**Imports:** js/model/texts.js, js/model/model.js, js/view/toolkit.js, js/view/treeview.js, js/tests/helpers_test.js, js/tests/behaviour/samples_test.js
**Imported by:** js/tests/behaviour/marking_test.js

The tests of buildings, groups and lines as a whole: what marking them marks and opens, the names of their filter buttons, and the white dash.

### 10.3.1 `branches()` EXPORTED lines 11–16

Runs the tests of buildings, groups and lines: `buildingMarks()`, `lineFilters()`, `singleLines()` and `groupMarks()`.

Called by in js/tests/behaviour/marking_test.js: `behaviourTests()`
Calls `buildingMarks()`, `lineFilters()`, `singleLines()`, `groupMarks()`

### 10.3.2 `buildingMarks()` INTERNAL lines 18–31

Tests that marking a building marks and opens everything in it and names its filter button "Full" with the building's name, and that marking one row in it then shows a white dash, which screen readers hear as mixed (Tests 18 to 21 and 7; Req. 24, 26, 31).

Called by `branches()`
Calls `closedInside()`; in js/model/texts.js: `fill()`, `nameOf()`; in js/model/model.js: `branchState()`; in js/view/treeview.js: `shown()`; in js/tests/helpers_test.js: `reset()`, `fixtures()`, `openTo()`, `tap()`, `nodeOf()`, `check()`, `same()`, `drawn()`; in js/tests/behaviour/samples_test.js: `chipNames()`, `boxOf()`

### 10.3.3 `lineFilters()` INTERNAL lines 33–45

Tests that a line's filter button is named after its deepest must-have item, and that clicking it clears its mark from the whole line but leaves the line's other kind of mark (Tests 22 to 24; Req. 25).

Called by `branches()`
Calls in js/model/texts.js: `nameOf()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `reset()`, `fixtures()`, `tap()`, `nodeOf()`, `check()`, `same()`, `keyDown()`; in js/tests/behaviour/samples_test.js: `chipNames()`

### 10.3.4 `singleLines()` INTERNAL lines 47–61

Tests that a fully marked building heading a single line is named after the line's last item, and one with nothing in it plainly (Req. 24), and that in a group of lines, such as Fortifications, a mark passes down to no item with other civilisations while the filter button is named after the deepest item (Tests 25 to 27 and 22; Req. 25, 30).

Called by `branches()`
Calls `markedAlone()`; in js/model/texts.js: `nameOf()`; in js/tests/helpers_test.js: `fixtures()`, `check()`, `same()`, `reset()`, `tap()`, `nodeOf()`; in js/tests/behaviour/samples_test.js: `chipNames()`

### 10.3.5 `markedAlone(item)` INTERNAL lines 63–67

Marks one item, from the start, and returns the names on the filter buttons.

Called by `singleLines()`
Calls in js/tests/helpers_test.js: `reset()`, `tap()`, `nodeOf()`; in js/tests/behaviour/samples_test.js: `chipNames()`

### 10.3.6 `groupMarks()` INTERNAL lines 69–79

Tests that marking a top-level group marks and opens everything in it, while the group holds no mark of its own, and greys out nothing in it (Tests 18, 19 and 28).

Called by `branches()`
Calls `closedInside()`; in js/model/model.js: `branchState()`; in js/view/treeview.js: `shown()`, `rowOf()`; in js/tests/helpers_test.js: `reset()`, `fixtures()`, `tap()`, `nodeOf()`, `check()`

### 10.3.7 `closedInside(top)` INTERNAL lines 81–85

Lists the items under a building or group, and throws if any of them is in sight before it is marked, which would leave nothing for the test of what marking opens.

Called by `buildingMarks()`, `groupMarks()`
Calls in js/model/texts.js: `nameOf()`; in js/view/treeview.js: `shown()`; in js/tests/behaviour/samples_test.js: `below()`

# 10.4 tree_test.js

**Folder:** js/tests/behaviour/ 115 lines 9 functions, 3 exported

**Imports:** js/model/civsets.js, js/model/texts.js, js/model/model.js, js/view/toolkit.js, js/view/treeview.js, js/tests/helpers_test.js
**Imported by:** js/tests/behaviour/marking_test.js

The tests of the tree as a whole: the panes with every branch open, the numbers on the items, and Expand all, Collapse all and the arrows.

### 10.4.1 `crowding()` EXPORTED lines 14–45

Tests that a full tree and the longest civilisation list never make the page scroll, push a pane's header out of it, make the panes overlap or differ in height, or move or resize a pane, and that the list scrolls inside its pane, in whatever window the tests run (Tests 29 to 35; Rules 12, 18; Req. 35).

Called by in js/tests/behaviour/marking_test.js: `behaviourTests()`
Calls `scrolls()`; in js/model/civsets.js: `size()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `reset()`, `box()`, `tap()`, `nodeOf()`, `listed()`, `check()`, `expect()`

### 10.4.2 `counts()` EXPORTED lines 47–59

Tests the numbers on the items, closed and open, and with filters, through `countKinds()` (Tests 65 to 69; Req. 3).

Called by in js/tests/behaviour/marking_test.js: `behaviourTests()`
Calls `countKinds()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `reset()`, `fixtures()`, `tap()`, `nodeOf()`, `keyDown()`

### 10.4.3 `countKinds()` INTERNAL lines 61–74

Checks each kind of row, an empty top-level row, a closed item with contents, an open item, an item with nothing under it and a group, and names the rows whose numbers differ from `wanted()`.

Called by `counts()`
Calls `ownCount()`, `miscounted()`; in js/model/texts.js: `nameOf()`; in js/view/treeview.js: `isOpen()`; in js/tests/helpers_test.js: `expect()`

### 10.4.4 `miscounted(item)` INTERNAL lines 76–79

Tells whether a row's number, or what screen readers hear of it, differs from `wanted()`.

Called by `countKinds()`
Calls `wanted()`; in js/view/treeview.js: `rowOf()`

### 10.4.5 `wanted(item)` INTERNAL lines 81–87

Works out a row's number and what screen readers hear of it: nothing on an empty top-level row, the row's own number when it is open, has nothing under it or is a group, and otherwise its own number, then the number with all under it.

Called by `miscounted()`
Calls `ownCount()`, `fitting()`; in js/model/civsets.js: `size()`; in js/model/texts.js: `plural()`, `fill()`; in js/view/treeview.js: `isOpen()`

### 10.4.6 `tree()` EXPORTED lines 89–108

Tests Expand all, Collapse all and the arrows, and that opening a branch also opens its only children (Tests 70 to 73; Req. 3, 18, 19).

Called by in js/tests/behaviour/marking_test.js: `behaviourTests()`
Calls in js/model/texts.js: `nameOf()`; in js/view/toolkit.js: `$()`; in js/view/treeview.js: `isOpen()`, `rowOf()`; in js/tests/helpers_test.js: `reset()`, `check()`, `twistyOf()`, `chainOf()`, `expect()`

### 10.4.7 `scrolls(el)` INTERNAL line 110

Tells whether an element's content is taller than the element, so that it scrolls.

Called by `crowding()`
Calls *no function of the code*

### 10.4.8 `ownCount(item)` INTERNAL line 112

Counts the civilisations that fit the marks and have an item, or for a group anything in it.

Called by `countKinds()`, `wanted()`
Calls `fitting()`; in js/model/civsets.js: `size()`

### 10.4.9 `fitting()` INTERNAL lines 114–115

The civilisations that fit every mark, worked out from the items apart from the page's code.

Called by `wanted()`, `ownCount()`
Calls *no function of the code*

# 10.5 lines_test.js

**Folder:** js/tests/behaviour/ 134 lines 8 functions, 3 exported

**Imports:** js/model/civsets.js, js/model/texts.js, js/model/model.js, js/view/toolkit.js, js/view/treeview.js, js/controller/optionshandler.js, js/tests/helpers_test.js, js/tests/behaviour/samples_test.js

**Imported by:** js/tests/behaviour/marking_test.js

The tests of lines: how a mark passes down a line and up to the items above it, and what marking a unit or technology opens.

## 10.5.1 `lineTops()` EXPORTED

lines 18–50

Tests that marking the top of a line passes the mark down only to the items with the same civilisations, must have or must lack, that marking it again clears it, that marking a line's last item marks the line before it with one filter button, named after that item, and that must lack on the top then clears the must-have marks below it (Tests 39 to 42, 27 and 22; Req. 25, 29, 30, 34).

Called by in js/tests/behaviour/marking_test.js: `behaviourTests()`
Calls `sameCivs()`; in js/model/texts.js: `nameOf()`; in js/tests/helpers_test.js: `reset()`, `fixtures()`, `tap()`, `nodeOf()`, `check()`, `keyDown()`; in js/tests/behaviour/samples_test.js: `listedAre()`, `chipNames()`

## 10.5.2 `parents()` EXPORTED

lines 52–72

Tests that must have passes up but never to a building, which then shows a dash, and that an item can be must lack while everything under it is must have (Tests 41, 43, 21, 7 and 5; Req. 4, 26, 34).

Called by in js/tests/behaviour/marking_test.js: `behaviourTests()`
Calls in js/model/texts.js: `nameOf()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `reset()`, `fixtures()`, `tap()`, `nodeOf()`, `check()`, `drawn()`, `keyDown()`; in js/tests/behaviour/samples_test.js: `below()`, `boxOf()`, `listedAre()`

## 10.5.3 `lines()` EXPORTED

lines 74–88

Tests that marking a closed unit or technology opens it to its first fork or its end (Test 59; Req. 28), then runs `evenLine()`, `unevenLine()` and `beyond()`.

Called by in js/tests/behaviour/marking_test.js: `behaviourTests()`
Calls `evenLine()`, `unevenLine()`, `beyond()`; in js/model/texts.js: `nameOf()`; in js/view/toolkit.js: `$()`; in js/view/treeview.js: `isOpen()`, `shown()`; in js/tests/helpers_test.js: `reset()`, `fixtures()`, `openTo()`, `tap()`, `nodeOf()`, `check()`, `twistyOf()`

## 10.5.4 `evenLine(even)` INTERNAL

lines 90–99

Tests an item below the top of a line whose item under it has the same civilisations: must have on it passes up to the items above it and down while the civilisations stay the same, and no further (Tests 41, 39 and 27; Req. 29, 30, 34).

Called by `lines()`
Calls `sameCivs()`, `marked()`; in js/model/texts.js: `nameOf()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `tap()`, `nodeOf()`, `check()`

## 10.5.5 `unevenLine(uneven)` INTERNAL

lines 101–109

Tests an item below the top of a line whose item under it has other civilisations: must lack on it passes down only while the civilisations stay the same, and up to nothing (Tests 39, 27 and 60; Req. 29, 30).

Called by `lines()`
Calls `sameCivs()`, `marked()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `keyDown()`, `nodeOf()`, `check()`; in js/tests/behaviour/samples_test.js: `below()`, `listedAre()`

## 10.5.6 `beyond()` INTERNAL

lines 111–130

With Include ticked, finds an item that some civilisations have without the unit or technology above it, and checks that a mark on the one above does not pass down to it, that must have on it does not pass up, that must have on it and must lack on the one above list just those civilisations, and that unticking Include, which may leave the one above out, puts it in that one's place with its mark kept (Tests 27, 61 to 64 and 5; Req. 30, 34).

Called by `lines()`
Calls in js/model/civsets.js: `size()`; in js/view/toolkit.js: `$()`; in js/controller/optionshandler.js: `chooseInclude()`; in js/tests/helpers_test.js: `openTo()`, `tap()`, `nodeOf()`, `check()`, `keyDown()`, `byPath()`; in js/tests/behaviour/samples_test.js: `listedAre()`

## 10.5.7 `sameCivs(item)` INTERNAL

line 132

Lists the items of a line, from an item down while each has one item under it that is not a group, that have the same civilisations as the item: those a mark on it passes down to (Req. 29, 30).

Called by `lineTops()`, `evenLine()`, `unevenLine()`
Calls in js/tests/helpers_test.js: `onlyChildren()`

### 10.5.8 `marked()` INTERNAL

line 134

Lists the marked items, leaving out the copies of shared rows, which carry their original's mark.

Called by `evenLine()`, `unevenLine()`
Calls *no function of the code*

# 10.6 greying_test.js

**Folder:** js/tests/behaviour/ 155 lines 12 functions, 2 exported

**Imports:** js/model/texts.js, js/model/model.js, js/model/data/datahandler.js, js/view/toolkit.js, js/view/treeview.js, js/controller/optionshandler.js, js/tests/helpers_test.js, js/tests/behaviour/samples_test.js

**Imported by:** js/tests/behaviour/marking_test.js

The tests of greying out: an item that excludes another, such as the Xolotl Warrior and the Stable, and the variants that replace others, such as the Pasture and the Farm, with their alternatives (Req. 37).

### 10.6.1 `rivals()` EXPORTED

lines 19–49

Tests an item that excludes another, such as the Xolotl Warrior and the Stable: must have on the one greys out and closes the other and everything in it, the one staying in sight and the other's arrow working; must have on any of those greys out the one (`excludedBy()`); what is greyed out cannot be marked; and must lack greys out nothing (Tests 44 to 51).

Called by in js/tests/behaviour/marking_test.js: `behaviourTests()`
Calls `side()`, `greyedItems()`, `excludedBy()`; in js/model/model.js: `branchState()`; in js/view/toolkit.js: `$()`; in js/view/treeview.js: `shown()`, `isOpen()`; in js/tests/helpers_test.js: `reset()`, `fixtures()`, `openTo()`, `tap()`, `nodeOf()`, `check()`, `same()`, `twistyOf()`, `keyDown()`; in js/tests/behaviour/samples_test.js: `listedAre()`

### 10.6.2 `excludedBy(by, rival)` INTERNAL

lines 51–60

Tests that must have on an item of the excluded branch greys out the item that excludes it, which then cannot be marked, by a click or by the model's `toggleMark()` itself (Tests 48 to 50).

Called by `rivals()`
Calls `greyedItems()`; in js/model/texts.js: `nameOf()`; in js/model/model.js: `toggleMark()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `tap()`, `nodeOf()`, `check()`, `same()`

### 10.6.3 `standIns()` EXPORTED

lines 62–83

Tests an item that replaces others, such as the Pasture and the Farm: a row they share (`sharedRow()`), must have on either greying out and closing the other (`shuts()`) until it is cleared, and, with Include ticked, the rows it borrows (`borrowedRows()`) and what each variant greys out (`variants()`) (Test 57; Req. 37).

Called by in js/tests/behaviour/marking_test.js: `behaviourTests()`
Calls `sharedRow()`, `shuts()`, `greyed()`, `borrowedRows()`, `variants()`; in js/view/treeview.js: `setOpen()`; in js/controller/optionshandler.js: `chooseInclude()`; in js/tests/helpers_test.js: `reset()`, `openTo()`, `tap()`, `nodeOf()`, `check()`

### 10.6.4 `sharedRow(item, other)` INTERNAL

lines 85–93

Tests that marking a row that an item holds of what it replaces marks it in both buildings, opens nothing in the other, and makes one plain filter button (Tests 52 to 54).

Called by `standIns()`
Calls in js/model/texts.js: `nameOf()`; in js/view/toolkit.js: `$()`; in js/view/treeview.js: `isOpen()`; in js/tests/helpers_test.js: `tap()`, `nodeOf()`, `check()`, `same()`; in js/tests/behaviour/samples_test.js: `below()`, `chipNames()`

### 10.6.5 `shuts(each, by)` INTERNAL

lines 95–106

Tests that must have on one item greys out and closes the other, which then can be marked neither by a click nor by X, and whose arrow does not open it (Tests 46, 48, 55 and 56; Req. 37).

Called by `standIns()`
Calls `greyed()`; in js/model/texts.js: `nameOf()`; in js/view/treeview.js: `setOpen()`, `isOpen()`; in js/tests/helpers_test.js: `tap()`, `nodeOf()`, `check()`, `keyDown()`, `twistyOf()`

### 10.6.6 `borrowedRows()` INTERNAL

lines 108–119

Tests that each item that replaces buildings holds, as copies, exactly their common and regional rows, but for those under `"own"`; a unit or technology it replaces lends it none (Test 58).

Called by `standIns()`
Calls `ownKeys()`; in js/tests/helpers_test.js: `check()`, `same()`; in js/tests/behaviour/samples_test.js: `below()`

### 10.6.7 `variants()` INTERNAL

lines 121–130

Tests that must have on each variant greys out exactly its alternatives, as `alternativesOf()` works them out from data.json (Test 55; Req. 37).

Called by `standIns()`
Calls `alternativesOf()`, `greyedItems()`; in js/model/texts.js: `nameOf()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `tap()`, `nodeOf()`, `check()`, `same()`

### 10.6.8 `ownKeys()` INTERNAL lines 132–138

Lists the items marked `"own"` and everything under them, which items standing in for theirs never show.

Called by `borrowedRows()`
Calls *no function of the code*

### 10.6.9 `alternativesOf(key)` INTERNAL lines 140–148

Lists the items that marking an item must have should grey out, from data.json alone: the items it names, the items that name it, and those that name what it names (Req. 37).

Called by `variants()`
Calls *no function of the code*

### 10.6.10 `side(top, skip)` INTERNAL line 150

Lists an item and everything under it but a given item and what is under that: what an item greys out of the branch it excludes, when it sits inside that branch.

Called by `rivals()`
Calls *no function of the code*

### 10.6.11 `greyedItems()` INTERNAL line 152

Lists the items whose rows are greyed out for screen readers, by `aria-disabled`.

Called by `rivals()`, `excludedBy()`, `variants()`
Calls in js/tests/helpers_test.js: `nodeOf()`

### 10.6.12 `greyed(item)` INTERNAL lines 154–155

Tells whether an item's row is greyed out both for screen readers and for the eye.

Called by `standIns()`, `shuts()`
Calls in js/tests/helpers_test.js: `nodeOf()`

# 10.7 finding_test.js

**Folder:** js/tests/behaviour/ 59 lines 4 functions, 1 exported

**Imports:** js/model/texts.js, js/model/model.js, js/view/toolkit.js, js/view/treeview.js, js/tests/helpers_test.js
**Imported by:** js/tests/behaviour/marking_test.js

---

The tests of Find: what it highlights, opens and counts, and that it drops a match that a mark greys out.

### 10.7.1 `finding()` EXPORTED lines 13–32

Tests that Find highlights exactly its matches, in yellow, opens the branches down to them and counts them beside the box, drops a match once a mark greys out its branch (`dropsGreyed()`), says when nothing matches, and clears its count when emptied (Tests 74 to 80; Req. 20, 21).

Called by in js/tests/behaviour/marking_test.js: `behaviourTests()`
Calls `found()`, `yellow()`, `dropsGreyed()`; in js/model/texts.js: `nameOf()`, `plain()`, `plural()`; in js/view/toolkit.js: `$()`; in js/view/treeview.js: `shown()`; in js/tests/helpers_test.js: `reset()`, `fixtures()`, `type()`, `check()`, `nodeOf()`, `beside()`

### 10.7.2 `dropsGreyed(rival)` INTERNAL lines 34–52

Tests that Find drops a match once a mark greys out its branch: marks a sample's rival, or that rival's own, until a branch is closed and greyed out, searches for an item in it without the mark, marks again, and checks that the match loses its yellow and that the count beside Find agrees (Test 79).

Called by `finding()`
Calls `found()`; in js/model/texts.js: `nameOf()`, `plural()`; in js/model/model.js: `behind()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `reset()`, `tap()`, `nodeOf()`, `type()`, `check()`

### 10.7.3 `yellow(colour)` INTERNAL lines 54–57

Tells whether a colour is yellow: red and green high, blue clearly lower (Req. 20).

Called by `finding()`
Calls *no function of the code*

### 10.7.4 `found(item)` INTERNAL line 59

Tells whether an item's row is highlighted as a match of Find.

Called by `finding()`, `dropsGreyed()`
Calls in js/tests/helpers_test.js: `nodeOf()`

# 11 js/tests/interface/

7 files 655 lines

The tests of the page's controls: the language menu, every language, the Options window and the popup about a language that could not load, the keyboard, the filters of unique items, the headings inside buildings and the report window. keys_test.js also runs the accessibility audit in many states.

**Subchapters:** 11.1 menu_test.js, 11.2 languages_test.js, 11.3 options_test.js, 11.4 keys_test.js, 11.5 uniques_test.js, 11.6 headings_test.js, 11.7 reporting_test.js

## 11.1 menu_test.js

**Folder:** js/tests/interface/ 107 lines 7 functions, 2 exported
**Imports:** js/model/texts.js, js/model/data/datahandler.js, js/view/toolkit.js, js/view/optionsview.js, js/tests/helpers_test.js, js/tests/interface/options_test.js
**Imported by:** js/tests/run_test.js, js/tests/interface/keys_test.js

The tests of the language menu in the Options window: opening, closing and choosing a language with the mouse and the keys, the start in English, and the choice kept in the address (Req. 8 to 12).

### 11.1.1 `menuTests()` EXPORTED lines 13–24

Tests the language menu with an item marked: opening it (`menuOpening()`), closing it (`menuClosing()`) and choosing a language (`menuChoosing()`), and that the mark and the civilisation list survive the change of language (Test 64). Then puts the first language back.

Called by in js/tests/run_test.js: `runAll()`
Calls `menuOpening()`, `menuClosing()`, `menuChoosing()`; in js/tests/helpers_test.js: `reset()`, `fixtures()`, `tap()`, `nodeOf()`, `check()`, `same()`, `listed()`, `expected()`, `pick()`

### 11.1.2 `menuKeys()` EXPORTED lines 26–41

Tests the language menu's keys: Down on its button opens it at the language in use, Down goes to the next language and End to the last (Tests 228 to 230; WCAG 2.1.1). Run with the accessibility tests.

Called by in js/tests/interface/keys_test.js: `accessibilityTests()`
Calls in js/view/toolkit.js: `$()`; in js/view/optionsview.js: `languageOptions()`; in js/tests/helpers_test.js: `reset()`, `keyDown()`, `check()`, `codes()`

### 11.1.3 `menuOpening()` INTERNAL lines 43–59

Tests opening the menu: it is in the Options window (Req. 8), opening the window puts focus on its button (WCAG 2.4.3), a click on the button opens it, it offers every language by its own name (Req. 9) and marks the language in use (WCAG 1.3.1) (Tests 81 to 85). Returns the window's size before and after, for `menuClosing()`.

Called by `menuTests()`
Calls `windowSize()`; in js/view/toolkit.js: `$()`; in js/view/optionsview.js: `languageOptions()`; in js/tests/helpers_test.js: `check()`, `codes()`

### 11.1.4 `menuClosing(sizes)` INTERNAL lines 61–78

Tests closing the menu: Escape closes it and hands focus back to its button, a left-click in its list or outside it closes it, and so does moving focus out of it, while the window never changes its size (Tests 86 to 90; Req. 12; Rule 18; WCAG 2.1.1, 2.4.3).

Called by `menuTests()`
Calls `windowSize()`; in js/view/toolkit.js: `$()`; in js/view/optionsview.js: `languageOptions()`; in js/tests/helpers_test.js: `keyDown()`, `check()`

### 11.1.5 `menuChoosing(first)` INTERNAL lines 80–93

Tests choosing a language: the menu closes with focus back on its button, which names the language chosen (WCAG 4.1.2), the page waits for OK, without a choice the page starts in English (`englishFirst()`; Req. 10), and the language chosen is kept in the address, except over `file://` (Tests 87, 89, 91 to 94).

Called by `menuTests()`
Calls `englishFirst()`; in js/view/toolkit.js: `$()`; in js/view/optionsview.js: `languageOptions()`; in js/tests/helpers_test.js: `codes()`, `check()`, `nameFor()`

### 11.1.6 `windowSize()` INTERNAL line 95

The Options window's size, as "width by height", to tell whether the language menu changes it.

Called by `menuOpening()`, `menuClosing()`
Calls in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `box()`

### 11.1.7 `englishFirst()` INTERNAL

lines 97–107

Tells whether a first visit, with no saved choice and a German browser, starts in the language the address names, or else in English (Req. 10). Puts the saved choice and the browser's languages back afterwards.

Called by `menuChoosing()`
Calls in js/model/data/datahandler.js: `preferredLanguage()`; in js/tests/helpers_test.js: `codes()`

# 11.2 languages_test.js

**Folder:** js/tests/interface/ 64 lines 4 functions, 1 exported

**Imports:** js/model/texts.js, js/model/model.js, js/model/data/datahandler.js, js/view/toolkit.js, js/tests/helpers_test.js
**Imported by:** js/tests/run_test.js

The tests of one language, run in every language in turn: the page's text, the tree's names, the footer and the civilisation list in that language (Req. 5, 11).

## 11.2.1 `languageTests(code)` EXPORTED

lines 17–36

Tests one language, picked through Options: the page's text and the tree's names are in it and no placeholders are left (Tests 95 to 97; Req. 11), then the footer (`footerTexts()`, and `footerOnce()` in the first language only) and the civilisation list (`civList()`). A failure names the first three offenders.

Called by in js/tests/run_test.js: `runAll()`
Calls `footerTexts()`, `footerOnce()`, `civList()`; in js/model/texts.js: `nameOf()`; in js/tests/helpers_test.js: `reset()`, `pick()`, `check()`, `labelOf()`, `visibleTexts()`, `codes()`

## 11.2.2 `footerTexts(text, code)` INTERNAL

lines 38–44

Tests that the footer names the game update in the chosen language (Req. 5) and gives Microsoft's notice in English (Tests 98, 99).

Called by `languageTests()`
Calls in js/model/texts.js: `fill()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `check()`, `tidy()`

## 11.2.3 `footerOnce()` INTERNAL

lines 46–55

Tests what the footer shows in every language alike, so only once: it names the data's source (Req. 5), its links lead where data.json says, Microsoft's notice is marked as English (WCAG 3.1.2), and the footer is as wide as the panes, or hidden on a phone (Tests 100 to 104).

Called by `languageTests()`
Calls in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `check()`, `expect()`, `footerAsWide()`, `footerHidden()`

## 11.2.4 `civList(code)` INTERNAL

lines 57–64

Tests that the civilisations are named in the chosen language (Req. 11) and sorted in its order (Tests 105, 106).

Called by `languageTests()`
Calls in js/model/texts.js: `civName()`; in js/tests/helpers_test.js: `listed()`, `check()`, `same()`

# 11.3 options_test.js

**Folder:** js/tests/interface/ 205 lines 20 functions, 3 exported

**Imports:** js/model/civsets.js, js/model/texts.js, js/model/model.js, js/model/data/datahandler.js, js/view/toolkit.js, js/view/optionsview.js, js/controller/optionshandler.js, js/tests/helpers_test.js, js/tests/audit_test.js

**Imported by:** js/tests/run_test.js, js/tests/interface/menu_test.js, js/tests/interface/keys_test.js

The tests of the Options window: opening and closing it, its two Include boxes by mouse and keyboard, and the choices remembered (Req. 7, 13 to 16), and the popup that says a language could not be loaded. Its helpers list the items the tree shows, those Include leaves out and the regional ones, the last two worked out from data.json.

### 11.3.1 `optionTests()` EXPORTED lines 23–33

Tests the Options window and its two Include boxes, with an item marked: what the tree leaves out by default (Tests 107, 108; Req. 15), opening the window (`openingOptions()`), ticking and unticking with OK (`confirmingOptions()`), and closing it without OK (`cancellingOptions()`).

Called by in js/tests/run_test.js: `runAll()`

Calls `keptPath()`, `leftOut()`, `someRegional()`, `openingOptions()`, `shownKeys()`, `confirmingOptions()`, `cancellingOptions()`; in js/tests/helpers_test.js: `reset()`, `expect()`, `check()`, `tap()`, `nodeOf()`, `byPath()`, `listed()`, `same()`

### 11.3.2 `optionKeys()` EXPORTED lines 35–57

Tests that Space ticks and unticks Include and Include Regionals (Test 232; WCAG 2.1.1), that nothing changes before OK, and that OK then shows or leaves out those items, keeping the mark (Tests 92, 107, 108, 64). Run with the accessibility tests; puts Include back unticked.

Called by in js/tests/interface/keys_test.js: `accessibilityTests()`

Calls `keptPath()`, `shownKeys()`, `withoutRegionals()`, `leftOut()`, `someRegional()`; in js/view/toolkit.js: `$()`; in js/controller/optionshandler.js: `chooseInclude()`; in js/tests/helpers_test.js: `reset()`, `tap()`, `nodeOf()`, `byPath()`, `listed()`, `keyDown()`, `check()`, `ticked()`, `same()`, `expect()`

### 11.3.3 `openingOptions(shown)` INTERNAL lines 59–74

Tests opening Options from its button: the window opens (Req. 7) with Include unticked (Req. 13) and Include Regionals ticked, its buttons are Cancel and OK (Req. 16), and a click on Include shows a green tick (Req. 14) but changes nothing before OK, which closes the window (`closedAndFocused()`) (Tests 92, 109 to 113).

Called by `optionTests()`

Calls `shownKeys()`, `closedAndFocused()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `check()`, `ticked()`, `same()`, `rendered()`, `drawn()`

### 11.3.4 `confirmingOptions(marksKept)` INTERNAL lines 76–86

Tests what OK does: with Include ticked the tree shows every item, and with Include Regionals unticked and ticked again the regional items go and come back, each time with the mark kept, and both choices are remembered, unless the browser allows no storage (Tests 64, 107, 108, 116).

Called by `optionTests()`

Calls `missing()`, `confirmBox()`, `withoutRegionals()`, `noStorage()`, `everyRegional()`; in js/tests/helpers_test.js: `expect()`, `check()`

### 11.3.5 `cancellingOptions()` INTERNAL lines 88–106

Tests closing Options without OK, by Cancel and by closing the window otherwise: the changes to Include and the language are thrown away, and the window opens again showing the settings in use (Tests 117, 118).

Called by `optionTests()`

Calls `closedAndFocused()`, `shownKeys()`; in js/view/toolkit.js: `$()`; in js/view/optionsview.js: `languageOptions()`; in js/tests/helpers_test.js: `keys()`, `check()`, `ticked()`

### 11.3.6 `closedAndFocused(by)` INTERNAL lines 108–111

Checks that Cancel or OK has closed the Options window and that focus is back on the Options button (Tests 114, 115; Req. 16; WCAG 2.4.3).

Called by `openingOptions()`, `cancellingOptions()`

Calls in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `check()`

### 11.3.7 `leftOut()` INTERNAL lines 113–120

Lists what the tree shows that Include, unticked, should leave out: the unique items (`uniqueKeys()`), and the groups and items with nothing under them that at most one civilisation has, or all but at most one (Req. 15).

Called by `optionTests()`, `optionKeys()`

Calls `uniqueKeys()`, `shownKeys()`; in js/model/civsets.js: `size()`; in js/model/texts.js: `nameOf()`

### 11.3.8 `missing()` INTERNAL lines 122–125

Lists the items of the full tree that the tree does not show, which with Include ticked must be none.

Called by `confirmingOptions()`
Calls `shownKeys()`; in js/tests/helpers_test.js: `keys()`

### 11.3.9 `someRegional()` INTERNAL lines 127–130

Tells whether the tree shows any regional item.

Called by `optionTests()`, `optionKeys()`
Calls `regionalKeys()`, `shownKeys()`

### 11.3.10 `everyRegional()` INTERNAL lines 132–135

Tells whether the tree shows every regional item.

Called by `confirmingOptions()`
Calls `shownKeys()`, `regionalKeys()`

### 11.3.11 `withoutRegionals()` INTERNAL line 137

Tells whether the tree shows just the items outside regional items and all under them (`outside()`), as it should with Include ticked and Include Regionals unticked.

Called by `optionKeys()`, `confirmingOptions()`
Calls `shownKeys()`, `regionalKeys()`; in js/tests/helpers_test.js: `same()`, `outside()`

### 11.3.12 `languagePopupTests()` EXPORTED lines 145–165

Tests the popup about a language that cannot load: with the next language's file made to fail (`failing()`), choosing it opens the modal popup with its message (Test 233), leaves the page and Options in the language in use (Test 234) and moves and resizes no pane, open or closed (Test 235; Req. 35); the audit runs with the popup open; and its OK closes it (Test 236) and gives focus back to the Options button (Test 237; WCAG 2.4.3).

Called by in js/tests/run_test.js: `runAll()`
Calls `paneBoxes()`, `failing()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `reset()`, `codes()`, `pick()`, `check()`, `same()`, `describe()`; in js/tests/audit_test.js: `audit()`

### 11.3.13 `failing(code, act)` INTERNAL lines 167–182

Makes a language's file fail to load while `act` runs: forgets the language's texts and sends the request for its file (`localeFile()`) to a file that does not exist, waits up to five seconds for the popup to open, then puts both back.

Called by `languagePopupTests()`
Calls in js/model/data/datahandler.js: `localeFile()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `wait()`

### 11.3.14 `paneBoxes()` INTERNAL lines 184–185

The place and size of each pane, to tell whether the popup moves or resizes one (Req. 35).

Called by `languagePopupTests()`
Calls *no function of the code*

### 11.3.15 `confirmBox(id)` INTERNAL line 187

Opens Options, clicks a box and presses OK, as a visitor changing one setting would.

Called by `confirmingOptions()`
Calls in js/view/toolkit.js: `$()`

### 11.3.16 `noStorage()` INTERNAL line 189

Tells whether the browser refuses the page its storage, so that no choice can be remembered.

Called by `confirmingOptions()`
Calls *no function of the code*

### 11.3.17 `shownKeys()` INTERNAL line 191

Lists the items currently in the tree, to compare the default tree with the full one.

Called by `optionTests()`, `optionKeys()`, `openingOptions()`, `cancellingOptions()`, `leftOut()`, `missing()`, `someRegional()`, `everyRegional()`, `withoutRegionals()`
Calls *no function of the code*

### 11.3.18 `uniqueKeys()` INTERNAL lines 193–200

Lists the items the default tree leaves out as unique: those inside unique groups and those aoe2techtree.net calls unique.

Called by `leftOut()`
Calls *no function of the code*

### 11.3.19 `regionalKeys()` INTERNAL

line 202

Lists the items aoe2techtree.net calls regional, which Include Regionals leaves out when unticked.

Called by `someRegional()`, `everyRegional()`, `withoutRegionals()`
Calls in js/tests/helpers_test.js: `keys()`

### 11.3.20 `keptPath()` INTERNAL

lines 204–205

Finds the path of a leaf below the top of the tree, and not regional, to mark while an Include box changes. Paths survive the tree being built again; item objects do not.

Called by `optionTests()`, `optionKeys()`
Calls *no function of the code*

# 11.4 keys_test.js

**Folder:** js/tests/interface/ 149 lines 11 functions, 2 exported

**Imports:** js/model/civsets.js, js/model/texts.js, js/model/model.js, js/view/toolkit.js, js/view/treeview.js, js/controller/optionshandler.js, js/tests/helpers_test.js, js/tests/audit_test.js, js/tests/interface/menu_test.js, js/tests/interface/options_test.js
**Imported by:** js/tests/run_test.js

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The keyboard tests of the tree, Filters and Find, and the runs of the accessibility audit (`audit()`) in many states and every language, which run with the other accessibility tests in the order TESTS.md lists them.

### 11.4.1 `keyboardTests()` EXPORTED lines 17–21

Runs the keyboard tests: `filterFocus()`, `treeMoves()` and `findKey()`.

Called by in js/tests/run_test.js: `runAll()`
Calls `filterFocus()`, `treeMoves()`, `findKey()`

### 11.4.2 `accessibilityTests()` EXPORTED lines 23–32

Runs the audits and the accessibility tests in the order TESTS.md lists them: the menu's audit and keys (`menuKeys()`), the first language, wider text spacing, the keys of Options (`optionKeys()`), then every other language and the Options window.

Called by in js/tests/run_test.js: `runAll()`
Calls `menuAudit()`, `languageAudits()`, `textSpacing()`, `optionAudits()`; in js/tests/helpers_test.js: `codes()`; in js/tests/interface/menu_test.js: `menuKeys()`; in js/tests/interface/options_test.js: `optionKeys()`

### 11.4.3 `filterFocus()` INTERNAL lines 34–50

Tests that a filter button clears its item and hands focus to the next filter button, and that clearing the last one hands it to the civilisations' heading, with every civilisation listed and the Filters list hidden (Tests 193 to 196; Req. 4, 22; WCAG 2.4.3).

Called by `keyboardTests()`
Calls in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `reset()`, `fixtures()`, `mustLack()`, `check()`, `same()`, `listed()`, `expected()`

### 11.4.4 `treeMoves()` INTERNAL lines 52–69

Tests the keys of the tree: Right opens a line all the way down, the steps of `stepKeys()` and `branchKeys()`, End and Home go to the last and the first item, and Down on an arrow goes to the next arrow (Tests 197, 204 to 206; WCAG 2.1.1).

Called by `keyboardTests()`
Calls `stepKeys()`, `branchKeys()`; in js/view/toolkit.js: `$()`; in js/view/treeview.js: `isOpen()`; in js/tests/helpers_test.js: `reset()`, `fixtures()`, `keyDown()`, `nodeOf()`, `check()`, `chainOf()`, `twistyOf()`

### 11.4.5 `stepKeys()` INTERNAL lines 71–80

Tests that Up on the first item keeps focus there, Down goes to the next item and Up to the item above (Tests 198 to 200; WCAG 2.1.1).

Called by `treeMoves()`
Calls in js/tests/helpers_test.js: `nodeOf()`, `keyDown()`, `check()`

### 11.4.6 `branchKeys()` INTERNAL lines 82–91

Tests that Right on an open branch goes to its first item, Left on an item goes back to the branch above it, and Left on an open branch closes it (Tests 201 to 203; WCAG 2.1.1).

Called by `treeMoves()`
Calls in js/view/treeview.js: `isOpen()`; in js/tests/helpers_test.js: `keyDown()`, `nodeOf()`, `check()`

### 11.4.7 `findKey()` INTERNAL lines 93–101

Tests that Enter in Find goes to the first match (Test 207; WCAG 2.1.1).

Called by `keyboardTests()`
Calls in js/model/texts.js: `nameOf()`, `plain()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `reset()`, `fixtures()`, `type()`, `keyDown()`, `check()`, `nodeOf()`

### 11.4.8 `menuAudit()` INTERNAL lines 103–110

Runs the accessibility audit with a mark and the language menu open. This is the audit's first run, so its checks are listed from here.

Called by `accessibilityTests()`
Calls in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `reset()`, `tap()`, `nodeOf()`, `fixtures()`; in js/tests/audit_test.js: `audit()`

### 11.4.9 `languageAudits(code)` INTERNAL

lines 112–125

Runs the accessibility audit in one language: as loaded, with filters, and with everything open. The focus checks run only in the first language.

Called by `accessibilityTests()`
Calls in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `reset()`, `fixtures()`, `pick()`, `codes()`, `mustHave()`, `mustLack()`; in js/tests/audit_test.js: `audit()`

### 11.4.10 `textSpacing()` INTERNAL

lines 127–137

Tests that wider text spacing, as WCAG 1.4.12 describes it, makes nothing stick out sideways, with an item marked (Test 231).

Called by `accessibilityTests()`
Calls in js/tests/helpers_test.js: `reset()`, `tap()`, `nodeOf()`, `fixtures()`, `expect()`, `everything()`, `box()`, `describe()`

### 11.4.11 `optionAudits()` INTERNAL

lines 139–149

Runs the audit with the Options window open, and with everything included and open.

Called by `accessibilityTests()`
Calls in js/view/toolkit.js: `$()`; in js/controller/optionshandler.js: `chooseInclude()`; in js/tests/helpers_test.js: `reset()`; in js/tests/audit_test.js: `audit()`

# 11.5 uniques_test.js

**Folder:** js/tests/interface/ 57 lines 7 functions, 1 exported

**Imports:** js/model/texts.js, js/model/model.js, js/model/data/datahandler.js, js/view/toolkit.js, js/controller/optionshandler.js, js/tests/helpers_test.js

**Imported by:** js/tests/run_test.js

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The tests of the filter buttons of unique items, which keep a button of their own while their line or group is marked too (Req. 36).

## 11.5.1 `uniqueFilterTests()` EXPORTED

lines 11–22

Tests the filter buttons of unique items, with Include ticked: in a line (`inLine()`) and in a full group (`inGroup()`), and that marking the end of a wholly unique line gives it one filter button (Test 188; Req. 36).

Called by in js/tests/run_test.js: `runAll()`

Calls `uniqueCases()`, `inLine()`, `inGroup()`, `names()`; in js/model/texts.js: `nameOf()`; in js/view/toolkit.js: `$()`; in js/controller/optionshandler.js: `chooseInclude()`; in js/tests/helpers_test.js: `reset()`, `openTo()`, `tap()`, `nodeOf()`, `check()`, `same()`

## 11.5.2 `uniqueCases()` INTERNAL

lines 24–29

Picks a unique item in a line with items that are not unique, and a line whose items are all unique. Throws when data.json lacks either, so that the tests fail rather than pass without checking.

Called by `uniqueFilterTests()`

Calls `unique()`

## 11.5.3 `inLine(item)` INTERNAL

lines 31–42

Tests a unique item marked in its line: it gets its own filter button beside the line's, which clears only it, while the line's button, once the item is marked again, leaves it marked (Tests 185 to 187; Req. 36).

Called by `uniqueFilterTests()`

Calls `names()`, `unmark()`; in js/model/texts.js: `nameOf()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `tap()`, `nodeOf()`, `check()`, `same()`

## 11.5.4 `inGroup(item)` INTERNAL

lines 44–50

Tests a unique item in a group marked whole: it keeps its own filter button beside the group's, and the group's button leaves it marked (Tests 185, 187; Req. 36).

Called by `uniqueFilterTests()`

Calls `names()`, `unmark()`; in js/model/texts.js: `nameOf()`; in js/tests/helpers_test.js: `tap()`, `nodeOf()`, `check()`, `same()`

## 11.5.5 `unique(item)` INTERNAL

line 52

Tells whether aoe2techtree.net calls an item unique.

Called by `uniqueCases()`

Calls *no function of the code*

## 11.5.6 `names()` INTERNAL

line 54

Lists the names on the filter buttons, sorted.

Called by `uniqueFilterTests()`, `inLine()`, `inGroup()`

Calls in js/view/toolkit.js: `$()`

## 11.5.7 `unmark(text)` INTERNAL

lines 56–57

Clicks the filter button with a given name, which clears its mark.

Called by `inLine()`, `inGroup()`

Calls in js/view/toolkit.js: `$()`

# 11.6 headings_test.js

**Folder:** js/tests/interface/ 33 lines 2 functions, 1 exported

**Imports:** js/model/model.js, js/view/treeview.js, js/controller/optionshandler.js, js/tests/helpers_test.js
**Imported by:** js/tests/run_test.js

The tests of the headings: a group inside a building, which has no checkbox and only opens and closes (Req. 3).

## 11.6.1 `headingTests()` EXPORTED

lines 10–31

Tests a group inside a building, such as the Castle's Unique Units, with Include ticked: it has no checkbox (Req. 3), a click on its name opens it, neither a click nor a right-click marks anything, and Down from the row above reaches its arrow (Tests 189 to 192; WCAG 2.1.1). Throws when data.json has no such group.

Called by in js/tests/run_test.js: `runAll()`
Calls `unmarked()`; in js/view/treeview.js: `isOpen()`, `shown()`; in js/controller/optionshandler.js: `chooseInclude()`; in js/tests/helpers_test.js: `reset()`, `openTo()`, `check()`, `nodeOf()`, `labelOf()`, `twistyOf()`, `keyDown()`

## 11.6.2 `unmarked()` INTERNAL

line 33

Tells whether no item is marked.

Called by `headingTests()`
Calls *no function of the code*

# 11.7 reporting_test.js

**Folder:** js/tests/interface/ 40 lines 3 functions, 1 exported

**Imports:** js/view/toolkit.js, js/view/report.js, js/controller/optionshandler.js, js/tests/helpers_test.js
**Imported by:** js/tests/run_test.js

The tests of the report window, in which the updater and the tests list what they found: what stays in place, what scrolls, and its Close button.

### 11.7.1 `reportTests()` EXPORTED

lines 10–34

Tests the report window with 200 lines in it: Close has focus (WCAG 2.4.3), only its list scrolls, its title, count and Close stay in place and in view (Rule 18), its download links scroll with the list, and Close sits on the right and is the plain button, with or without Simple UI (Tests 289 to 295).

Called by in js/tests/run_test.js: `runAll()`
Calls `tops()`, `look()`; in js/view/toolkit.js: `$()`; in js/view/report.js: `openReport()`, `report()`; in js/controller/optionshandler.js: `chooseSimple()`; in js/tests/helpers_test.js: `check()`, `box()`

### 11.7.2 `tops(pinned)` INTERNAL

line 36

The tops of the report's title, count and Close, rounded, to tell whether scrolling the list moves them.

Called by `reportTests()`
Calls in js/tests/helpers_test.js: `box()`

### 11.7.3 `look(close)` INTERNAL

lines 38–40

Sums up how Close is drawn, with its `::before` and `::after`: background, border, corners, colour, font and content, to compare it with and without Simple UI.

Called by `reportTests()`
Calls *no function of the code*

# 12 js/tests/pictures/

7 files 802 lines

The tests of the picture look, which run with Simple UI off: the screen's pictures, the boxes and icons, the Find box, the tooltip, the tree's buttons and the scroll bars. pictures_test.js runs them all, and picturetools_test.js holds the tools the others share.

**Subchapters:** 12.1 pictures_test.js, 12.2 picturetools_test.js, 12.3 icons_test.js, 12.4 findbox_test.js, 12.5 tooltips_test.js, 12.6 treebuttons_test.js, 12.7 scrollbars_test.js

## 12.1 pictures_test.js

**Folder:** js/tests/pictures/ 175 lines 15 functions, 1 exported
**Imports:** js/model/civsets.js, js/model/model.js, js/model/data/datahandler.js, js/view/toolkit.js, js/controller/optionshandler.js, js/tests/helpers_test.js, js/tests/pictures/picturetools_test.js, js/tests/pictures/icons_test.js, js/tests/pictures/findbox_test.js, js/tests/pictures/tooltips_test.js, js/tests/pictures/treebuttons_test.js, js/tests/pictures/scrollbars_test.js
**Imported by:** js/tests/run_test.js

The tests of the picture look, run with Simple UI off, and the runner of every test in js/tests/pictures/. There the page need not meet WCAG (Rule 22), nor the UI particulars of REQUIREMENTS.md (Req. 0). This file tests the screen's pictures, the cursors, the Options button, switching Simple UI and the Civ view.

### 12.1.1 `pictureTests()` EXPORTED lines 25–43

Runs the tests of the picture look with Simple UI off: the screen (`screenTests()`), the icons (`iconTests()`), the Find box (`findBoxTests()`), the tooltip (`tooltipTests()`), the tree's buttons once the fonts have loaded (`treeButtonTests()`), Simple UI (`simpleUiTests()`), the Civ view (`civViewTests()`) and the scroll bars (`scrollBarTests()`), then turns Simple UI on again (Rules 9, 18, 22; Req. 0, 32, 33).

Called by in js/tests/run_test.js: `runAll()`
Calls `screenTests()`, `simpleUiTests()`, `civViewTests()`; in js/controller/optionshandler.js: `chooseSimple()`; in js/tests/helpers_test.js: `reset()`; in js/tests/pictures/icons_test.js: `iconTests()`; in js/tests/pictures/findbox_test.js: `findBoxTests()`; in js/tests/pictures/tooltips_test.js: `tooltipTests()`; in js/tests/pictures/treebuttons_test.js: `treeButtonTests()`; in js/tests/pictures/scrollbars_test.js: `scrollBarTests()`

### 12.1.2 `screenTests()` INTERNAL lines 45–63

Tests the pictures as a whole: no page scrolling and the footer without Simple UI, the choice for each screen (`misjudged()`), the background, the sizes of the pictures, the parchment and the cursors, then the Options button (`optionsButtonTests()`) (Tests 29, 103, 104, 119 to 124; Rule 12).

Called by `pictureTests()`
Calls `wrongSizes()`, `offParchment()`, `strayCursors()`, `bigCursors()`, `optionsButtonTests()`; in js/tests/helpers_test.js: `check()`, `footerAsWide()`, `footerHidden()`, `expect()`, `need()`; in js/tests/pictures/picturetools_test.js: `fitted()`, `misjudged()`, `withGrades()`, `picture()`, `grade()`

### 12.1.3 `wrongSizes()` INTERNAL lines 65–72

Lists the pictures that do not suit the parchment chosen, trying every parchment in turn (`withGrades()`): `_sd` and `_x1` pictures with fhd and qhd, `_4k` and `_x2` ones with uhd and ultrawide. Options is open meanwhile, so that its pictures and its backdrop count too (Test 121).

Called by `screenTests()`
Calls in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `everything()`, `reset()`; in js/tests/pictures/picturetools_test.js: `styles()`, `urlsOf()`, `withGrades()`, `fitting()`

### 12.1.4 `offParchment()` INTERNAL lines 74–94

Lists the header, panes and footer that stick out of the parchment, worked out from the five numbers `--ratio`, `--left`, `--top`, `--right` and `--bottom` in pictures.css and the window's size (Test 122).

Called by `screenTests()`
Calls in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `rendered()`, `describe()`

### 12.1.5 `strayCursors()` INTERNAL lines 96–101

Lists the elements without the right cursor: the pointing hand over the links of the civilisations and of the footer, and the sword elsewhere (Test 123).

Called by `screenTests()`
Calls in js/tests/helpers_test.js: `everything()`, `describe()`; in js/tests/pictures/picturetools_test.js: `graded()`

### 12.1.6 `bigCursors()` INTERNAL lines 103–107

Lists the elements whose `_x2` cursor is not set at 1.5x, which draws it at two thirds of its size, as large as the `_x1` ones. Run with the 4K pictures only (Test 124).

Called by `screenTests()`
Calls in js/tests/helpers_test.js: `everything()`, `describe()`

### 12.1.7 `optionsButtonTests()` INTERNAL lines 109–116

Tests the Options button: it shows its picture, at rest its normal one, and its gold ring (Tests 125 to 127).

Called by `screenTests()`
Calls in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `check()`; in js/tests/pictures/picturetools_test.js: `graded()`

### 12.1.8 `simpleUiTests()` INTERNAL lines 118–141

Tests ticking and unticking Simple UI in Options, with an item marked: its box never moves (Rule 18), OK takes every image and the civilisations' links away (Req. 33), the choice is remembered, the title is the full sentence alone (Req. 2), and unticking brings every image and link back (Tests 116, 164 to 168). Throws if the pictures or links are missing to begin with.

Called by `pictureTests()`
Calls `imagesShown()`, `simpleSpot()`, `picturesOn()`, `civLinks()`; in js/model/civsets.js: `size()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `tap()`, `nodeOf()`, `check()`, `expect()`, `rendered()`, `same()`, `ui()`

### 12.1.9 `picturesOn()` INTERNAL lines 143–144

Tells whether the main pictures show: those of the page, of its body and of the Options button.

Called by `simpleUiTests()`
Calls `imagesShown()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `describe()`

### 12.1.10 `imagesShown()` INTERNAL lines 146–147

Names every element that shows a picture: images and everything CSS draws with a picture, the page's own background included. Simple UI must leave it empty (Req. 33).

Called by `simpleUiTests()`, `picturesOn()`
Calls in js/tests/helpers_test.js: `everything()`, `describe()`; in js/tests/pictures/picturetools_test.js: `pictured()`

### 12.1.11 `simpleSpot()` INTERNAL lines 149–152

Returns where the Simple UI box is, to check that switching Simple UI does not move it (Rule 18).

Called by `simpleUiTests()`
Calls in js/view/toolkit.js: `$()`

### 12.1.12 `civLinks()` INTERNAL line 154

Counts the civilisations in the list that are links.

Called by `simpleUiTests()`
Calls in js/view/toolkit.js: `$()`

### 12.1.13 `civViewTests()` INTERNAL lines 156–164

Tests that OK shows the civilisations in the Civ view chosen, as tiles side by side after Tile and not after List, and that the choice is remembered (Tests 116, 169; Req. 16). Throws with fewer than two civilisations listed.

Called by `pictureTests()`
Calls `pickView()`, `sideBySide()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `check()`

### 12.1.14 `pickView(choice)` INTERNAL lines 166–170

Chooses a Civ view, Tile or List, through Options and OK, as a visitor would.

Called by `civViewTests()`
Calls in js/view/toolkit.js: `$()`

### 12.1.15 `sideBySide()` INTERNAL lines 172–175

Tells whether the first two civilisations sit side by side, as tiles do, rather than one under the other.

Called by `civViewTests()`
Calls in js/view/toolkit.js: `$()`

# 12.2 picturetools_test.js

**Folder:** js/tests/pictures/ 70 lines 15 functions, 13 exported

**Imports:** js/view/pictures/pictures.js, js/tests/helpers_test.js
**Imported by:** js/tests/audit_test.js, js/tests/pictures/pictures_test.js, js/tests/pictures/icons_test.js, js/tests/pictures/findbox_test.js, js/tests/pictures/tooltips_test.js, js/tests/pictures/treebuttons_test.js, js/tests/pictures/scrollbars_test.js

Small tools the tests of the picture look share: the size of pictures the page chose, telling a picture by its address, the pictures an element shows, the parchment a screen should get, and checks run for every parchment. `GRADES` gives the size of pictures each parchment goes with.

### 12.2.1 `grade()` EXPORTED line 16

The size of pictures the page chose, `sd` or `4k`, from `data-assets`.

Called by `scale()`; in js/tests/pictures/pictures_test.js: `screenTests()`; in js/tests/pictures/scrollbars_test.js: `comboArrows()`
Calls *no function of the code*

### 12.2.2 `scale()` EXPORTED line 18

The scale of the pictures that come as `_x1` and `_x2`, such as the cursors and the scroll bars: `x2` with the `4k` pictures, `x1` otherwise.

Called by in js/tests/pictures/icons_test.js: `civIcons()`; in js/tests/pictures/scrollbars_test.js: `wrongSprites()`
Calls `grade()`

### 12.2.3 `picture(style, path)` EXPORTED line 20

Tells whether the background in an element's style is a given picture.

Called by in js/tests/pictures/pictures_test.js: `screenTests()`; in js/tests/pictures/icons_test.js: `treeIcon()`; in js/tests/pictures/findbox_test.js: `findBoxTests()`
Calls `at()`

### 12.2.4 `at(url, path)` EXPORTED line 22

Tells whether a CSS `url()` points at a given picture in `assets/images/`.

Called by `picture()`; in js/tests/pictures/findbox_test.js: `framed()`
Calls *no function of the code*

### 12.2.5 `graded(value, base)` EXPORTED line 24

Tells whether a CSS value names a given picture in any of its sizes, `_sd`, `_4k`, `_x1` or `_x2`.

Called by in js/tests/pictures/pictures_test.js: `strayCursors()`, `optionsButtonTests()`; in js/tests/pictures/icons_test.js: `offSprite()`, `offRadio()`, `civIcons()`; in js/tests/pictures/treebuttons_test.js: `cut()`
Calls *no function of the code*

### 12.2.6 `pictured(el)` EXPORTED lines 26–28

Tells whether an element shows a picture through CSS: a background, list marker, border image, content or cursor, in itself or its parts.

Called by in js/tests/audit_test.js: `audit()`; in js/tests/pictures/pictures_test.js: `imagesShown()`
Calls `styles()`

### 12.2.7 `styles(el)` EXPORTED line 30

Gets an element's styles and those of its `::before`, `::after` and `::marker` parts, so decorations are checked too.

Called by `pictured()`; in js/tests/pictures/pictures_test.js: `wrongSizes()`; in js/tests/pictures/treebuttons_test.js: `cut()`
Calls *no function of the code*

### 12.2.8 `urlsOf(style)` EXPORTED lines 32–33

Lists the picture addresses in an element's style: background, border image, list marker, content and cursor.

Called by in js/tests/pictures/pictures_test.js: `wrongSizes()`; in js/tests/pictures/scrollbars_test.js: `sheetOf()`, `sprite()`
Calls `files()`

### 12.2.9 `files(value)` INTERNAL line 35

Reads the addresses out of a CSS value.

Called by `urlsOf()`
Calls *no function of the code*

### 12.2.10 misjudged() EXPORTED lines 37–40

Lists the screens, from phones to ultrawide and from 100% to 300%, for which `screenPictures()` picks another parchment or picture size than `fitted()`.

Called by in js/tests/pictures/pictures_test.js: `screenTests()`
Calls `fitted()`; in js/view/pictures/pictures.js: `screenPictures()`

### 12.2.11 fitted(width = screen.width, height = screen.height, ratio = devicePixelRatio) EXPORTED lines 42–48

Works out on its own, apart from the page's code, which parchment a screen should get from its size and scale, to judge `screenPictures()`.

Called by `misjudged()`; in js/tests/pictures/pictures_test.js: `screenTests()`
Calls `beyond()`

### 12.2.12 beyond(wide, high, [most, tallest]) INTERNAL line 50

Tells whether a screen, in its own pixels, is wider or taller than a given size.

Called by `fitted()`
Calls *no function of the code*

### 12.2.13 fitting(src, background) EXPORTED lines 52–56

Tells whether a picture's address suits a parchment: its own parchment, or the `_sd` and `_x1` pictures with fhd and qhd and the `_4k` and `_x2` ones with uhd and ultrawide.

Called by in js/tests/pictures/pictures_test.js: `wrongSizes()`
Calls *no function of the code*

### 12.2.14 withGrades(look) EXPORTED lines 58–68

Runs a check once for each parchment, with the page's `data-background` and `data-assets` set to it, and returns what the runs found, each once. Puts the page's own choice back afterwards, so that one window can judge every parchment.

Called by in js/tests/pictures/pictures_test.js: `screenTests()`, `wrongSizes()`; in js/tests/pictures/scrollbars_test.js: `scrollBarTests()`
Calls *no function of the code*

### 12.2.15 fitBars() EXPORTED line 70

Fits every scroll bar to its list at once, as resizing the window does, so that the tests need not wait for the browser to draw the page again.

Called by in js/tests/pictures/tooltips_test.js: `everyTipTests()`; in js/tests/pictures/scrollbars_test.js: `scrollBarTests()`, `barStates()`
Calls *no function of the code*

# 12.3 icons_test.js

**Folder:** js/tests/pictures/ 153 lines 18 functions, 1 exported

**Imports:** js/model/icons.js, js/model/model.js, js/model/data/datahandler.js, js/view/toolkit.js, js/controller/optionshandler.js, js/tests/helpers_test.js, js/tests/pictures/picturetools_test.js

**Imported by:** js/tests/pictures/pictures_test.js

The tests of the checkboxes, the radio buttons and the icons in the picture look: each box shows its state's picture, each row and civilisation link its own cell of its atlas, outlined as `colour_key.csv` says, and each atlas fits its grid.

### 12.3.1 `iconTests()` EXPORTED lines 25–40

Runs the tests of the boxes (`boxTests()`) and, with Include ticked so that every item is in the tree, of the icons: a row has an icon exactly when its item or group has a picture (Test 130), the tree's icons (`treeIconTests()`), the civilisations' (`civIconTests()`) and the atlases (`misfit()`; Test 137). Throws unless `colour_key.csv` gives nine outlines.

Called by in js/tests/pictures/pictures_test.js: `pictureTests()`
Calls `boxTests()`, `edges()`, `ownPicture()`, `treeIcon()`, `treeIconTests()`, `civIcons()`, `civIconTests()`, `misfit()`, `atlases()`; in js/controller/optionshandler.js: `chooseInclude()`; in js/tests/helpers_test.js: `reset()`, `nodeOf()`, `expect()`

### 12.3.2 `boxTests()` INTERNAL lines 42–52

Tests, with one item must have and one must lack, that every checkbox shows its state's picture from the checkbox sheet (`offSprite()`) and that the Civ view's radio buttons show theirs (`offRadio()`) (Tests 128, 129). Throws when the tree shows no box of some state.

Called by `iconTests()`
Calls `offSprite()`, `offRadio()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `fixtures()`, `mustHave()`, `mustLack()`, `expect()`, `describe()`

### 12.3.3 `offSprite(el)` INTERNAL lines 54–60

Tells whether a checkbox, those of Options included, does not show the sheet's picture for its state, empty, ticked or crossed, at the sheet's size and place, or has a drawn mark other than the dash of a mixed one.

Called by `boxTests()`
Calls in js/tests/helpers_test.js: `drawn()`; in js/tests/pictures/picturetools_test.js: `graded()`

### 12.3.4 `offRadio(input)` INTERNAL lines 62–67

Tells whether a radio button of the Civ view does not show its picture from its sheet, the lower one when chosen.

Called by `boxTests()`
Calls in js/tests/pictures/picturetools_test.js: `graded()`

### 12.3.5 `edges()` INTERNAL lines 69–74

Reads `colour_key.csv` into the outline colour of each kind and reach of tree icon, so the tests compare the page with the file.

Called by `iconTests()`
Calls `hex()`; in js/model/data/datahandler.js: `load()`

### 12.3.6 `hex(code)` INTERNAL line 76

Turns a colour written as six hex digits into the `rgb(r, g, b)` form the browser reports.

Called by `edges()`
Calls *no function of the code*

### 12.3.7 `ownPicture(item)` INTERNAL line 78

The picture data.json gives a group or an item, if any: from `icons.groups` for a group, from `pictures` for an item.

Called by `iconTests()`
Calls *no function of the code*

### 12.3.8 `treeIcon({ item, own, icon })` INTERNAL lines 80–83

Describes a row's icon for the checks: its item, its picture, the cell of its atlas it should show (`cellIn()`) and how to tell that atlas.

Called by `iconTests()`
Calls `cellIn()`; in js/tests/pictures/picturetools_test.js: `picture()`

### 12.3.9 `treeIconTests(icons, outlines)` INTERNAL lines 85–99

Tests every tree icon: it shows its own cell of its atlas (`offCell()`), sits between its checkbox and its name, is 44 pixels square, and has the outline `colour_key.csv` gives it, a group's none (Tests 131 to 134).

Called by `iconTests()`
Calls `offCell()`, `square()`, `outlined()`; in js/tests/helpers_test.js: `expect()`, `nodeOf()`, `labelOf()`

### 12.3.10 `square(look)` INTERNAL line 101

Tells whether an icon is 44 pixels square.

Called by `treeIconTests()`
Calls *no function of the code*

### 12.3.11 `outlined(look, colour)` INTERNAL lines 103–104

Tells whether an icon has a solid outline of 2 pixels in a given colour.

Called by `treeIconTests()`
Calls *no function of the code*

### 12.3.12 `civIcons()` INTERNAL lines 106–115

Describes the icon of each civilisation link for the checks, as `treeIcon()` does a row's: its cell of the civilisations' atlas at the screen's scale. Throws when no civilisation is listed.

Called by `iconTests()`
Calls `cellIn()`; in js/model/icons.js: `civFile()`; in js/view/toolkit.js: `$()`; in js/tests/pictures/picturetools_test.js: `scale()`, `graded()`

### 12.3.13 `civIconTests(civs)` INTERNAL lines 117–122

Tests that every civilisation link begins with its icon, and that the icon shows its own cell of the atlas (`offCell()`) (Tests 135, 136).

Called by `iconTests()`
Calls `offCell()`; in js/tests/helpers_test.js: `expect()`

### 12.3.14 `cellIn(names, name, side)` INTERNAL lines 124–129

Works out where a picture's cell is in its atlas: the files sorted by name, row by row, in a square grid.

Called by `treeIcon()`, `civIcons()`
Calls *no function of the code*

### 12.3.15 `offCell({ icon, cell, shows })` INTERNAL lines 131–138

Tells whether an icon does not show its cell of its atlas: the wrong file, repeat, box, size or position.

Called by `treeIconTests()`, `civIconTests()`
Calls `share()`

### 12.3.16 `share(part, of)` INTERNAL line 140

Turns a column or row into the percentage CSS needs to show that cell of an atlas.

Called by `offCell()`
Calls *no function of the code*

### 12.3.17 `atlases(icons)` INTERNAL lines 142–143

Collects the atlases that the icons show, each with its grid, for `misfit()`.

Called by `iconTests()`
Calls *no function of the code*

### 12.3.18 `misfit(grids)` INTERNAL lines 145–153

Lists the atlases whose size does not fit their grid of cells, 48 pixels for the tree's and 60 or 104 for the civilisations', which would shift every icon.

Called by `iconTests()`
Calls in js/tests/helpers_test.js: `need()`

# 12.4 findbox_test.js

**Folder:** js/tests/pictures/ 51 lines 4 functions, 1 exported

**Imports:** js/model/texts.js, js/model/model.js, js/view/toolkit.js, js/tests/helpers_test.js, js/tests/pictures/picturetools_test.js
**Imported by:** js/tests/pictures/pictures_test.js

The tests of the Find box in the picture look: its frame, its search icon and clearing cross, its width, and the number found beside it.

### 12.4.1 `findBoxTests()` EXPORTED

lines 13–34

Tests the Find box with the pictures: its frame (`framed()`), the search icon while empty and the clearing cross otherwise, its width of up to 325 pixels, the number found aligned right and on one line beside the box, and a width that typing never changes (Tests 78, 138 to 142; Req. 21; Rule 18). Then tests the cross (`crossTests()`).

Called by in js/tests/pictures/pictures_test.js: `pictureTests()`
Calls `framed()`, `lens()`, `crossTests()`; in js/model/texts.js: `nameOf()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `box()`, `check()`, `rendered()`, `type()`, `beside()`; in js/tests/pictures/picturetools_test.js: `picture()`

### 12.4.2 `lens()` INTERNAL

line 36

The style of the search icon, drawn as the `::after` of the box around Find.

Called by `findBoxTests()`
Calls in js/view/toolkit.js: `$()`

### 12.4.3 `framed(field)` INTERNAL

lines 38–40

Tells whether Find has the text field's frame: its border image, cut 15 pixels in and repeated, not stretched.

Called by `findBoxTests()`
Calls in js/tests/pictures/picturetools_test.js: `at()`

### 12.4.4 `crossTests()` INTERNAL

lines 42–51

Tests that the cross clears the search and hands focus to the Find box, and that the focused box shows no focus outline (Tests 143 to 145).

Called by `findBoxTests()`
Calls in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `check()`, `rendered()`, `describe()`

# 12.5 tooltips_test.js

**Folder:** js/tests/pictures/ 100 lines 9 functions, 1 exported

**Imports:** js/model/model.js, js/model/data/datahandler.js, js/view/toolkit.js, js/view/optionsview.js, js/view/pictures/tooltips.js, js/tests/helpers_test.js, js/tests/pictures/picturetools_test.js

**Imported by:** js/tests/pictures/pictures_test.js

The tests of the tooltip in the picture look: over the tree and in the Options window, Include's, the Tooltips setting, and a tooltip on every control.

### 12.5.1 `tooltipTests()` EXPORTED lines 17–22

Runs the tests of the tooltip: `treeTipTests()`, `includeTipTests()`, `silentTipTests()` and `everyTipTests()`.

Called by in js/tests/pictures/pictures_test.js: `pictureTests()`
Calls `treeTipTests()`, `includeTipTests()`, `silentTipTests()`, `everyTipTests()`

### 12.5.2 `treeTipTests()` INTERNAL lines 24–38

Tests the tooltip over a tree row: it says how to mark it, has its parchment colour, starts at the cursor's bottom-right corner where there is room, and stays inside the window (Tests 146 to 149).

Called by `tooltipTests()`
Calls `inside()`, `hover()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `nodeOf()`, `box()`, `check()`, `ui()`

### 12.5.3 `includeTipTests()` INTERNAL lines 40–51

Tests the tooltip over Include, inside the Options window, with Include's own tooltip, while the description stays hidden but still describes the box to screen readers (Tests 150 to 153).

Called by `tooltipTests()`
Calls `hover()`, `inside()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `check()`, `ui()`, `box()`

### 12.5.4 `silentTipTests()` INTERNAL lines 53–62

Tests that no tooltip shows once Tooltips is unticked, and that the choice is remembered (Tests 116, 154). Then ticks it again.

Called by `tooltipTests()`
Calls `hover()`, `inside()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `nodeOf()`, `check()`

### 12.5.5 `everyTipTests()` INTERNAL lines 64–82

Tests that every control has a tooltip of at most 100 characters, but for Find's, and that a link's tooltip is its address (Tests 155, 156): on the page with every branch open, a mark and a search, then in the Options window and with its language list open (`tipsOver()`).

Called by `tooltipTests()`
Calls `tipsOver()`, `unique()`; in js/view/toolkit.js: `$()`; in js/view/optionsview.js: `setMenu()`; in js/tests/helpers_test.js: `tap()`, `nodeOf()`, `fixtures()`, `type()`, `reset()`, `expect()`, `describe()`; in js/tests/pictures/picturetools_test.js: `fitBars()`

### 12.5.6 `tipsOver(scope)` INTERNAL lines 84–93

Moves the mouse over every control and link in a part of the page that can be seen, is not disabled and is outside the report, and notes the tooltip each shows, if any.

Called by `everyTipTests()`
Calls `hover()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `rendered()`, `box()`

### 12.5.7 `inside(el)` INTERNAL line 95

A point 5 pixels inside an element's top left corner, where the mouse is put over it.

Called by `treeTipTests()`, `includeTipTests()`, `silentTipTests()`
Calls in js/tests/helpers_test.js: `box()`

### 12.5.8 `hover(el, x, y)` INTERNAL lines 97–98

Moves the mouse over an element, to a given point, as a visitor would.

Called by `treeTipTests()`, `includeTipTests()`, `silentTipTests()`, `tipsOver()`
Calls *no function of the code*

### 12.5.9 `unique(names)` INTERNAL line 100

Drops repeated names from a list, as some controls are passed over in more than one part of the page.

Called by `everyTipTests()`
Calls *no function of the code*

# 12.6 treebuttons_test.js

**Folder:** js/tests/pictures/ 74 lines 8 functions, 1 exported

**Imports:** js/model/texts.js, js/model/model.js, js/view/toolkit.js, js/view/treeview.js, js/tests/helpers_test.js, js/tests/pictures/picturetools_test.js
**Imported by:** js/tests/pictures/pictures_test.js

The tests of the two buttons above the tree in the picture look: Clear, and Expand all or Collapse all, side by side, drawn whole, staying put and keeping focus, in every language.

### 12.6.1 `treeButtonTests()` EXPORTED lines 12–30

Tests the tree's buttons: Clear and Expand all, with Collapse all in Expand all's place exactly while a top branch is open (`showing()`), side by side across the pane (`sideBySide()`), drawn whole (`looks()`), the toggle button neither moving nor losing focus when it changes (Rule 18), and all of it in every language (`everyLanguage()`) (Tests 157, 161, 162).

Called by in js/tests/pictures/pictures_test.js: `pictureTests()`
Calls `spots()`, `showing()`, `sideBySide()`, `looks()`, `everyLanguage()`; in js/view/toolkit.js: `$()`; in js/view/treeview.js: `setOpen()`; in js/tests/helpers_test.js: `tap()`, `nodeOf()`, `twistyOf()`, `check()`

### 12.6.2 `showing(ids, when)` INTERNAL lines 32–33

Checks that the buttons shown are the ones given, such as Clear and Collapse all (Test 157).

Called by `treeButtonTests()`
Calls `shown()`; in js/tests/helpers_test.js: `check()`, `same()`, `describe()`

### 12.6.3 `sideBySide()` INTERNAL lines 35–44

Tests that the two buttons fill the pane's width side by side, equally wide (Test 158).

Called by `treeButtonTests()`
Calls `shown()`, `spots()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `box()`, `check()`

### 12.6.4 `looks()` INTERNAL lines 46–52

Tests that each button shows the whole button picture (`cut()`) and that no caption is cut (Tests 159, 160).

Called by `treeButtonTests()`, `everyLanguage()`
Calls `shown()`, `cut()`; in js/tests/helpers_test.js: `expect()`, `describe()`

### 12.6.5 `everyLanguage(sideways)` INTERNAL lines 54–64

Picks every other language in turn, ending with the first, and tests that the buttons keep their size and place (Test 163; Rule 18) and are drawn whole, with Expand all and with Collapse all (`looks()`).

Called by `treeButtonTests()`
Calls `spots()`, `looks()`; in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `codes()`, `pick()`, `check()`

### 12.6.6 `spots(top = true)` INTERNAL lines 66–67

The place and size of each button shown, rounded, with or without their tops.

Called by `treeButtonTests()`, `sideBySide()`, `everyLanguage()`
Calls `shown()`; in js/tests/helpers_test.js: `box()`

### 12.6.7 `shown()` INTERNAL line 69

Lists the buttons above the tree that can be seen.

Called by `showing()`, `sideBySide()`, `looks()`, `spots()`
Calls in js/view/toolkit.js: `$()`; in js/tests/helpers_test.js: `rendered()`

### 12.6.8 `cut(el)` INTERNAL lines 71–74

Tells whether a button is not drawn as the whole button picture: the wrong file or sheet size, drawn parts left on, or not 290 by 50 in proportion.

Called by `looks()`
Calls in js/tests/helpers_test.js: `box()`; in js/tests/pictures/picturetools_test.js: `graded()`, `styles()`

# 12.7 scrollbars_test.js

**Folder:** js/tests/pictures/ 179 lines 19 functions, 1 exported

**Imports:** js/model/model.js, js/view/toolkit.js, js/view/optionsview.js, js/view/pictures/scrollbars.js, js/controller/optionshandler.js, js/tests/helpers_test.js, js/tests/pictures/picturetools_test.js

**Imported by:** js/tests/pictures/pictures_test.js

The tests of the game's scroll bars in the picture look, and the helpers that judge a bar. Each check of a bar is a function named in BAR_TESTS with its test, run on every bar shown in each state barStates() is given.

### 12.7.1 scrollBarTests() EXPORTED

lines 26–52

Tests the game's scroll bars: they hide the browser's own (ownBars()), every bar is right (barStates()) with every branch closed, with every branch open and a filter, scrolled to the end, in Options and with the language list open, the tree keeps its width as its bar comes, the down arrow and the wheel scroll (scrollTests()), the language list's bar has the combo box's arrows (comboArrows()) and its wheel scrolls a language at a time (languageWheel()), and with Simple UI the lists have the browser's own bars and none of the game's (Tests 170, 174, 181, 183, 184). Throws when the tree does not scroll with every branch open.

Called by in js/tests/pictures/pictures_test.js: pictureTests()
Calls ownBars(), barStates(), scrollTests(), comboArrows(), languageWheel(); in js/view/toolkit.js: $(); in js/view/optionsview.js: setMenu(); in js/view/pictures/scrollbars.js: barsByBox(); in js/controller/optionshandler.js: chooseSimple(); in js/tests/helpers_test.js: box(), expect(), tap(), nodeOf(), check(), reset(), describe(); in js/tests/pictures/picturetools_test.js: fitBars(), withGrades()

### 12.7.2 ownBars(width) INTERNAL

lines 54–55

Lists the lists whose own browser scroll bar is not as wanted: hidden with the pictures and back with Simple UI.

Called by scrollBarTests()
Calls in js/view/pictures/scrollbars.js: barsByBox(); in js/tests/helpers_test.js: describe()

### 12.7.3 barStates(when) INTERNAL

lines 57–68

Tests the scroll bars in one state of the page: a list has the game's bar exactly while it scrolls, and never with Simple UI (Test 171), and every bar shown passes each check of BAR_TESTS (Tests 172 to 178).

Called by scrollBarTests()
Calls partsOf(); in js/view/toolkit.js: simple(); in js/view/pictures/scrollbars.js: barsByBox(); in js/tests/helpers_test.js: rendered(), expect(), describe(); in js/tests/pictures/picturetools_test.js: fitBars()

### 12.7.4 partsOf({ list, bar }) INTERNAL

lines 70–80

Gathers what the checks of a bar need: its list, arrows, track and knob and their boxes, the list's style, whether it is the language list, the pictures that did not load, and the proportions of an arrow and of the knob in their sheets (frameShape()).

Called by barStates()
Calls sheetOf(), loaded(), frameShape(); in js/view/toolkit.js: $(); in js/tests/helpers_test.js: describe(), box()

### 12.7.5 sheetOf(el) INTERNAL

line 82

Loads the picture sheet that an element draws from, to read its size.

Called by partsOf()
Calls in js/tests/pictures/picturetools_test.js: urlsOf()

### 12.7.6 loaded(images) INTERNAL

lines 84–91

Waits until every image has loaded or failed, or ten seconds have passed, so the tests check pictures only once they are there, then stops its own timer and lists the images that did not load.

Called by partsOf()
Calls settled()

### 12.7.7 settled(img) INTERNAL

lines 93–94

Waits until an image has loaded or failed.

Called by loaded()
Calls *no function of the code*

### 12.7.8 frameShape(sheet, columns, rows) INTERNAL

line 96

The height of one picture in a sheet divided by its width, from the sheet's own size, so a test can tell a picture drawn in proportion from a stretched one.

Called by partsOf()
Calls *no function of the code*

### 12.7.9 `unloadedSheets({ name, unloaded })` INTERNAL lines 98–100

Names a bar whose picture sheets did not load (Test 172).

Called by `the top-level code`
Calls in js/tests/helpers_test.js: `need()`

### 12.7.10 `offEdge({ name, list, style, outer, inner })` INTERNAL lines 102–106

Names a bar that does not run along its list's right edge, inside its border and as high as its inside (Test 173).

Called by `the top-level code`
Calls in js/tests/helpers_test.js: `need()`

### 12.7.11 `outOfRoom({ name, style, outer, inner })` INTERNAL lines 108–111

Names a bar that is not in the room kept for it in its list's right padding, where it covers nothing and moves nothing (Test 174; Rule 18).

Called by `the top-level code`
Calls in js/tests/helpers_test.js: `need()`

### 12.7.12 `wrongSprites({ name, menu, up, track, knob, down })` INTERNAL lines 113–122

Names the parts of a bar that do not show their own pictures, unstretched: the arrows from the endings' sheet, or the combo box's on the language list, the track repeated downwards, and the knob (Test 175).

Called by `the top-level code`
Calls `sprite()`; in js/tests/pictures/picturetools_test.js: `scale()`

### 12.7.13 `sprite(el)` INTERNAL lines 124–128

Describes how an element draws its picture: the file, its size, its repeat and its position.

Called by `wrongSprites()`
Calls in js/tests/pictures/picturetools_test.js: `urlsOf()`

### 12.7.14 `unequal({ name, menu, up, track, knob, down, outer })` INTERNAL lines 130–133

Names a bar whose arrows, body and knob are not equally wide; the language list's arrows, the combo box's, are left out (Test 176).

Called by `the top-level code`
Calls in js/tests/helpers_test.js: `need()`, `box()`

### 12.7.15 `misshapen({ name, menu, up, knob, down, ends, grip })` INTERNAL lines 135–140

Names a bar whose arrows or knob are out of their pictures' proportions; the language list's arrows are left out (Test 177).

Called by `the top-level code`
Calls in js/tests/helpers_test.js: `box()`, `need()`

### 12.7.16 `strayKnob({ name, list, track, knob })` INTERNAL lines 142–146

Names a bar whose knob is not where its list is scrolled to (Test 178).

Called by `the top-level code`
Calls in js/tests/helpers_test.js: `box()`, `need()`

### 12.7.17 `scrollTests(list, bar)` INTERNAL lines 148–162

Tests that a press on the down arrow scrolls its list a line at once, and that the wheel over the bar scrolls it at once, without animation (Tests 179, 180; Rule 11).

Called by `scrollBarTests()`
Calls in js/tests/helpers_test.js: `check()`

### 12.7.18 `comboArrows(board)` INTERNAL lines 164–171

Names the arrows of the language list's bar that are not of the combo box's size, 19 by 14 pixels, or on a 4K screen 35 by 28 of the screen's own pixels, or not centred on the bar, for one parchment (Test 181).

Called by `scrollBarTests()`
Calls in js/view/toolkit.js: `$()`; in js/view/pictures/scrollbars.js: `barsByBox()`; in js/tests/helpers_test.js: `box()`; in js/tests/pictures/picturetools_test.js: `grade()`

### 12.7.19 `languageWheel()` INTERNAL lines 173–179

Tests that the wheel scrolls the language list by one language at once (Test 182; Rule 11).

Called by `scrollBarTests()`
Calls in js/view/toolkit.js: `$()`; in js/view/optionsview.js: `languageOptions()`; in js/tests/helpers_test.js: `check()`

# 13 js/tests/static/

5 files 519 lines

The tests that read the files instead of using the page: scanner_test.js reads the scripts and style sheets for the others, rules_test.js and layers_test.js check the rules of RULES.md and of model, view and controller, data_test.js looks for mistakes in data.json and the language files, and updates_test.js checks the updater.

**Subchapters:** 13.1 scanner_test.js, 13.2 rules_test.js, 13.3 layers_test.js, 13.4 data_test.js, 13.5 updates_test.js

## 13.1 scanner_test.js

**Folder:** js/tests/static/ 155 lines 14 functions, 8 exported
**Imports:** js/model/data/datahandler.js
**Imported by:** js/tests/run_test.js, js/tests/static/rules_test.js, js/tests/static/layers_test.js

The scanner that the rules tests read the code with: it reads every script and style sheet of the page as text, follows the scripts' imports, and finds the functions of a script by pairing its braces outside its strings and regular expressions. It holds no tests itself.

### 13.1.1 `scripts()` EXPORTED lines 18–38

Reads every script reachable from `js/script.js`, following its imports, with the path and the names of each (`importsIn()`), and notes which ones load only through a later import, as the tests do.

Called by in js/tests/run_test.js: `runAll()`
Calls `importsIn()`; in js/model/data/datahandler.js: `load()`

### 13.1.2 `importsIn(code)` INTERNAL lines 40–44

Lists the imports of a script: each static `import` and `export … from`, with the names it brings in, and each dynamic import, which loads later.

Called by `scripts()`
Calls `namesIn()`

### 13.1.3 `namesIn(clause = "")` INTERNAL lines 46–47

Reads the names in the braces of an import as they are exported, without their `as` renamings, so the tests can tell which file imports what.

Called by `importsIn()`
Calls *no function of the code*

### 13.1.4 `styles()` EXPORTED lines 49–50

Reads every style sheet that index.html links, with its address.

Called by in js/tests/run_test.js: `runAll()`
Calls in js/model/data/datahandler.js: `load()`

### 13.1.5 `guardScanner(files)` EXPORTED lines 52–56

Makes sure the scanner can be trusted before the rules tests use it: it throws when the function finder misreads its sample, or when a script has an import that `scripts()` could not follow, so that the tests fail "the tests ran to the end" instead of passing over code they never read.

Called by in js/tests/static/rules_test.js: `sourceTests()`
Calls `functionsOf()`, `importLines()`

### 13.1.6 `importLines(code)` INTERNAL lines 58–59

Counts the import lines of a script and its `import(` calls, to compare with the imports `scripts()` followed.

Called by `guardScanner()`
Calls *no function of the code*

### 13.1.7 `functionsOf(code)` EXPORTED lines 61–77

Finds where every function body in a script starts and ends, by pairing its braces in the blanked text.

Called by `guardScanner()`; in js/tests/static/rules_test.js: `longFunctions()`
Calls `blanked()`, `opensBody()`

### 13.1.8 `opensBody(text, at, paren)` INTERNAL lines 79–85

Tells whether a brace opens a function's body: after =>, or after a parameter list that follows a name which is not `if`, `for`, `while`, `switch`, `catch` or `with`.

Called by `functionsOf()`
Calls *no function of the code*

### 13.1.9 `blanked(code)` EXPORTED lines 87–131

Blanks out the insides of strings, regular expressions and template texts, keeping the code inside `${}` and every line break, so that only real braces and parentheses are left to count.

Called by `functionsOf()`; in js/tests/static/rules_test.js: `codeTests()`
Calls `regexMayStart()`, `closing()`

### 13.1.10 `closing(code, at)` INTERNAL lines 133–146

Finds where a string or regular expression that starts at a place ends, minding escapes and, for a regular expression, character sets.

Called by `blanked()`
Calls *no function of the code*

### 13.1.11 `regexMayStart(last, word)` INTERNAL lines 148–149

Tells whether a / after a given character or word starts a regular expression rather than a division, so `blanked()` skips the right text.

Called by `blanked()`
Calls *no function of the code*

### 13.1.12 `linesOfCode(code)` EXPORTED line 151

Counts the lines that are not blank, as Rules 23 and 25 count a function or a file.

Called by in js/tests/static/rules_test.js: `sizeTests()`, `longFunctions()`
Calls *no function of the code*

### 13.1.13 `named(files, test)` EXPORTED line 153

Lists the paths of the files a test picks out, the offenders a rules test names.

Called by in js/tests/static/rules_test.js: `codeTests()`, `loadingTests()`, `dependencies()`, `assetTests()`; in js/tests/static/layers_test.js: `pageTests()`, `modelTests()`, `viewTests()`
Calls *no function of the code*

### 13.1.14 `testing({ path })` EXPORTED line 155

Tells whether a file is one of the tests, in `js/tests/`.

Called by in js/tests/static/rules_test.js: `loadingTests()`; in js/tests/static/layers_test.js: `layerTests()`, `modelTests()`, `viewTests()`
Calls *no function of the code*

# 13.2 rules_test.js

**Folder:** js/tests/static/ 134 lines 14 functions, 1 exported
**Imports:** js/model/data/datahandler.js, js/tests/helpers_test.js, js/tests/static/scanner_test.js, js/tests/static/layers_test.js
**Imported by:** js/tests/run_test.js

The tests of the rules for the code: comments, CSS and HTML in the scripts, how the scripts load, file names, dependencies, the sizes of files, functions and lines, pictures, sounds and animations, and game data in the scripts. They read the page's files as text through scanner_test.js; the tests of model, view and controller are in layers_test.js.

### 13.2.1 `sourceTests(page, sheets, files)` EXPORTED

lines 28–36

Runs the rules tests on index.html, up to the code that Live Server injects, the style sheets and the scripts, once `guardScanner()` has made sure the scanner reads them right: `codeTests()`, `loadingTests()`, `sizeTests()`, `layerTests()` and `assetTests()`.

Called by in js/tests/run_test.js: `runAll()`
Calls `codeTests()`, `loadingTests()`, `sizeTests()`, `assetTests()`; in js/tests/static/scanner_test.js: `guardScanner()`; in js/tests/static/layers_test.js: `layerTests()`

### 13.2.2 `codeTests(own, sheets, files)` INTERNAL

lines 38–48

Tests that index.html, the style sheets and the scripts have no comments, with `//` looked for only outside strings (Rule 7), and that the scripts set no styles but numbers for CSS custom properties and build no markup (Rule 2; Tests 238 to 240).

Called by `sourceTests()`
Calls in js/tests/helpers_test.js: `expect()`, `need()`; in js/tests/static/scanner_test.js: `named()`, `blanked()`

### 13.2.3 `loadingTests(sheets, files)` INTERNAL

lines 50–57

Tests that the tests load only when testing and every other script with the page (Rules 4, 6), that no two files share a name (Rule 29), and that the code has no dependencies (Rule 14; Tests 241 to 244).

Called by `sourceTests()`
Calls `dependencies()`; in js/tests/helpers_test.js: `expect()`; in js/tests/static/scanner_test.js: `named()`, `testing()`

### 13.2.4 `dependencies(sheets, files)` INTERNAL

lines 59–70

Lists what the code takes from outside the project: a style sheet linked from elsewhere, an import that is not relative or leads out of the project, `require`, an `@import` in a style sheet, and a `url()` outside the project.

Called by `loadingTests()`
Calls `outward()`, `urlsIn()`; in js/tests/static/scanner_test.js: `named()`

### 13.2.5 `outward({ specifier, target })` INTERNAL

line 72

Tells whether an import is not relative or leads out of the project.

Called by `dependencies()`
Calls *no function of the code*

### 13.2.6 `sizeTests(own, sheets, files)` INTERNAL

lines 74–88

Tests that no file has over 400 lines of code, no function over 60 and no line of a script over 120 characters (Rules 23 to 25), and that a top-level function with statements is a `function` declaration (Rule 6; Tests 245 to 248).

Called by `sourceTests()`
Calls `longFunctions()`, `wide()`, `widths()`, `arrows()`; in js/tests/helpers_test.js: `expect()`; in js/tests/static/scanner_test.js: `linesOfCode()`

### 13.2.7 `longFunctions(code, lines)` INTERNAL

lines 90–100

Names every function over 60 lines of code, with its line (Rule 23).

Called by `sizeTests()`
Calls in js/tests/static/scanner_test.js: `linesOfCode()`, `functionsOf()`

### 13.2.8 `wide(lines)` INTERNAL

line 102

Lists the lines of a script that are over 120 characters (Rule 24).

Called by `sizeTests()`, `widths()`
Calls *no function of the code*

### 13.2.9 `widths({ path, lines })` INTERNAL

lines 104–105

Says how many lines of a script are over 120 characters, and from which line, for a failure's detail.

Called by `sizeTests()`
Calls `wide()`

### 13.2.10 `arrows(lines)` INTERNAL

line 107

Lists the lines of a script where a top-level arrow function opens a block of statements, which should be a `function` declaration (Test 248).

Called by `sizeTests()`
Calls *no function of the code*

### 13.2.11 `assetTests(own, sheets, files)` INTERNAL

lines 109–125

Tests that no image in a template has a source, that only the picture look's style rules name pictures (Rules 9, 22), that the page plays no sounds (Rule 10), that nothing moves but the loading spin (Rule 11), and that the scripts name no civilisation or item of the game (Rule 1; Tests 262 to 266).

Called by `sourceTests()`
Calls `urlsIn()`, `inPictureLook()`, `unspun()`; in js/tests/helpers_test.js: `expect()`, `keys()`; in js/tests/static/scanner_test.js: `named()`

### 13.2.12 `inPictureLook(selector)` INTERNAL

line 127

Tells whether every selector of a style rule is under `.pictures`, the class of the picture look, so that Simple UI shows none of its pictures.

Called by `assetTests()`
Calls *no function of the code*

### 13.2.13 `unspun(sheet)` INTERNAL

line 129

Takes the loading picture's spin out of a style sheet's text, the only animation allowed, before it is searched for others.

Called by `assetTests()`
Calls *no function of the code*

### 13.2.14 `urlsIn({ path, code })` INTERNAL

lines 131–134

Lists every `url()` in a style sheet, with the selector of its rule and the full address it names.

Called by `dependencies()`, `assetTests()`
Calls *no function of the code*

# 13.3 layers_test.js

**Folder:** js/tests/static/ 86 lines 9 functions, 1 exported
**Imports:** js/model/model.js, js/tests/helpers_test.js, js/tests/static/scanner_test.js
**Imported by:** js/tests/static/rules_test.js

The tests that model, view and controller stay apart (Rule 6), read from the scripts' text and imports through scanner_test.js, and from the model's items once the other tests have run.

### 13.3.1 `layerTests(files)` EXPORTED lines 24–29

Runs the tests of model, view and controller: `pageTests()` on the scripts of the page, then `importTests()`, `modelTests()` and `viewTests()` on every script.

Called by in js/tests/static/rules_test.js: `sourceTests()`
Calls `pageTests()`, `importTests()`, `modelTests()`, `viewTests()`; in js/tests/static/scanner_test.js: `testing()`

### 13.3.2 `pageTests(scripts)` INTERNAL lines 31–38

Tests that only the views touch the page, and that only js/model/data/ touches the address and loads or stores data (Tests 249 to 251).

Called by `layerTests()`
Calls in js/tests/helpers_test.js: `expect()`; in js/tests/static/scanner_test.js: `named()`

### 13.3.3 `importTests(files)` INTERNAL lines 40–45

Tests that the model imports only the model, and that no view imports a controller (Tests 252, 253).

Called by `layerTests()`
Calls `within()`, `reaching()`; in js/tests/helpers_test.js: `expect()`

### 13.3.4 `modelTests(files)` INTERNAL lines 47–69

Tests that no script outside js/model/ and the tests works on the model's sets or imports civsets.js, that only page.js observes the model, that model.js shares `subscribe` but not `notify`, that model.js and itemtree.js hold no texts, and that no item holds a page element (Tests 254 to 258).

Called by `layerTests()`
Calls `importing()`, `exporting()`; in js/tests/helpers_test.js: `expect()`, `need()`; in js/tests/static/scanner_test.js: `testing()`, `named()`

### 13.3.5 `viewTests(files)` INTERNAL lines 71–77

Tests that no view stores data, that only the model saves or restores its own state, Include, Include Regionals and the language, and that no view names a mark in a call to its controller (Tests 259 to 261).

Called by `layerTests()`
Calls `within()`; in js/tests/helpers_test.js: `expect()`; in js/tests/static/scanner_test.js: `named()`, `testing()`

### 13.3.6 `within(folder)` INTERNAL line 79

Picks the files in a folder.

Called by `importTests()`, `viewTests()`
Calls *no function of the code*

### 13.3.7 `reaching(file, wrong)` INTERNAL lines 81–82

Lists the imports of a file that lead where they should not, as "... imports ...".

Called by `importTests()`
Calls *no function of the code*

### 13.3.8 `importing(file, name)` INTERNAL line 84

Tells whether a file imports a name.

Called by `modelTests()`
Calls *no function of the code*

### 13.3.9 `exporting(code, name)` INTERNAL line 86

Tells whether a script exports a name as a `const`, `let` or `function`.

Called by `modelTests()`
Calls *no function of the code*

# 13.4 data_test.js

**Folder:** js/tests/static/ 115 lines 15 functions, 2 objects, 2 exported
**Imports:** js/model/itemtree.js, js/model/updater.js, js/model/data/datahandler.js, js/tests/helpers_test.js
**Imported by:** js/tests/run_test.js

The tests of the data: that every language has every text and name, and that data.json holds its tree, its items, its texts and its source as the page and the updater expect.

### 13.4.1 FOOTER OBJECT, INTERNAL lines 9–14

The fields of `source` in data.json that the footer names, each with its check: the name, the address, and the game update's number and address (Test 281; Req. 5).

Called by `sourceDataTests()`
Calls *no function of the code*

### 13.4.2 UPDATER OBJECT, INTERNAL lines 15–18

The fields of `source` that the updater needs, each with its check: the address of the game's data and the offset of its help texts (Test 282).

Called by `sourceDataTests()`
Calls *no function of the code*

### 13.4.3 textTests() EXPORTED lines 20–28

Tests that every language has all of the page's text, and names every group, item and civilisation (Tests 1 to 3; Req. 11).

Called by in js/tests/run_test.js: `runAll()`
Calls `missing()`; in js/model/itemtree.js: `treeKeys()`; in js/tests/helpers_test.js: `codes()`, `expect()`, `keys()`

### 13.4.4 missing(code, part, names) INTERNAL lines 30–31

Lists the names a language lacks in one part of its texts, such as `ui` or `civs`.

Called by `textTests()`
Calls *no function of the code*

### 13.4.5 dataTests() EXPORTED lines 33–39

Runs the checks of data.json: its tree, its items, its texts and its source.

Called by in js/tests/run_test.js: `runAll()`
Calls `treeTests()`, `itemTests()`, `textDataTests()`, `sourceDataTests()`; in js/tests/helpers_test.js: `keys()`

### 13.4.6 treeTests(all) INTERNAL lines 41–55

Tests the tree in data.json: every item once, only known fields, `"unique"` and `"lines"` only on groups and `"own"` only on items, and `"excludes"` and `"replaces"` naming other items of the tree (Tests 267 to 272).

Called by `dataTests()`
Calls `nodesOf()`, `misflagged()`, `linksOf()`; in js/tests/helpers_test.js: `expect()`

### 13.4.7 nodesOf(list) INTERNAL line 57

Lists every node of a tree, those under the others included.

Called by `treeTests()`
Calls *no function of the code*

### 13.4.8 misflagged(flag, kind) INTERNAL line 59

Makes the question whether a node holds a flag wrongly: as anything but `true`, or on the wrong kind of node.

Called by `treeTests()`
Calls *no function of the code*

### 13.4.9 linksOf(node, link) INTERNAL line 61

Lists the items a node's `"excludes"` or `"replaces"` names, one or several.

Called by `treeTests()`
Calls *no function of the code*

### 13.4.10 itemTests(all) INTERNAL lines 63–78

Tests every item of data.json: its availability names only known civilisations, it has an icon, and it is common, regional or unique; and each entry of `extras` is a tree item with a name number and known civilisations, stored as the updater writes it (Tests 273 to 277).

Called by `dataTests()`
Calls `knownCivs()`, `kept()`; in js/tests/helpers_test.js: `expect()`, `need()`

### 13.4.11 `knownCivs(rule)` INTERNAL

lines 80–81

Tells whether an availability, `"all"` or a list under `has` or `lacks`, names only known civilisations.

Called by `itemTests()`
Calls *no function of the code*

### 13.4.12 `kept(key, extra)` INTERNAL

lines 83–87

Tells whether data.json stores an extra's availability and icon as the updater makes them of it, through `fromExtra()`.

Called by `itemTests()`
Calls `alike()`; in js/model/updater.js: `fromExtra()`, `availabilityOf()`, `pictureOf()`

### 13.4.13 `alike(a, b)` INTERNAL

line 89

Tells whether two values are equal, by their JSON.

Called by `kept()`
Calls *no function of the code*

### 13.4.14 `textDataTests(all)` INTERNAL

lines 91–101

Tests the texts in data.json: counted texts with an "other" form, the same placeholders in every language, and overrides of known names (Tests 278 to 280).

Called by `dataTests()`
Calls `holes()`; in js/tests/helpers_test.js: `codes()`, `expect()`

### 13.4.15 `holes(value)` INTERNAL

line 103

Lists the placeholders of a text, such as `{n}`, once each and sorted, to compare a translation with English.

Called by `textDataTests()`
Calls *no function of the code*

### 13.4.16 `sourceDataTests()` INTERNAL

lines 105–112

Tests that the source the footer names, with its game update's number and address, the updater's source, the tool texts and the ignore list are filled in (Tests 281 to 284).

Called by `dataTests()`
Calls `unfilled()`, `FOOTER()`, `UPDATER()`; in js/tests/helpers_test.js: `expect()`, `need()`

### 13.4.17 `unfilled(fields, source)` INTERNAL

lines 114–115

Lists the fields of the source that their checks, in `FOOTER` or `UPDATER`, find empty.

Called by `sourceDataTests()`
Calls *no function of the code*

# 13.5 updates_test.js

**Folder:** js/tests/static/ 29 lines 3 functions, 1 exported

**Imports:** js/model/civsets.js, js/model/updater.js, js/model/data/datahandler.js, js/tests/helpers_test.js
**Imported by:** js/tests/run_test.js

---

The tests of the updater's writing, run without going online.

### 13.5.1 `updaterTests()` EXPORTED lines 6–19

Tests the updater's writing without going online: each language goes in its own file, the files it writes read back the same, availability is stored in its shortest form, and stored names are already tidy (Tests 285 to 288).

Called by in js/tests/run_test.js: `runAll()`
Calls `readBack()`, `shortest()`; in js/model/updater.js: `toJson()`, `joinLines()`; in js/model/data/datahandler.js: `pack()`, `localeFile()`; in js/tests/helpers_test.js: `codes()`, `check()`, `same()`, `expect()`, `keys()`

### 13.5.2 `readBack(written, languages)` INTERNAL lines 21–27

Reads the files the updater would write back into the page's data, as the page reads them at the start.

Called by `updaterTests()`
Calls in js/model/data/datahandler.js: `unpack()`, `addLanguage()`

### 13.5.3 `shortest(key)` INTERNAL line 29

Works out the shortest form of an item's availability, as the updater stores it.

Called by `updaterTests()`
Calls in js/model/civsets.js: `civsOf()`, `setOf()`; in js/model/updater.js: `availabilityOf()`

# 14 locales/

1 file 612 lines

One file for each language but English, all laid out alike, so only de.json is described here.

**Subchapters:** 14.1 de.json

---

## 14.1 de.json

**Folder:** locales/ 612 lines 2 keys

The German texts. Every other language has a file like it in `locales/`, which the page loads when that language is chosen (`localeFile()`) and the updater writes. It holds the two keys of data.json that differ by language, with the same keys inside.

| Lines | Key | What is there |
|---|---|---|
| 2–6 | `overrides` | German names to use instead of the game's own. |
| 7–611 | `text` | The page's texts in German, in the same four parts as data.json's `text`. |
| 8–73 | `ui` | The page's own texts. |
| 74–80 | `groups` | The name of each group. |
| 81–138 | `civs` | The name of each civilisation. |
| 139–610 | `items` | The name of each item. |

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